Categories for the Working Mathematician - Self Study

Grant Talbert

Summer, 2025

BOSTON UNIVERSITY

Abstract

This text contains my notes on Categories for the Working Mathematician, by Saunders Mac Lane. I read this book independently, starting in mid spring semester of my sophomore year, 2025. My notes largely paraphrase the text, and I work out whatever examples I see fit.

Contents

Chapter 1

Categories, Functors, and Natural Transformations

1.1 Axioms for Categories

We start by describing categories using *metagraphs*, which do not employ the use of set theory. A metagraph consists of objects, arrows, and two operatons:

- Domain, which maps arrows to an object $\operatorname{dom} f = a$;
- Codomain, which maps arrows to an object $\operatorname{cod} f = b$.

We write $f : a \to b$ to indicate f is an arrow with $\operatorname{dom} f = a$ and $\operatorname{cod} f = b$.

A *metacategory* is a metagraph together with two additional operations:

- Identity, assigning each item a an arrow 1_a;
- Composite, assigning each pair (f, g) of arrows with $\operatorname{dom} g = \operatorname{cod} f$, an arrow $g \circ f : \operatorname{dom} f \to \operatorname{cod} g$ called their composite;

such that composition is associative and the identity arrows 1_a are identities with respect to composition. Alternative notations for the identity 1_a include id_a, and simply a itself. Examples of metacategories include the usual category **Set** of sets and **Grp** of groups, topological spaces with continuous maps as arrows, and compact Hausdorff spaces with the same as arrows.

Since identity arrows are uniquely determined for each object, it's possible to dispense entirely of the objects in a metacategory and define an arrows-only metacategory C. There exist a collection of arrows, for which some pairs (f, g) of arrows are called *composable*. For any composable arrows, we say the composition $g \circ f$ is defined, and we can then define an identity arrow to be any arrow u satisfying $f \circ u = f$ and $u \circ g = u$ for any f, g such that the composition is defined. The data must then satisfy the axioms

- $(k \circ g) \circ f$ is defined iff $k \circ (g \circ f)$ is defined, and they are equal;
- The composite kgf is defined if and only if the composites kg and gf are defined;
- For each arrow g of C, there exist identity arrows u, u' such that gu and $u'g$ are defined.

The final axiom indeed shows that identity arrows are unique in this regard, and allows defining of a domain and codomain for the arrow g. An arrows-only metacategory thus satisfies the usual metacategory axioms, if identity arrows are taken as objects.

1.2 Categories

Definition 1.1: Directed Graph

A directed graph is a set O of objects, a set A of arrows, and two functions

$$A \underset{\mathrm{cod}}{\overset{\mathrm{dom}}{\rightrightarrows}} O$$

where the set of composable pairs is the set

$$A \times_O A = \{(g, f) \mid g, f \in A, \operatorname{dom} g = \operatorname{cod} f\}.$$

Definition 1.2: Category

A category is a directed graph with two additional functions

$$O \xrightarrow{\mathrm{id}} A, \qquad A \times_O A \xrightarrow{\circ} A,$$

$$c \longmapsto 1_c, \qquad (g, f) \longmapsto g \circ f,$$

where $g \circ f$ is usually written gf, such that

$$\operatorname{dom} 1_a = \operatorname{cod} 1_a, \qquad \operatorname{dom} gf = \operatorname{dom} f, \qquad \operatorname{cod} gf = \operatorname{cod} g.$$

for all $a \in O$ and $(g, f) \in A \times_O A$, such that a metacategory is formed.

That is, a category is an interpretation of a metacategory within set theory. If C is a category, we usually write $a \in C$ or $f \in C$ to say a is an object of C, or f is an arrow of C. We also write

$$\operatorname{Hom}(b, c) = \{f \in C \mid \operatorname{dom} f = b, \operatorname{cod} f = c\}$$

for the set of arrows $b \to c$. We will show later that this definition of a category is the same as saying a category is a monoid for the product $\times_O$.

Examples of categories include

- $\mathbf{0}$, the empty category;
- $\mathbf{1}$, the category with one object and one arrow;
- $\mathbf{2}$, the category with two objects and precisely one non-identity arrow $a \to b$;
- $\mathbf{3}$, the category with three objects whose non-identity arrows form a triangle

$$\begin{tikzcd}
& \bullet \arrow[dr] & \\
\bullet \arrow[ur] & & \bullet \arrow[ll]
\end{tikzcd}$$

- $\downdownarrows$, the category with two objects and two non-identity arrows $a \rightrightarrows b$; we call such arrows *parallel arrows*.

A category is called *discrete* when all arrows are identities. Any set forms a discrete category, and any discrete category is completely determined by it's arrows.

A *monoid* is a category with one object. A monoid is thus fully described by it's arrows, and it's composition operation. A monoid can thus be described as a set M (of arrows) together with a binary operation $M \times M \to M$ and an identity element; such an object is precisely a semigroup with unity. Thus, for any category C and any object $a \in C$, the category $\mathrm{Hom}(a, a)$ forms a monoid. In this vein of thought, a group is thus a category where every arrow has a two-sided inverse.

Given a set V of sets, the category $\mathbf{Ens}_V$ takes V as the objects, and morphisms as any regular functions between sets in V. **Ens** refers to an arbitrary category of this type.

An ordinal number n can be seen as the set $\{0, \ldots, n-1\}$, and hence each ordinal is a category when viewed as a preorder. We can also consider the category ω, the first infinite cardinal, which when viewed as a category looks like

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \ldots$$

The category Δ takes all finite ordinals as objects and all order-preserving functions $f : m \to n$ as arrows. This category is generally called the *simplical category*, and it is extremely important.

We want an actual category **Set**, which is clearly not a set itself. We will circumvent this by allowing there to exist a "universe" $\mathcal{U}$, and will call a set "small" if it belongs to the universe. We will construct this explicitly later (and I will be calling small sets "sets" and large sets "classes" like a normal fucking person). This allows us to construct

- **Set**, objects are sets and morphisms are functions;
- $\mathbf{Set}_*$, objects are sets with a selected point, and morphisms are functions preserving these points;
- **Ens**, category of all classes and functions within a variable set V;
- **Cat**, category of small categories and morphisms as functors;
- **Grp**, category of groups and morphisms as group homomorphisms;
- **Ab**, category of abelian groups;
- **Mon**, category of (small) monoids;
- **Ring**, category of rings;
- **CRing**, category of commutative rings;
- R-**Mod**, category of left R-modules;
- **Mod**-R, category of right R-modules;
- **Top**, category of topological spaces;
- **Toph**, category of topological spaces with morphisms as homotopy classes of maps;
- $\mathbf{Top}_*$, category of topological spaces with a distinguised base point; base preserving continuous maps.

1.3 Functors

Definition 1.3: Functor

Let C, D be categories. A functor $T : C \to D$ consists of an object function T mapping each $c \in C$ to some $Tc \in D$, and an arrow function mapping each $f : c \to c'$ in C to $Tf : Tc \to Tc'$ in D, in such a way that

$$T1_c = 1_{Tc}, \qquad T(g \circ f) = T(g) \circ T(f)$$

whenever the composite $g \circ f$ is defined in C.

For example, the power set functor $\mathscr{P} : \mathbf{Set} \to \mathbf{Set}$, defined such that for any $f : A \to B$, the new map $\mathscr{P}f : \mathscr{P}A \to \mathscr{P}B$ is such that for any $S \subseteq A$, $\mathscr{P}f(S) = f(S)$.

Functors first arose in algebraic topology. For any n, a topological space X can be assigned an abelian group $H_n(X)$ called the n-th homology group of X. However, for any $: X \to Y$, there exists an induced $H_n(f) : H_n(X) \to H_n(Y)$ morphism of groups, and thus H_n becomes a functor $\mathbf{Top} \to \mathbf{Ab}$. We can in fact show homologygroups are invariant under homotopy, and so we can equally define a functor $\mathbf{Toph} \to \mathbf{Grp}$. The homotopy groups π_n are also functors; since they depend on a base point they define a functor $\mathbf{Top}_* \to \mathbf{Grp}$.

A functor which "forgets" underlying structure of an object is called a *forgetful functor*. For example, there is an apparent forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$, another $U : \mathbf{Ring} \to \mathbf{Ab}$, and so on.

Clearly functors may be composed, and moreover there is an identity functor 1_C for any category C. We may thus consider the metacategory of all categories, or the category $\mathbf{Cat}$ of all small categories.

A functor is called an *isomorphism* if it is two-sided invertible; in general isomorphism is too strong a condition, and weaker conditions are more useful. A functor is *full* if for every $c, c' \in C$ and every arrow $g : Tc \to Tc'$ in D, there exists $f : c \to c'$ in C such that $g = Tf$. A functor is *faithful*, or an embedding, if for all $c_1, c_2 \in C$ and parallel arrows $f_1, f_2 : c_1 \to c_2$ such that $Tf_1 = Tf_2$, we have $f_1 = f_2$.

These properties can be realized more intuitively. A functor defines functions $T : \mathrm{Hom}(c, c') \to \mathrm{Hom}(Tc, Tc')$. A functor is full when these functions are surjective; it is faithful when they are injective. A fully faithful functor is then a functor where these functions are bijections.

A subcategory S of a category C is a subset of the objects and arrows in C in such a way that S forms a category. There is automatically a faithful inclusion functor $S \hookrightarrow C$; we say a subcategory is *full* when the inclusion functor is full. Intuitively, S contains all arrows in C for the specified subcollection of objects.

1.4 Natural Transformations

Definition 1.4: Natural Transformation

Let $T, S : C \to D$ be functors. A natural transformation $\tau : S \Rightarrow T$ is a function which assigns to each $c \in C$ an arrow $\tau_c = \tau c : Sc \to Tc$ of D, called the *component of* τ *at* c, such that for all $f : c \to c'$ in C, the diagram

$$\begin{tikzcd} Sc \arrow[r, "\tau c"] \arrow[d, "Sf"'] & Tc \arrow[d, "Tf"] \\ Sc' \arrow[r, "\tau c'"] & Tc' \end{tikzcd}$$

commutes.

We say any component $\tau c : Sc \to Tc$ is *natural* in c. We can think of a natural transformation $\tau : S \Rightarrow T$ as a way of transforming the image of a functor S into that of a functor T. For example, given a triangle

$$\begin{tikzcd} a \arrow[dr, "f"] \arrow[dd, "h"'] & \\ & b \arrow[dl, "g"] \\ c & \end{tikzcd}$$

we would expect the diagram

$$
\begin{tikzcd}
Sa \arrow[rr, "\tau a"] \arrow[dr, "Sf"] \arrow[dd, "Sh"'] & & Ta \arrow[dr, "Tf"] \arrow[dd] & \\
& Sb \arrow[rr] \arrow[dl, "Sg"] & & Tb \arrow[dl, "Tg"] \\
Sc \arrow[rr, "\tau c"'] & & Tc &
\end{tikzcd}
$$

commute. Indeed, it does.

Natural transformations define *morphisms of functors*. If a natural transformation has every component invertible, we call it a *natural isomorphism* and write $S \cong T$.

For example, the determinant is a natural transformation $\det_K : \mathrm{GL}_n \Rightarrow ()^*$, where $K \mapsto K^*$ maps a ring to it's group of units. These are clearly two functors, both maps **CRing** $\to$ **Grp**, and it's not that difficult to see.

Definition 1.5: Equivalent of Categories

An equivalence of categories $\mathcal{C}, \mathcal{D}$ is a pair of functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$, which are inverses up to natural isomorphism; meaning there exist natural isomorphisms

$$1_{\mathcal{C}} \cong G \circ F, \qquad 1_{\mathcal{D}} \cong F \circ G.$$

1.5 Monics, Epis, and Zeros

This section formalizes some familiar notions category-theoretically. Isomorphisms are defined as invertible arrows.

Definition 1.6: Monic/Epi

A morphism $f : a \to b$ is monic if for all morphisms $g_1, g_2 : z \to a$, if $f \circ g_1 = f \circ g_2$, then $g_1 = g_2$. In other words, monic morphisms (monomorphisms) are left-cancellable. Similarly, epimorphisms are right-cancellable; if $f : a \to b$ and $g_1, g_2 : b \to z$, then f is epi iff $g_1 \circ f = g_2 \circ f$ implies $g_1 = g_2$.

In **Set**, monomorphisms are precisely injections, and epimorphisms are precisely surjections. Monics and epis need not have left/right inverses, but if a morphism has a left/right inverse, it is necessarily monic/epi.

Definition 1.7: Section/Retraction

A right-inverse of a morphism f is a morphism g such that $gf = 1$, and a left-inverse is a morphism h such that $fh = 1$. Right-inverses of f are called *sections* of f, and left-inverses are called *retractions* of f. If f has a left-inverse, we say it is a *split* monic, and if it has a right inverse, it is a *split* epi.

Idempotent arrows are endomorphisms f such that $f^2 = f$, and split idempotent morphisms are idempotent morphisms f such that $f = hg$, and $gh = 1$.

Definition 1.8: Initial/Terminal

An initial object I is an object such that there is a unique arrow $I \to x$ for all objects x, and a terminal object T is an object such that there is a unique arrow $x \to T$ for all x. An object that is initial and terminal is called a null object or a 0 object, and the unique arrows are called zero-morphisms.

Any two initial objects are easily isomorphic, and the same with terminal objects. Any composition with a zero morphism is easily a zero morphism.

Definition 1.9: Groupoid

A groupoid is a category where every morphism is an isomorphism. A group is then a groupoid with one element.

A typical example of a groupoid is the fundamental groupoid $\pi(X)$ of a topological space X. Objects are points in X, and morphisms are homotopy classes of paths between points. Since every arrow is invertible, endomorphism sets $\mathrm{End}_G(x)$ are groups for any groupoid G. The first fundamental group $\pi_1(X, x)$ can then be defined as $\mathrm{End}_{\pi(X)}(x)$.

1.6 Foundations

Herein, we have assumed the existence of many classes, such as the class of all groups, all sets, etc. For example, consider the class of all groups. Let φ be the property that a set x is a group. Then the way we describe $\mathrm{Obj}(\mathbf{Grp})$ is given by $\{x \mid \varphi(x)\}$. This amounts to the axiom of comprehension, which cannot be adopted in generality due to paradoxes such as Russel's paradox, $\{x \mid x \notin x\}$. ZFC assumes the axiom of restricted comprehension, which requires that the set-builder notation for constucting sets be defined as a subset of some other set, such as $\{x \in X \mid \varphi(x)\}$. However, there can be no set of all sets, for similar reasons, so this makes it impossible to construct $\mathrm{Obj}(\mathbf{Set})$, or $\mathrm{Obj}(\mathbf{Grp})$.

To avoid these paradoxes, we assume in addition to ZFC the existence of a *universe*, sometimes known as a Grothendieck universe.

Definition 1.10: Universe

A universe U is a set such that

1. $x \in u \in U$ implies that $x \in U$;
2. $u, v \in U$ implies that $\{u, v\} \in U$, $(u, v) \in U$, and $u \times v \in U$;
3. $x \in U$ implies that $\mathscr{P}(x) \in U$ and $\cup x \in U$ ($\cup x$ is the union over all elements of x);
4. $\omega \in U$ for ω the set of all finite ordinals;
5. if $f : a \to b$ is surjective with $a \in U$, then $b \in U$.

Some of these properties are clearly redundant.

These properties ensure that U remains closed under the standard operations of set theory, and thus we can safely perform set theory within U, and thus, all of ordinary mathematics can be performed safely within U, since $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ can all be derived from $\omega \cong \mathbb{N}$.

This means that we can only consider sets within the universe when performing math, and thus simply write $\mathrm{Obj}(\mathbf{Grp}) = \{x \in U \mid \varphi(x)\}$. Usually we will not write $x \in U$, since it becomes repetitive. Moreover, we choose $\mathrm{Obj}(\mathbf{Set}) = U$. In other words, we think as if the only sets that exist are elements of U.

This leads us to some terminology. Small sets are sets that are contained within U; i.e. x is a small set iff $x \in U$; and large sets are sets that are not elements of U. This includes U itself; we often call large sets *proper classes*. We use the word *class* to denote any subset of U, and all small sets are easily shown to be classes. Functions between small sets can also be regarded as small sets; we call them *small functions*. For my notes, I reserve the word "set" to mean exclusively "small set", and will write "proper class" instead of "large set" when possible, since this appears to be more modern terminology.

We can then similarly append the word "small" and "large" to various other constructs, considering for example *small groups* and *small rings*, although I will simply call these structures by their standard name, unless a distinction needs to be made. Distinctions are usually made for categories, since it is often important to discuss when a category is small vs when one is large. Note that $U \notin U$, so **Set** is not a small category, as this would contradict the axiom of regularity. Similarly, it can be shown that **Grp** is not small.

It seems prudent at this point to make precise what list of axioms we are using. The axioms of ZFC on a membership relation $\in$ guarentee

- the set $\varnothing$ exists;
- for any sets u, v, the sets $\{u, v\}, u \times v, \mathscr{P}(u), \cup u, (u, v)$ exist;
- sets with the same elements are equal (Extensionality);
- there exists an inductive set (Infinity);
- choice functions exist (Choice)
- there are no infinite chains $\cdots x_n \in \cdots \in x_0$, and no set contains itself (Regularity);
- the image of a function is a set (Replacement).

We will on occasion need to consider large sets which are not classes, such as the category **Cls** of all classes, defined as $\mathrm{Obj}(\mathbf{Cls}) = \mathscr{P}(U)$. We will make distiction between classes and large sets at such a time, if necessary.

This allows discussion of all categories, but not all metacategories. Attempting to describe all of these leads to some strange issues, and as such there has been discussion as to whether set theory is the proper foundational device for category theory, and in some cases, mathematics itself.

1.7 Large Categories

We just list a lot of large categories I am already familiar with, and their monics. Since I know all of them already, I will simply list the relevant information given. All of these are assumed to contain all objects of the given type which can be formed from sets.

- **Ab**: monics are injectives, epis are surjectives;
- **Ring**: monics are injectives, and surjectives are epis, but not the converse. $\mathbb{Z}$ is initial and 0 is terminal;
- **Top**: monics are injections and epis are surjections, the singleton is terminal and the empty space is initial;
- **Toph**: homotopy classes of injectives *need not be monics*;
- $\mathbf{Set}_*$: the singleton is initial and terminal (zero object), monics are left-invertible maps, epis are right-invertible maps, and isomorphisms are precisely those maps that are monic and epi;
- $\mathbf{Top}_*$: no information given as to the monics;
- **Rel**: the category of relations; $\mathrm{Obj}(\mathbf{Rel}) = U$, and morphisms are given by relations:

  $$\mathrm{Hom}(X, Y) = \{R \subseteq X \times Y\}.$$

  Composition $S \circ R$ for $S : Y \to Z$ and $R : X \to Y$ is given by

  $$S \circ R = \{(x, z) \mid (x, y) \in R \wedge (y, z) \in S\}.$$

  The identity arrow is the identity relation, and **Rel** holds **Set** as a subcategory.

Definition 1.11: Concrete Category

A *concrete category* is a pair (C, U) with C a category and U a forgetful functor $U : C \to \mathbf{Set}$. This is understood intuitively as being a category where each object has an underlying set, and morphisms are functions between these sets.

All of the above categories, except **Toph** and **Rel**, are concrete categories. Many categories we will encounter are concrete, but many are not. This makes our notion of a category more "abstract" than that of a concrete category.

1.8 Hom-Sets

Recall that $\mathrm{Hom}_C(x, y) = \{f \in \mathrm{Mor}(C) \mid f : x \to y\}$. This is often also written as $\hom_C(x, y)$, $C(x, y)$, or $\mathrm{Hom}(x, y)$. These are called the hom-sets, and as we have seen before, categories can be defined entirely via their hom-sets. In this definition (of small categories), composition is given by functions $\circ : \mathrm{Hom}(b, c) \times \mathrm{Hom}(a, b) \to \mathrm{Hom}(a, c)$ for each triple (a, b, c). Associativity is then described by the commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}(c,d) \times \mathrm{Hom}(b,c) \times \mathrm{Hom}(a,b) \arrow[r] \arrow[d] & \mathrm{Hom}(b,d) \times \mathrm{Hom}(a,b) \arrow[d] \\
\mathrm{Hom}(c,d) \times \mathrm{Hom}(a,c) \arrow[r] & \mathrm{Hom}(a,d)
\end{tikzcd}$$

Functors and natural transformations can be formulated similarly.

An important thing to note is that oftentimes the hom-sets can have additional structure. One simple case (that I've already seen) is where hom-sets are abelian groups; we call such a category an **Ab**-category or a *preadditive* category. In these categories, hom-sets come with an abelian addition operation, and composition is bilinear with respect to addition, meaning

$$f \circ (g + h) = (f \circ g) + (f \circ h),$$

and likewise for the other side.

Since composition in an **Ab**-category A is bilinear

$$A(b, c) \times A(a, b) \to A(a, c),$$

it can be written with a tensor product as a linear map

$$A(b, c) \otimes_{\mathbb{Z}} A(a, b) \to A(a, c).$$

We can thus describe an **Ab**-category as

- A collection of objects $\mathrm{Obj}(A)$;
- A function which assigns pairs to abelian groups $(a, b) \mapsto A(a, b)$;
- For each triple (a, b, c), a linear morphism of abelian groups $A(b, c) \otimes_{\mathbb{Z}} A(a, b) \to A(a, c)$;

such that the associative and unit laws are satisfied. The definition of a preadditive category is the same as that of a category, except $\times$ is replaced with $\otimes_{\mathbb{Z}}$, and the singleton is replaced with $\mathbb{Z}$ as the unit under tensor.

This evidently generalizes; given a category M with a kind of tensor product $\otimes$ and a unit I, we may be able to view hom-sets in a category A as objects in M. Such categories are called M-enriched categories, and we will see later how to formalize this notion.

Finally, additive functors are functors $F : A \to B$ which determine for each abelian group $A(a, b)$ and $B(F(a), F(b))$, an abelian group homomorphism $F : A(a, b) \to B(F(a), F(b))$. Denote by **Ab-cat** the category of all small **Ab**-categories, with morphisms as additive functors.

Chapter 2

Constructions on Categories

2.1 Duality

Duality is the process of reversing the arrows. In this section, we will describe duality axiomatically, independent of sets and functions.

The elementary theory of an abstract category (ETAC) consists of statements Σ with letters $a, b, c \dots$ for objects and letters $f, g, h \dots$ for arrows. *Atomic* statements consist of statements of the form

- a is the (co)domain of f;
- i is the identity arrow of a;
- $a = b$;
- $f = g$;
- $f = g \circ h$.

A statement Σ is any well-defined formula that can be constructed from atomic statements using standard logical operators and quantifiers. For example, the statement $f : a \to b$, which is a shorthand for "a is the domain of f *and* b is the codomain of f", is a statement Σ.

Duals of statements in ETAC are formed by making the following (somewhat redundant) replacements:

Statement Σ	Dual statement Σ^*
$f : a \to b$	$f : b \to a$
$a = \operatorname{dom} f$	$a = \operatorname{cod} f$
$b = \operatorname{cod} f$	$b = \operatorname{dom} f$
$h = g \circ f$	$h = f \circ g$
f is monic	f is epi
u is a right-inverse of f	u is a left-inverse of f
t is terminal	t is initial

If a statement involves a diagram, the dual of that statement is the diagram given by reversing all arrows.

The dual of each axiom of a category is also an axiom, and thus replacing a proof with it's dual gives a valid proof for the dual statement. This is known as *duality*: given a statement Σ which follows from the axioms of a category, it's dual Σ^* also follows.

Duality also applies to statements about several categories and functors between them. This does *not* reverse the direction of the functor, it simply reverses the direction of the arrows in the categories.

2.2 Contravariance and Opposites

We consider the opposite category C^{op}, which flips all the arrows. It is easily shown a statement is true in C if and only if it's dual is true in C^{op}, and thus C^{op} can be viewed as a manifestation of the principle of duality.

We can also consider opposite functors; for any functor $F : C \to D$, the assignment $F^{\text{op}} : C^{\text{op}} \to D^{\text{op}}$ given by $x \mapsto F(x)$ and $f^{\text{op}} \mapsto F(f)^{\text{op}}$ is also clearly covariantly functorial. This means there is an assignment $C \mapsto C^{\text{op}}$ and $F \mapsto F^{\text{op}}$, which is in fact functorial, and defines a (covariant) functor $\mathbf{Cat} \to \mathbf{Cat}$.

We can also consider contravariant functors $G : C \to D$, where if $f : a \to b$ in C, then $G(f) : G(b) \to G(a)$. That is, contravariant functors flip the direction of arrows. They can equivalently be viewed as covariant functors $G : C^{\text{op}} \to D$, which this text and my notes will do throughout.

Hom-sets provide important examples of co- and contravariant functors. For each $a \in C$, the assignment $x \mapsto C(a, x)$ gives a covariant functor $C \to \mathbf{Set}$ by mapping morphisms as follows: for any $f : x \to y$, the induced morphism is given by

$$C(a, f) : C(a, x) \to C(a, y), \qquad \varphi \mapsto f \circ \varphi.$$

This works since $\varphi : a \to x$, so $f \circ \varphi : a \to x \to y$. This is easily shown to be functorial.

Similarly, the contravariant hom-functor $x \mapsto C(x, a)$, written as $C(-, a) : C^{\text{op}} \to \mathbf{Set}$, has action on morphisms given by

$$C(f, a) : C(y, a) \to C(x, a), \qquad \varphi \mapsto \varphi \circ f.$$

This works since $\varphi : y \to a$, so $\varphi \circ f : x \to y \to a$.

Another example is the *sheaf of germs of continuous functions on* X, for X a topological space. Let $\mathbf{Open}(X)$ be the poset category given by all open subsets of X under the partial order $\subseteq$. Let $\overline{C}(U)$ denote the set of all continuous functors $f : U \to \mathbb{R}$. The restriction assignment $f \mapsto f|_V$ is a function $\overline{C}(U) \to \overline{C}(V)$ for all $V \subseteq U$. This makes $\overline{C}$ a contravariant functor $\mathbf{Open}(X) \to \mathbf{Set}$.

Another example is the assignment $\mathbf{Ring} \to \mathbf{CAT}$ given by $R \mapsto \mathbf{Mod}\text{-}R$. For any ring homomorphism $\rho : R \to S$, the functor $\mathbf{Mod}\text{-}\rho$ is given by mapping S-modules M to R-modules M, with R-action $m \cdot r$ given by $m \cdot \rho(r)$.

2.3 Products of Categories

Definition 2.1: Product Category

Let C, D be categories. Define the *product category* $C \times D$ as the category where

$$\text{Obj}(C \times D) = \{(c, d) \mid c \in C, d \in D\},$$

and morphisms $(c_1, d_1) \to (c_2, d_2)$ are given by pairs (f, g), where

$$f : c_1 \to c_2, \qquad g : d_1 \to d_2$$

in C and D, respectively. Composition is defined pointwise.

Product categories come woth projections $\pi_C : C \times D \to C$ and $\pi_D : C \times D \to D$ satisfying the universal property of products:

$$\begin{tikzcd}
 & Z \arrow[dl, "F"'] \arrow[d, dashed] \arrow[dr, "G"] & \\
C & C \times D \arrow[l, "\pi_C"] \arrow[r, "\pi_D"'] & D
\end{tikzcd}$$

Two functors $U : C \to C'$ and $V : D \to D'$ also have a product $U \times V : C \times C' \to D \times D'$, given by

$$(U \times V)(x, y) = (U(x), V(y)); \qquad (U \times V)(f, g) = (U(f), V(g)).$$

This is equivalently the unique arrow in the diagram

$$\begin{tikzcd}
 & C \times D \arrow[dl, "U \circ \pi_C"'] \arrow[d, dashed, "U \times V"] \arrow[dr, "V \circ \pi_D"] & \\
C' & C' \times D' \arrow[l] \arrow[r] & D'
\end{tikzcd}$$

One clearly also has that you can compose products of functors pointwise, by a simple diagram chase, so product is infact a functor

$$\times : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}.$$

Similar functors exist $\mathbf{Grp} \times \mathbf{Grp} \to \mathbf{Grp}$, $\mathbf{Top} \times \mathbf{Top} \to \mathbf{Top}$, and many more. Functors from a product category to a regular category are called *bifunctors*. Bifunctors satisfy the property that if one of the arguments is held fixed and the other is allowed to vary, this induces a functor as well.

Proposition 2.1

Let B, C, D be categories. For all $b \in B$ and $c \in C$, let $L_c : B \to D$ and $M_b : C \to D$ be functors such that $M_b(c) = L_c(b)$ for all b and c. There exists a bifunctor $X : B \times C \to D$ with $S(-, c) = L_c$ and $S(b, -) = M_b$ for all b and c, if and only if for all $f : b \to b'$ and $g : c \to c'$, we have

$$M_{b'}(g) \circ L_c(f) = L_{c'}(f) \circ M_b(g).$$

These equal arrows are then the value $S(f, g)$.

Proof. First, let S be a bifunctor. Let $f : c \to \overline{c}$ and $g : b \to \overline{b}$. Then since $L_c(b) = M_b(c)$ for all pairs (b, c), we have the equalities

$$M_b(c) = L_c(b), \qquad L_c(\overline{b}) = M_{\overline{b}}(c), \qquad M_b(\overline{c}) = L_{\overline{c}}(b), \qquad M_{\overline{b}}(\overline{c}) = L_{\overline{c}}(\overline{b}).$$

Functorality of M and L then gives us the commutative diagram

$$\begin{tikzcd}
L_c(b) \arrow[r, "L_c(g)"] \arrow[d, equal] & L_c(\overline{b}) = M_{\overline{b}}(c) \arrow[r, "M_{\overline{b}}(f)"] & M_{\overline{b}}(\overline{c}) \arrow[d, equal] \\
M_b(c) \arrow[r, "M_b(f)"'] & M_b(\overline{c}) = L_{\overline{c}}(b) \arrow[r, "L_{\overline{c}}(g)"'] & L_{\overline{c}}(\overline{b})
\end{tikzcd}$$

This gives us the desired equality. The converse gives us this equality, allowing S to be well-defined, and functoriality is then manifest from functorality of M and L. ■

Hom also gives an example of a bifunctor; if C is locally small, then Hom is a bifunctor $\mathrm{Hom} : C^{\mathrm{op}} \times C \to \mathbf{Set}$. There is also an isomorphism $(B \times C)^{\mathrm{op}} \cong B^{\mathrm{op}} \times C^{\mathrm{op}}$.

Natural transformations between bifunctors $\alpha : S \Rightarrow S'$ are viewed as follows. α is a function which assigns to each pair $(b, c) \in B \times C$, an arrow $\alpha(b, c) : S(b, c) \to S'(b, c)$ (the component of the natural transformation at (b, c)) satisfying the naturality condition.

We call α *natural in b for each $c \in C$* if the components define another natural transformation

$$\alpha(-, c) : S(-, c) \Rightarrow S'(-, c)$$

for each $c \in C$.

Proposition 2.2

Let $S, \overline{S} : B \times C \to D$ be bifunctors. A function α is a natural transformation $\alpha : S \Rightarrow \overline{S}$ if and only if $\alpha(b, c)$ is natural in b for each c, and natural in c for each b.

Proof. First, let $\alpha : S \Rightarrow \overline{S}$. Since this defines a naturality condition between *all* pairs (b, c), simply fixing a given c and letting b vary still induces a naturality condition, since action on morphisms can be viewed as pairs $(f, 1_c)$. For the converse, I will denote by $\alpha(b, -)_c$ the component of $\alpha(b, -)$ at c, and likewise for $\alpha(-, c)$. Note that $\alpha(b, -)_c = \alpha(b, c) = \alpha(-, c)_b$. Let $(f, g) : (b, c) \to (\overline{b}, \overline{c})$. This gives us the commutative diagram

$$\begin{tikzcd}
S(b,c) \arrow[r, "S(1_b,g)"] \arrow[d, "\alpha(b,-)_c"'] \arrow[rrr, bend left, "S(f,g)"] & S(b,\overline{c}) \arrow[r, equal] \arrow[d, "\alpha(b,-)_{\overline{c}} = \alpha(-,\overline{c})_b"] & S(b,\overline{c}) \arrow[r, "S(f,1_{\overline{c}})"] \arrow[d] & S(\overline{b},\overline{c}) \arrow[d, "\alpha(-,\overline{c})_{\overline{b}}"] \\
\overline{S}(b,c) \arrow[r, "\overline{S}(1_b,g)"'] \arrow[rrr, bend right, "\overline{S}(f,g)"'] & \overline{S}(b,\overline{c}) \arrow[r, equal] & \overline{S}(b,\overline{c}) \arrow[r, "\overline{S}(f,1_{\overline{c}})"'] & \overline{S}(\overline{b},\overline{c})
\end{tikzcd}$$

By commutativity, we have that $\alpha : S \Rightarrow \overline{S}$. ■

2.4 Functor Categories

Let $F, G, H : C \to D$ be functors, and let $\sigma : F \Rightarrow G$, $\tau : G \Rightarrow H$ be natural transformations. We can compose natural transformations to obtain a natural transformation $\tau \cdot \sigma : F \Rightarrow H$, given by $(\tau \cdot \sigma)_c = \tau_c \circ \sigma_c$. Diagramatically, this is given by the commutative diagram

$$\begin{tikzcd}
F(c) \arrow[r, "F(f)"] \arrow[d, "\sigma_c"] \arrow[dd, bend right, "(\tau\cdot\sigma)_c"'] & F(d) \arrow[d, "\sigma_d"'] \arrow[dd, bend left, "(\tau\cdot\sigma)_d"] \\
G(c) \arrow[r, "G(f)"] \arrow[d, "\tau_c"] & G(d) \arrow[d, "\tau_d"'] \\
H(c) \arrow[r, "H(f)"'] & H(d)
\end{tikzcd}$$

Composition in this manner is manifestly associative, and it also has as an identity the identity transformation $1_F : F \Rightarrow F$, where each component is an identity morphism. This allows us to define the functor category.

Definition 2.2: Functor Category

Let C, D be categories. The *functor category* $\mathrm{Fun}(C, D)$ takes as objects all functors $C \to D$, and as morphisms natural transformations. We often write

$$\mathrm{Nat}(F, G) = \mathrm{Hom}_{\mathrm{Fun}(C,D)}(F, G).$$

Hom-sets in this category need not be small, and moreover functor categories need not be classes. They can be large sets which are not classes.

2.5 The Category of All Categories

We have defined what we call *vertical composition* of natural transformations, where we compose them along functors. There is, however, another way to compose natural transformations, known as *horizontal composition*. Let A, B, C be categories, and let

$$\begin{tikzcd} A \arrow[r, bend left, "F_1"] \arrow[r, bend right, "G_1"'] \arrow[r, phantom, "\Downarrow\sigma"] & B \arrow[r, bend left, "F_2"] \arrow[r, bend right, "G_2"'] \arrow[r, phantom, "\Downarrow\tau"] & C \end{tikzcd}$$

commute. That is, $F_1, G_1 : A \to B$ and $F_2, G_2 : B \to C$ are functors, and $\sigma : F_1 \Rightarrow G_1$, $\tau : F_2 \Rightarrow G_2$ are natural transformations. We want to compose these natural transformations.

To do this, first form the composite functors $F_2 \circ F_1$ and $G_2 \circ G_1$. From hereon out, using components of natural transformations as subscripts will become impractical, so I will swap to the book's function notation, where $\sigma(c)$ is the component of σ at c.

Note that $\sigma(c) : F_1(c) \to G_1(c)$, and thus

$$F_2(\sigma(c)) : (F_2 \circ F_1)(c) \to (F_2 \circ G_1)(c).$$

Similarly, we have

$$G_2(\sigma(c)) : (G_2 \circ F_1)(c) \to (G_2 \circ G_1)(c).$$

Since τ is a natural transformation $F_2 \Rightarrow G_2$, it has components at $F_1(c)$ and $G_1(c)$, and thus by the naturality condition the square

$$\begin{tikzcd} (F_2 \circ F_1)(c) \arrow[r, "\tau(F_1(c))"] \arrow[d, "F_2(\sigma(c))"'] & (G_2 \circ F_1)(c) \arrow[d, "G_2(\sigma(c))"] \\ (F_2 \circ G_1)(c) \arrow[r, "\tau(G_1(c))"'] & (G_2 \circ G_1)(c) \end{tikzcd}$$

commutes. Define *horizontal composition* as the diagonal of this square, meaning

$$(\tau \circ \sigma)(c) = G_2(\sigma(c)) \circ \tau(F_1(c)) = \tau(G_1(c)) \circ F_2(\sigma(c)).$$

To show naturality of $\tau \circ \sigma$, consider the commutative diagram

$$\begin{tikzcd} (F_2 \circ F_1)(c) \arrow[r, "F_2(\sigma(c))"] \arrow[d, "(F_2 \circ F_1)(c)"'] & (F_2 \circ G_1)(c) \arrow[r, "\tau(G_1(c))"] \arrow[d, "(F_2 \circ G_1)(f)"'] & (G_2 \circ G_1)(c) \arrow[d, "(G_2 \circ G_1)(f)"] \\ (F_2 \circ F_1)(d) \arrow[r, "F_2(\sigma(d))"'] & (F_2 \circ G_1)(c) \arrow[r, "\tau(G_1(d))"'] & (G_2 \circ G_1)(d) \end{tikzcd}$$

The right square commutes by naturality of τ, and the left square commutes by naturality of σ and functorality of F_2. Thus, the diagram commutes. Note that the horizontal composites are precisely the components $(\tau \circ \sigma)(c)$ and $(\tau \circ \sigma)(d)$. Therefore, $\tau \circ \sigma : F_2 \circ F_1 \Rightarrow G_2 \circ G_1$ is natural.

Horizontal composition is once again manifestly associative, and moreover the identity natural transformation on the identity functor is an identity for horizontal composition. The book at this point adopts the notation of using F not only to denote a functor, but also to denote the identity natural transformation 1_F. I will not use this notation.

With notation as above, we also have that there are natural transformations

$$1_{F_2} \circ \sigma : F_2 \circ F_1 \Rightarrow F_2 \circ G_1, \qquad \tau \circ 1_{G_1} : F_2 \circ G_1 \Rightarrow G_2 \circ G_1.$$

We can then redefine horizontal composition in generality as

$$\tau \circ \sigma = (1_{G_2} \circ \sigma) \cdot (\tau \circ 1_{F_1}) = (\tau \circ 1_{G_1}) \cdot (1_{F_2} \circ \sigma).$$

One may see by virtue of a short diagram chase (which I did on my whiteboard and do not wish to type) that given the situation

$$\begin{tikzcd}
A \arrow[r, bend left=70, "F_1"{name=F1}] \arrow[r, "F_2"{name=F2}] \arrow[r, bend right=70, "F_3"'{name=F3}] & B \arrow[r, bend left=70, "G_1"{name=G1}] \arrow[r, "G_2"{name=G2}] \arrow[r, bend right=70, "G_3"'{name=G3}] & C
\arrow[Rightarrow, from=F1, to=F2, "\sigma_F"]
\arrow[Rightarrow, from=F2, to=F3, "\tau_F"]
\arrow[Rightarrow, from=G1, to=G2, "\sigma_G"]
\arrow[Rightarrow, from=G2, to=G3, "\tau_G"]
\end{tikzcd}$$

we have the equality

$$(\tau_G \cdot \sigma_G) \circ (\tau_F \cdot \sigma_F) = (\tau_G \circ \tau_F) \cdot (\sigma_G \circ \sigma_F).$$

This is known as the *interchange law*, which turns out to somewhat classify natural transformations (for small categories):

Theorem 2.1

The collection of all natural transformations is the set of arrows of two different categories under two different operations of composition, $\cdot$ and $\circ$, which satisfy the interchange law. Moreover, any identity for $\circ$ is an identity for $\cdot$.

We often omit the $\circ$ when composing along categories, similarly to how we often omit $\circ$ when composing functors or morphisms. To make the distinction clear, we will generally retain the $\cdot$ for vertical composition.

A double category is a categpry which has arrows with two different compositions, composable along morphisms and objects, which satisfy the interchange law. A 2-category is a double category for which identities for $\circ$ are also identities for $\cdot$.

2.6 Comma Categories

Comma categories are categories where objects are certain kinds of arrows. For example, given an object $b \in C$, the category $(b \downarrow C)$ of objects *under* b, wherein objects are pairs (f, c) with $c \in C$ and $f : b \to c$; morphisms $(f, c) \to (g, d)$ are morphisms $\varphi : c \to d$ in C, such that the diagram

$$\begin{tikzcd}
& b \arrow[dl, "f"'] \arrow[dr, "g"] & \\
c \arrow[rr, "\varphi"'] & & d
\end{tikzcd}$$

commutes.

A similar example is the category $(C \downarrow a)$ of objects *over* a, where objects are pairs (f, c) with $f : c \to a$, and morphisms $(f, c) \to (g, d)$ are commutative triangles

$$\begin{tikzcd} c \arrow[rr, "\varphi"] \arrow[dr, "f"'] & & d \arrow[dl, "g"] \\ & a & \end{tikzcd}$$

in C. For example, $\{*\}$ is terminal in **Set**, so there is always a unique map $X \to \{*\}$, and thus $(\mathbf{Set} \downarrow \{*\})$ is isomorphic to **Set**.

Let $F : C \to D$ and $b \in D$. The category $(b \downarrow S)$ of objects S-under b, is formed of pairs (f, c) with $c \in C$ and $f : b \to F(c)$. Morphisms $(c, f) \to (d, g)$ in $(b \downarrow S)$ are morphisms $\varphi : c \to d$ in C, such that the diagram

$$\begin{tikzcd} & b \arrow[dl, "f"'] \arrow[dr, "g"] & \\ F(c) \arrow[rr, "F(\varphi)"'] & & F(d) \end{tikzcd}$$

commutes. The category $(S \downarrow b)$ of objects S-over b is constructed analogously.

For example, let $U : \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor. For some $x \in \mathbf{Set}$, the category $(x \downarrow U)$ is given by functions $x \to U(g)$ for g a group; for example the function mapping x into the free group on x is one such object. Clearly this object is of some importance, and will be treated extensively later.

Here is the general construction of a comma category. Let A, B, C be categories, and let

$$A \xrightarrow{F} C \xleftarrow{G} B$$

be functors. The comma category $(F \downarrow G)$ takes triples (a, b, f) as objects with $a \in A$, $b \in B$, and $f : F(a) \to G(b)$. Arrows $(a_1, b_1, f_1) \to (a_2, b_2, f_2)$ are pairs (k, h) of arrows $k : a_1 \to a_2$ and $h : b_1 \to b_2$, such that

$$f_2 \circ F(k) = G(h) \circ f_1.$$

Equivalently, the diagram

$$\begin{tikzcd} F(a_1) \arrow[r, "F(k)"] \arrow[d, "f_1"'] & F(a_2) \arrow[d, "f_2"] \\ G(b_1) \arrow[r, "G(h)"'] & G(b_2) \end{tikzcd}$$

must commute. Composition of morphisms is done pointwise.

One recovers all of the previous examples by replacing objects b with functors $\mathbf{1} \to C$ mapping $1 \mapsto b$, and by replacing categories C with identity functors 1_C. Many categorical constructions can be recovered using comma categories. For example, we can consider $(a \downarrow b)$ for some $a, b \in C$, which takes as objects pairs (a, b, f), where $f : a \to b$. Equivalently, $\mathrm{Obj}(a \downarrow b) \cong \mathrm{Hom}_C(a, b)$. Since $\mathbf{1}$ has no non-identity morphisms, there are no non-identity morphisms in this category, and it is thus a discrete category.

2.7 Graphs and Free Categories

Recall the construction of the free monoid $F(X)$ on the set X, which is formed by all finite strings in X, has multiplication given by juxtaposition, and for which any set-function $X \to M$ with M any monoid extends to a unique morphism of monoids $F(X) \to M$. To construct a free *category*, we replace a set with a directed graph.

Recall that a directed graph G is a pair (O, A) of sets of objects and arrows, together with functions $\operatorname{dom}, \operatorname{cod} : A \to O$. Morphisms $G \to G'$ are a pair of functions $D_O : O \to O'$ and $D_A : A \to A'$ such that

$$D_O \operatorname{dom} f = \operatorname{dom} D_A f, \qquad D_O \operatorname{cod} f = \operatorname{cod} D_A f$$

for all $f \in A$. The category **Grph** takes as objects all small graphs, as morphisms all morphisms of graphs, and has a forgetful functor $U : \mathbf{Cat} \to \mathbf{Grph}$.

An O-graph is any graph for which O is the set of objects, and morphisms of O-graphs are morphisms of graphs for which D_O is the identity. We define the product over O of two O-graphs A, B to be

$$A \times_O B = \{(g, f) \mid \operatorname{dom} g = \operatorname{cod} f, g \in A, f \in B\};$$

the set of composable pairs. We then define $\operatorname{dom}(g, f) = \operatorname{dom} f$, and $\operatorname{cod}(g, f) = \operatorname{cod} g$. This product is associative, and the O-graph O, which takes dom and cod the identity functions, is a unit for $\times_O$. A category with $\operatorname{Obj}(C) = O$ is then an O-graph together with functions c, i (composition and identity), satisfying the necessary coherence diagrams to ensure associativity of c and unity of i with respect to c.

In this sense, categories are similar to monoids, with $\times_O$ the product. O-graphs can then be used to generate free categories, similarly to monoids.

Theorem 2.2

Let $G = \{A \rightrightarrows O\}$ be a small graph. There is a small category $C = C_G$ with $\operatorname{Obj}(C) = O$ and a morphism $P : G \to U(C)$ of graphs, for $U : \mathbf{Cat} \to \mathbf{Grph}$ the forgetful functor, such that for any category Z and any morphism $D : G \to U(Z)$ of graphs, there is a unique functor $D' : C \to Z$ making the diagram

$$\begin{tikzcd} G \arrow[r, "P"] \arrow[dr, "D"'] & C \arrow[d, dashed, "D'"] \\ & Z \end{tikzcd}$$

commute (with the obvious regards as to forgetful functors and the like). In particular, if $U(Z)$ is an O-graph, then D' is the identity on objects.

Proof. Note that the comma category $(G \downarrow U)$ takes as objects pairs (C, F), where $F : G \to U(C)$ is a morphism of graphs, and C is a category. Morphisms $\varphi : (C, F_C) \to (D, F_D)$ are thus functors $\varphi : C \to D$ such that the diagram

$$\begin{tikzcd} & G \arrow[dl, "F_C"'] \arrow[dr, "F_D"] & \\ U(C) \arrow[rr, "U(\varphi)"'] & & U(D) \end{tikzcd}$$

commutes. It thus follows that the free category C_G must be initial in the comma category $(G \downarrow U)$.

To define C, take $\operatorname{Obj}(C) = O$ and arrows of C to be finite paths

$$a_1 \to a_2 \to \cdots \to a_{n-1} \to a_n.$$

Since composition is not necessarily defined in a graph, this is formalized as a pair $(a_1, f_1, \ldots, a_n)$. We take each pair $(a_1, f_1, \ldots, a_n)$ to be a morphism $a_1 \to a_n$ in C. Composition is defined via concatenation, meaning

$$(a_1, f_1, \ldots, a_m) \circ (a_1, f_1, \ldots, a_n) = (a_1, f_1, \ldots, a_{n+m}) : a_1 \to a_{n+m}.$$

The morphism P of graphs simply maps arrows $f : a_1 \to a_2$ to the string (a_1, f, a_2). Showing the existence of D' is fairly simple; the definition of D' is just $(a_1, f, a_2) \mapsto f$ for the smallest arrows, given by composition otherwise, and agrees with D on object mappings. The uniqueness and commutivity of the diagram is manifest/forced. ■

When O is a singleton, the graph reduces to a set X of arrows, and the theorem provides for us the construction of a free monoid.

Corollary 2.1

For any set X, there is a monoid M and a function $p : X \to U(M)$, with U the forgetful functor, satisfying the universal property of free monoids.

Graphs can be used to describe diagrams. For any graph G, a diagram in the shape of G in the category B is a morphism $D : G \to B$ of graphs. This corresponds to a functor $D' : C_G \to B$ via the bijection $D' \mapsto D = U(D') \circ P$. This bijection

$$\mathbf{Cat}(C_G, B) \cong \mathbf{Grph}(G, U(B))$$

arises from an adjunction relation between F and U.

2.8 Quotient Categories

Proposition 2.3

Let C be a category, and let R be a function mapping each pair (a, b) in C to a binary relation $R_{a,b}$ on $C(a, b)$. Then there exists a category C/R and a functor $\pi : C \to C/R$ such that

- If $fR_{a,b}f'$ in C, then $\pi(f) = \pi(f')$;
- If $H : C \to D$ is any functor such that $fR_{a,b}f'$ implies $H(f) = H(f')$, then there is a unique functor $H' : C/R \to D$ such that $H = H' \circ \pi$ - in other words, π is universal with respect to this property. The unique functor is a bijection on objects.

Proof. R is a congruence if

- $R_{a,b}$ is an equivalence relation for all pairs;
- If $f, f' : a \to b$ have $fR_{a,b}f'$, then for all $g : a \to a'$ and all $h : b \to b'$, the compositions satisfy $(h \circ f \circ g)R_{a',b'}(h \circ f' \circ g)$ (composition preserves R).

There exists a minimal congruence R' containing R (which I refuse to prove), so we define C/R as the category where $\mathrm{Obj}(C/R) = \mathrm{Obj}(C)$, and $C/R(x, y) = C(x, y)/R'$. After this, anyone who has done proofs about universality of quotients can do this with their eyes closed, and it would be a waste of time to type it up instead of doing exercises. ■

Remark

Any relation that is already a congruence, such as homotopy, can be directly quotiented by without finding the minimal containing congruence. Additionally, in a similar but not equivalent vein to group presentations, if C is generated by graph G as a free category, we call C/R the group with generators G and relations R.

Chapter 3

Universals and Limits

3.1 Universal Arrows

We showed that the arrow $G \to U(C)$ from the graph into the free category formed on that graph is a *universal* arrow. This section makes precise universality in generality.

Definition 3.1: Universal Arrow

If $F : D \to C$ is a functor and $c \in C$, a universal arrow from c to F is a pair (r, u) with $r \in D$ and $u : c \to F(r)$, such that for all other pairs (d, f) with $d \in D$ and $f : c \to F(d)$, there is a unique arrow $f' : r \to d$ making the diagram

$$\begin{tikzcd}
c \arrow[r, "u"] \arrow[dr, "f"'] & F(r) \arrow[d, dashed, "F(f')"] & r \arrow[d, dashed, "f'"] \\
& F(d) & d
\end{tikzcd}$$

commute. That is, all other pairs (d, f) factor uniquely through the universal arrow under F.

Another way to view universal arrows from c to F is as initial objects in the comma category $(c \downarrow F)$: recall that objects are pairs (d, f) with $d \in D$ and arrows $f : c \to F(d)$ in C, and morphisms $(d_1, f_1) \to (d_2, f_2)$ are arrows $\varphi : d_1 \to d_2$ in D such that the diagram

$$\begin{tikzcd}
c \arrow[d, "f_1"'] \arrow[dr, "f_2"] & \\
d_1 \arrow[r, "F(\varphi)"'] & d_2
\end{tikzcd}$$

commutes. It then becomes clear that an initial object generates the same diagram as in definition 3.1.

An example of universal arrows arises in any situation with a free object and a forgetful functor. For the purposes of formalism, I will use the fact that free objects give rise to the free functor. Let $F : C \to D$ be a free functor with $U : D \to C$ a forgetful functor, and let $x \in C$. The free object $F(x)$ on x satisies the universal property of free objects:

$$\begin{tikzcd}
x \arrow[r, "i"] \arrow[dr, "f"'] & U(F(x)) \arrow[d, dashed, "U(f')"] & x \arrow[d, dashed, "f'"] \\
& U(z) & z
\end{tikzcd}$$

Note that this is precisely the same as the statement that $(F(x), i)$ is a universal arrow from x to U.

Another example is with quotients. The universal property of quotients states that the quotient of a set S by a relation R satisfies the property

$$\begin{tikzcd} S \arrow[r, "f"] \arrow[d, "\pi"'] & Z \\ S/R \arrow[ur, dashed, "f'"'] & \end{tikzcd}$$

We can view this as a universal arrow as well. Let $H : \mathbf{Set} \to \mathbf{Set}$ be the functor mapping sets X to the subset of $\mathrm{Hom}_{\mathbf{Set}}(S, X)$ consisting of functions $f : S \to X$ for which sRs' implies $fs = fs'$. The quotient $(S/R, \pi)$ is then immediately seen to be a universal arrow from S to H. This process works in other categories as well. This universal property is quite useful; for example in **Grp**, all isomorphism theorems and other properties of quotient groups follow easily without so much as mention of cosets.

This characterization allows for construction of universal objects which are initial with respect to some property, but the duality of universal arrows allows for the construction of terminal objects as well.

Definition 3.2: Universal Arrows, Again

Let $F : D \to C$ be a functor, with $c \in C$. A universal arrow from F to c is a pair (r, v) with $r \in D$ and $v : F(r) \to c$, which is terminal with respect to this property: for each pair (f, d) with $d \in D$ and $f : F(d) \to c$, there is a unique morphism $f' : d \to r$ such that the diagram

$$\begin{tikzcd} F(d) \arrow[r, "f"] \arrow[d, dashed, "F(f')"'] & c & d \arrow[d, dashed, "f'"] \\ F(r) \arrow[ur, "v"'] & & r \end{tikzcd}$$

commutes. Equivalently, a universal arrow from F to c is a terminal object in $(F \downarrow c)$.

An important example of a universal arrow in this regard is a product. Let C be a category with finite products, and let $x, y \in C$. The product is well-known to satisfy the universal property

$$\begin{tikzcd} & z \arrow[dl, "f"'] \arrow[d, dashed] \arrow[dr, "g"] & \\ x & x \times y \arrow[l, "\pi_x"] \arrow[r, "\pi_y"'] & y \end{tikzcd}$$

To view this as a universal arrow, let $\Delta : C \to C \times C$ be the diagonal functor

$$x \mapsto (x, x), \qquad f \mapsto (f, f).$$

This is manifestly functorial. Let $r \in C$ and $p : r \to x$, $q : r \to y$ be morphisms in C. Then we have that $(p, q) : (r, r) \to (x, y)$. Let $(r, (p, q))$ be a universal arrow from Δ to (x, y). We then have that the diagram

$$\begin{tikzcd} (r, r) \arrow[r, "(p,q)"] \arrow[d, dashed, "(f',f')"'] & (x, y) & r \arrow[d, dashed, "f'"] \\ (z, z) \arrow[ur, "(f,g)"'] & & z \end{tikzcd}$$

commutes for all $z \in C$ and for all pairs $f : z \to x$ and $g : z \to y$. Using our knowledge of product categories, the universal property of products follows immediately, and thus $(r, (p, q))$ satisfy the universal property of products.

3.2 The Yoneda Lemma

Proposition 3.1

Let $F : D \to C$ be a functor. A pair $(r, u : c \to F(r))$ is universal from c to F if and only if for each pair (d, f), the function mapping the induced morphism f' to the composition $F(f') \circ u : c \to F(d)$ is a bijection of hom-sets

$$D(r, d) \cong C(c, F(d)).$$

This bijection is natural in d, meaning the function is a natural isomorphism. Conversely, any natural isomorphism $D(r, d) \cong C(s, F(d))$ in d is determined by a universal arrow $u : c \to F(r)$.

Proof. Universality immediately guarentees that the function is a bijection. Let f, g both have f' as the induced morphism. By commutivity of the relevant diagram, $F(f') \circ u = f$ and $F(f') \circ u = g$, and thus $f = g$. Thus, the function is injective, and surjectivity follows by definition, as for any morphism $g : c \to F(d)$, universality says there exists a morphism $g' : r \to d$ which maps to g under this function. For the converse direction, surjectivity guarentees that such a morphism f' always exists, and from this injectivity guarentees universality in much the same way.

Naturality of the function in d, which I will denote by ν, means that

$$\nu : D(r, -) \Rightarrow C(c, -) \circ F = C(c, F(-)).$$

Proof of this amounts to a proof that for all $\varphi : d \to d'$, the diagram

$$\begin{tikzcd} D(r,d) \arrow[r, "D(r,\varphi)"] \arrow[d, "\nu(d)"'] & D(r,d') \arrow[d, "\nu(d')"] \\ C(r,F(d)) \arrow[r, "C(r,F(\varphi))"'] & C(r,F(d')) \end{tikzcd}$$

commutes. Working elementwise, we have that the following must be equivalent:

$$f' \longmapsto \varphi \circ f' \longmapsto F(\varphi \circ f) \circ u$$
$$f' \longmapsto F(f') \circ u \longmapsto F(\varphi) \circ F(f') \circ u$$

This follows from functorality of F, and thus ν is natural in d.

For the converse, let $f' : r \to d$. Naturality of ν guarentees commutivity of the diagram

$$\begin{tikzcd} D(r,r) \arrow[r, "\nu(r)"] \arrow[d, "D(r,f')"'] & C(c,F(r)) \arrow[d, "C(c,F(f'))"] \\ D(r,d) \arrow[r, "\nu(d)"'] & C(c,F(d)) \end{tikzcd}$$

Consider in particular the image of 1_r under both of these paths. Define $u := \nu(r)(1_r)$. We have

$$1_r \xmapsto{\nu(r)} u \xmapsto{C(c,F(f'))} F(f') \circ u$$
$$1_r \underset{D(r,f')}{\longmapsto} f' \underset{\nu(d)}{\longmapsto} \nu(d)(f')$$

We thus have that $\nu(d)(f') = F(f') \circ u$. Since $\nu(d)$ is a bijection for all d, it follows that f' is necessarily unique (which is equivalent here to the fact that f is unique), and thus (r, u) is universal. ∎

If C and D are locally small, it follows that the functor $C(c, -) \circ F$ is naturally isomorphic to a hom-functor $D(r, -)$. This is known as a representation.

Definition 3.3: Representation

Let D be locally small. A representation of a functor $F : D \to \mathbf{Set}$ is a pair (r, ψ) with $r \in D$ and ψ a natural isomorphism

$$\psi : D(r, -) \cong F.$$

The object r is called the representing object, and the functor F is said to be *representable* if a representation exists.

Representable functions are just hom-functors up to isomorphism. They can be connected to universal arrows through the following result.

Proposition 3.2

Let $*$ denote a singleton, and let D be locally small. If $(r, u : * \to F(r))$ is universal from $*$ to $F : D \to \mathbf{Set}$, then the function $\psi : f' \mapsto F(f')(u(*)) \in F(d)$ is a representation of F, with r the representing object. Every representation of F is obtained in this way from exactly one such universal arrow.

Proof. I will begin by making clear what the proposition says. u is universal from $*$ to F, and the object obtained by evaluating $F(f') \circ u$ at $*$ is the representing object.

Since functions $f : * \to X$ are fully determined by $f(*)$, there is a bijection $\mathrm{Hom}_{\mathbf{Set}}(*, X) \to X$ given by $f \mapsto f(*)$. This is clearly a natural isomorphism $\alpha\, \mathrm{Hom}_{\mathbf{Set}}(*, -) \cong 1_{\mathbf{Set}}$, so composing with F gives a natural isomorphism $\mathrm{Hom}_{\mathbf{Set}}(*, F(-)) \cong F$ by horizontal composition $\alpha \circ 1_F$. By proposition 3.1, we also have $\varphi : \mathrm{Hom}_{\mathbf{Set}}(*, F(-)) \cong C(r, -)$, and thus vertical composition $\varphi \cdot \alpha$ gives a natural isomorphism $F(-) \cong C(r, -)$. ∎

Note that a universal arrow from c to $F : D \to C$ amounts to an isomorphism $D(r, d) \cong C(c, F(d))$, and hence a representation $D(r, -) \cong C(c, F(-))$, or equally a universal element of the functor. This discussion now leads us naturally to the Yoneda lemma, a result of fundamental importance in category theory.

Lemma 3.1 (Yoneda)

Let C be locally small, $F : C \to \mathbf{Set}$ be a functor, and let $c \in C$. Then there is a bijection

$$y : \mathrm{Nat}(C(c, -), F) \cong F(c)$$

given by mapping natural transformations $\alpha : C(c, -) \Rightarrow F$ to $\alpha_c(1_c)$.

Proof. The map is well-defined; if $\alpha = \alpha'$, then clearly $\alpha_c = \alpha'_c$, and thus $\alpha_c(1_c) = \alpha'_c(1_c)$.

Let $\alpha : C(c, -) \Rightarrow F$. Then for each morphism $f : c \to d$, there is a commutative diagram

$$\begin{tikzcd} C(c,c) \arrow[r, "\alpha_c"] \arrow[d, "C(c,f)"'] & F(c) \arrow[d, "F(f)"] \\ C(c,d) \arrow[r, "\alpha_d"'] & F(d) \end{tikzcd}$$

Denote $\alpha_c(1_c)$ by u. Then by commutivity of the diagram, we have the equality

$$F(f)(u) = \alpha_d(f).$$

Thus, each component α_d is determined by the value of u. This shows injectivity, as if $u = u'$, then the corresponding natural transformations α, α' are necessarily equal, since for each d, for each $f : c \to d$, the value $\alpha_d(f) = F(f)(u) = F(f)(u') = \alpha'_d(f)$.

Finally, to show surjectivity, let $u \in F(c)$. Suppose there is a function α defined such that $\alpha_c(1_c) = u$, and for all $f : c \to d$, we have $\alpha_d(f) = F(f)(u)$. This makes the above diagram commute, and thus α is natural, concluding the proof. ■

Corollary 3.1

For all $r, s \in C$, there exists a unique arrow $u : s \to r$ such that each natural transformation $\alpha : C(r, -) \Rightarrow C(s, -)$ is given by $C(u, -)$.

Proof. First, I will break down the meaning of this corollary. The natural transformation $\alpha : C(r, -) \Rightarrow C(s, -)$ has components $\alpha_x : C(r, x) \to C(s, x)$ given by $f \mapsto f \circ u$.

By the Yoneda lemma,

$$\mathrm{Nat}(C(r, -), C(s, -)) \cong C(s, r).$$

Thus, for each α, there exists a unique morphism $h : s \to r$ such that for each $f : c \to x$, we define $\alpha_x(f) = C(s, f)(h) = f \circ h$, where h is the element $\alpha_r(1_r)$ which uniquely determines α. Thus, each α_x is simply given by composing on the right with h, or thus $\alpha_x = C(x, h)$, and thus $\alpha = C(-, h)$. ■

The isomorphism y is natural in both F and c, which is done by viewing y as a natural transformation bewteen the evaluation functor $(F, c) \mapsto F(c)$ and the functor $N(F, c) = \mathrm{Nat}(C(c, -), F)$. This is done by viewing them both as functors $\mathrm{Fun}(C, \mathbf{Set}) \to \mathbf{Set}$.

Lemma 3.2 (Yoneda)

With notation as previously, the bijections y give a natural transformation $y : N \Rightarrow E$.

This lemma is easily enough proven. Note that we thus have a fully faithful functor $Y : C^{\mathrm{op}} \to \mathrm{Fun}(C, \mathbf{Set})$ given by $r \mapsto C(r, -)$. It's dual is another functor $Y' : D \to \mathrm{Fun}(D^{\mathrm{op}}, \mathbf{Set})$, also fully faithful, and often known as the *Yoneda Embedding*.

If C is not locally small, these restrictions can be bypassed by instead considering the category **Ens** whose objects at least contain the hom-sets of C, and whose morphisms are functions between those hom-sets.

3.3 Coproducts and Colimits

Colimits will be introduced with a collection of special cases, which will all be universal constrictions.

Let $\Delta : C \to C \times C$ be the diagonal functor $x \mapsto (x, x)$ and $f \mapsto (f, f)$. A universal arrow from an object (a, b) to Δ is a *coproduct diagram.* This universal arrow consists of an object $c \in C$, and arrows $i_a : a \to c$ and $i_b : b \to c$ in C; that is, an arrow $(i_a, i_b) : (a, b) \to \Delta(c)$; satisfying the universal property that for any object $z \in C$ and any morphism $(f, g) : (a, b) \to (x, y)$, there exists a unique arrow $h : c \to z$ in C making the diagram

$$\begin{tikzcd}
(a,b) \arrow[r, "{(i_a,i_b)}"] \arrow[dr, "{(f,g)}"'] & (c,c) \arrow[d, dashed, "{(h,h)}"] & c \arrow[d, dashed, "h"] \\
& (z,z) & z
\end{tikzcd}$$

commute. We call the object c the *coproduct* of a and b, and write it as $a \amalg b$. We can also write the universal property as a diagram in C rather than in $C \times C$:

$$\begin{tikzcd}
a \arrow[r, "i_a"] \arrow[dr, "f"'] & a \amalg b \arrow[d, dashed, "h"] & b \arrow[l, "i_b"'] \arrow[dl, "g"] \\
& z &
\end{tikzcd}$$

This gives us the familiar universal property of coproducts. We also sometimes write the morphism h as $f \amalg g$. If the coproducts exists, it are necessarily unique up to isomorphism, and thus the assignment $(f, g) \mapsto f \amalg g$ becomes a bijection

$$C(a, z) \times C(b, z) \cong C(a \amalg b, z),$$

which is natural in z. If all coproducts exist in C, then $\amalg$ becomes a bifunctor $C \times C \to C$. In a preorder, the coproduct of two elements a, b is any least upper bound $\inf(a, b)$ in the preorder.

Infinite coproducts are similar. We can view a finite coproduct as a universal arrow to the diagonal functor into $C \times C = C^2$. Given an infinite coproduct indexed over a set X, we can take X to define a discrete category and consider a diagonal functor to the functor category C^X, which can be viewed as the category of X-indexed families of objects in C. The X-ary diagonal functor $\Delta : C \to C^X$ maps objects c to the constant functor $x \mapsto c$ for all $x \in X$, which is viewed as the constant family $c_x = c$ for all $x \in X$. The action of the constant functor on morphisms is to map $f \mapsto 1_c$ for all f, and thus we can define constant natural transformations $f : c \to d$ where each component $f_x = f$. This is the action of Δ on morphisms, and we can view them as X-indexed families of morphisms. A coproduct

$$c = \coprod_{x \in X} c_x$$

is then a universal arrow $(c, \{i_x\}_{x \in X})$ from $\{c_x\}_{x \in X}$ to Δ. The universal property gives that the assignment $f \mapsto \{f \circ i_x\}_{x \in X}$ is a bijection

$$C\left(\coprod_{x \in X} c_x, z\right) \cong \prod_{x \in X} C\left(c_x, z\right),$$

natural in z.

If the entries in a coproduct are constant; that is $c_x = c$ for all x, then we call it a *copower* and write it as $X \cdot c$, so that

$$C(X \cdot c, z) \cong C(c, z)^X.$$

For example, in **Set**, the copower is the cartesian product $X \cdot Y = X \times Y$.

Coequalizers are a generalization of cokernels in categories where zero objects may not exist. Given a pair $f, g : a \to b$ of objects, a coequalizer $\mathrm{coeq}(a, b)$ of a and b is a pair $(e, u : b \to e)$ such that $uf = ug$, and u is universal with respect to this propery:

$$\begin{tikzcd}
a \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & b \arrow[r, "u"] \arrow[dr, "h"'] & e \arrow[d, dashed, "h'"] \\
& & c,
\end{tikzcd}$$

which commutes with the standard rules (meaning it is not implied that $f = g$).

We can view coequalizers as univeral arrows as follows. Let $\downdownarrows$ denote the category

$$\begin{tikzcd}
1 \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & 2.
\end{tikzcd}$$

A functor $\downdownarrows \to C$ is then a choice of two elemets, and two parallel arrows. Morphisms in the category $C^{\downdownarrows}$ (natural transformations between these functors) then simply become pairs (h, k) which commute the two diagrams

$$\begin{tikzcd}
a \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] \arrow[d, "h"'] & b \arrow[d, "k"] \\
a' \arrow[r, shift left, "f'"] \arrow[r, shift right, "g'"'] & b'
\end{tikzcd}$$

where the images of the two implicit functors are as implied.

Define the $\downdownarrows$-ary diagonal functor $\Delta : C \to C^{\downdownarrows}$ once again as mapping objects to the constant functor, and morphisms to the constant natural transformations between these functors:

$$\begin{tikzcd}
c \arrow[d, "f"] & & c \arrow[r, shift left, "1"] \arrow[r, shift right, "1"'] \arrow[d, "f"'] & c \arrow[d, "f"] \\
c' & {} \arrow[r, mapsto] & c' \arrow[r, shift left, "1"] \arrow[r, shift right, "1"'] & c'
\end{tikzcd}$$

Now return to the pair $f, g : a \to b$. An arrow h with $hf = hg$ is the same as an arrow $(hf = hg, h) : (f, g) \to (1_c, 1_c)$ in the functor category:

$$\begin{tikzcd}
a \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] \arrow[d, "hf"'] & b \arrow[d, "h"] \\
c \arrow[r, shift left, "1"] \arrow[r, shift right, "1"'] & c
\end{tikzcd}$$

A coequalizer is just universal with respect to this, so a coequalizer is simply a univeersal arrow (e, u) from the functor identifying (f, g) to the functor Δ.

Given a pair of arrows $f, g : a \to b, c$ with common domain, a pushout of (f, g) is a triple (u, v, r), which is universal with respect to commutative squares:

$$\begin{tikzcd}
a \arrow[r, "f"] \arrow[d, "g"'] & b \arrow[d, "u"] \arrow[ddr, bend left, "h"] & \\
c \arrow[r, "v"'] \arrow[drr, bend right, "k"'] & r \arrow[dr, dashed] & \\
& & z
\end{tikzcd}$$

We usually write $b \amalg_a c$ or $b \amalg_{(f,g)} c$ for r, and also call it the *fibered sum*. Let $\cdot \leftarrow \cdot \rightarrow \cdot$ be a category, and let the diagonal functor be exactly what intuition dictates. The pushout is then a universal arrow from (f, g) to Δ.

All of these constructions involve a universal arrow to some diagonal functor. Now is time to define colimits in generality. For any categories C, J, let $\Delta : C \to C^J$ be the diagonal functor, where $\Delta(c)$ is the constant functor $j \mapsto c$ for all $j \in J$, and $f : i \to j$ maps to $1_c : c \to c$. This allows us to define, for all $g : c \to d$ in C, the constant natural transformation $\Delta(g) : \Delta(c) \Rightarrow \Delta(d)$, where each component $\Delta(g)_j = g$.

Definition 3.4: Colimit

With notation as above, the colimit of a functor $F : J \to C$ is a universal arrow (r, u) from F to Δ, usually written as $r = \varinjlim F$.

Recall that this implies u is a natural transformation from F to $\Delta(r)$, such that for all $c \in C$ and all natural transformations $\alpha : F \Rightarrow \Delta(c)$, the diagram

$$\begin{tikzcd} F \arrow[r, "u"] \arrow[dr, "\alpha"'] & \Delta(r) \arrow[d, dashed, "\Delta(\alpha')"] \\ & \Delta(c) \end{tikzcd}$$

commutes.

Looking only at the fact that there is a natural transformation $u : F \Rightarrow \Delta(r)$, we have that for any morphism $f : i \to j$ in J, the diagram

$$\begin{tikzcd} F(i) \arrow[r, "F(f)"] \arrow[d, "u_i"'] & F(j) \arrow[d, "u_j"] \\ \Delta(r)(i) \arrow[r, "\Delta(r)(f)"'] & \Delta(r)(j) \end{tikzcd}$$

commutes. Of course, $\Delta(r)(-) = r$ and $\Delta(r)(f) = 1_r$, so this reduces to the diagram

$$\begin{tikzcd} F(i) \arrow[r, "F(f)"] \arrow[d, "u_i"'] & F(j) \arrow[d, "u_j"] \\ r \arrow[r, equal] & r \end{tikzcd}$$

This is typically rewritten as

$$\begin{tikzcd} F(i) \arrow[rr, "F(f)"] \arrow[dr, "u_i"'] & & F(j) \arrow[dl, "u_j"] \\ & r & \end{tikzcd}$$

commuting for all $f : i \to j$ in J. Such an object r, together with a natural transformation u, is knwon as an *co-cone* for F. The universality of r and u means that for any other co-cone (c, α) for F, the diagram

$$\begin{tikzcd} F(i) \arrow[rr, "F(f)"] \arrow[dr, "u_i"'] \arrow[ddr, bend right, "\alpha_i"'] & & F(j) \arrow[dl, "u_j"] \arrow[ddl, bend left, "\alpha_j"] \\ & r = \varinjlim F \arrow[d, dashed, "\alpha'"] & \\ & c & \end{tikzcd}$$

commutes for any $f : i \to j$ in J.

3.4 Products and Limits

Definition 3.5: Limit

Let $F : J \to C$ be a functor. A limit of F is a universal arrow $(\varprojlim F, u)$ from the diagonal functor $\Delta : C \to C^J$ to F. Equivalently, it is a *cone* which is universal with respect to cones, where a cone is the dual of a co-cone.

More explicitly, a cone is a pair $(r, u : \Delta(r) \Rightarrow F$, meaning that for all $f : i \to j$ in J, the diagram

$$\begin{tikzcd} & r \arrow[dl, "u_i"'] \arrow[dr, "u_j"] & \\ F(i) \arrow[rr, "F(f)"'] & & F(j) \end{tikzcd}$$

commutes. As with any universals, if they exist, limits, colimits, and the limiting (co-)cones are uniquely determined up to isomorphism by F.

When limits exist, there are natural isomorphisms

$$C(c, \varprojlim F) \cong \mathrm{Nat}(\Delta(c), F) = \mathrm{Cone}(c, F);$$

$$\mathrm{Cone}(F, c) = \mathrm{Nat}(F, \Delta(c)) \cong C(\varinjlim F, c).$$

Examples of limits include duals to all the constructions mentioned in section 3.3, and more that were not mentioned. Examples include products, kernels, pullbacks, and equalizers.

3.5 Categories with Finite Products

A category has *finite products* if for all finite collections of objects (and by induction, all pairs of objects), there exists a product diagram with a product object and projections. In particular, C has a product of *no* objects, which would be the limit of the null functor, and thus simply a terminal object. Define the diagonal map $\delta_c : c \to c \times c$ as the induced morphism in the product diagram $1_c \times 1_c$; meaning $\pi_1 \delta_c = \pi_1 \delta_c = 1_c$.

Proposition 3.3

If a category C has a terminal object t and a product diagram for any pair of objects, then it has finite products. The diagrams provide a bifunctor $\times : C \times C \to C$. For any $a, b, c \in C$, there is an isomorphism $\alpha_{a,b,c} : a \times (b \times c) \to (a \times b) \times c$ called the *associator*, natural in a, b, c. Equivalently, this is a natural isomorphism $\times \circ (1_C \otimes \times) \Rightarrow \times \circ (\times \otimes 1_C)$. I have here denoted by $\otimes$ the product of functors, simply for clarity of notation. Additionally, for all $a \in C$ there are natural isomorphisms

$$\lambda_a : t \times a \cong a, \qquad \rho_a : a \times t \cong t$$

called the *left-* and *right-unitors*, respectively. These are equivalently viewed as natural isomorphisms

$$\lambda : t \times - \Rightarrow 1_C, \qquad \rho : - \times t \Rightarrow 1_C.$$

Proof. Existence of products is immediate by induction, and the existence of the morphisms is easily shown via a short diagram chase. I decline to include the proof in these notes to avoid drawing many diagrams. ■

If a category C has both finite products and coproducts, then the arrow

$$\coprod_{i\in[m]} a_i \to \prod_{j\in[n]} b_j$$

is determined uniquely by an $m \times n$ matrix of arrows $f_{jk} = p_k f i_j : a_j \to b_k$, with j ranging over $[m]$ and k ranging over $[n]$. In categories of finite dimensional vector spaces, coproducts and products coencide, and thus this is a usual matrix for a change of basis.

More generally, if C has a zero object 0, then there is a *canonical* arrow

$$\coprod_i a_i \to \prod_i a_i.$$

3.6 Groups in Categories

Given a category with finite products with t the terminal object, a monoid (this definition will be upgraded later) is a triple (c, μ, η) with $\mu : c \times c \to c$ and $\eta : t \to c$, such that the diagrams

$$\begin{tikzcd}
c \times (c \times c) \arrow[r, "\alpha"] \arrow[d, "1\times\mu"'] & (c \times c) \times c \arrow[r, "\mu\times 1"] & c \times c \arrow[d, "\mu"] \\
c \times c \arrow[rr, "\mu"'] & & c,
\end{tikzcd}$$

$$\begin{tikzcd}
t \times c \arrow[r, "\eta\times 1"] \arrow[dr, "\lambda"'] & c \times c \arrow[d, "\mu"] & c \times t \arrow[l, "1\times\eta"'] \arrow[dl, "\rho"] \\
& c &
\end{tikzcd}$$

commute. Here, α is the associator, λ is the left unitor, and ρ is the right unitor, as given in proposition 3.3. By viewing μ as multiplication in the monoid, and by viewing η as the unitor morphism taking the place of the identity element of a monoid, this definition makes perfect sense. Taking δ_c to once again be the diagonal morphism $c \to c \times c$, given by the product $1_c \times 1_c$, we can extend this definition to encompass that of a group by introducting an "inversion" morphism $i : c \to c$ (the book uses ζ, I think thats stupid).

Definition 3.6: Group Object

A group object in a category C with finite products is a pair (c, μ, η, i) such that (c, μ, η) form a monoid with respect to the induced monoidal structure by $\times$ (a monoid as described above), and such that the diagram

$$\begin{tikzcd}
c \arrow[r, "\delta_c"] \arrow[d, dashed] & c \times c \arrow[r, "1\times i"] & c \times c \arrow[d, "\mu"] \\
t \arrow[rr, "\eta"'] & & c
\end{tikzcd}$$

commutes. We call μ multiplication, η the unitor, and i inversion.

A group object in **Set** is easily shown to simply be a group, if we take μ is the standard multiplication, the image $\eta(*)$ to be the identity (since in **Set** $t = *$ is the singleton), and i to be the map $c \mapsto c$. It is easily shown that given a group object, these definitions satisfy the group axioms.

It is easily shown that given any group G and any set X, the set $G^X = \mathrm{Hom}_{\mathbf{Set}}(X, G)$ has a group structure given by pointwise multiplication. This generalizes via the Yoneda lemma.

Proposition 3.4

Let C be a category with finite products. Then an object c is a group object if and only if the hom-functor $\mathrm{Hom}(-, c)$ is a group in the functor category $\mathbf{Set}^{C^{\mathrm{op}}}$.

Proof. Recall that by universality of product, for any $f : z \to a \times b$, by proposition 3.1 there is a natural bijection $C(z, a) \times C(z, b) \cong C(z, a \times b)$ given by $(\pi_a f, \pi_b f) \mapsto h$, and vice versa. We thus have that multiplication $\mu : c \times c \to c$ induces a multiplication $\overline{\mu}$ on $C(-, c) \times C(-, c)$ by the composition

$$C(-, c) \times C(-, c) \longrightarrow C(-, c \times c) \xrightarrow{C(-,\mu)} C(-, c),$$

which for any pair $(f : a \to c, g : b \to c)$ maps

$$(f, g) \longmapsto f \times g \longmapsto \mu \circ (f \times g).$$

I don't feel like finishing this proof. ∎

3.7 Colimits of Representable Functors

Theorem 3.1

Let $F : C \to \mathbf{Set}$ with D small. Then F can be represented canonically as a colimit of a diagram of representable functors $C \to \mathbf{Set}$.

Proof. Let $J = (1 \downarrow F)$, where we remark that 1 is a singleton (so we are changing notation from $*$ to 1). Since morphisms $1 \to z$ identify elements of z, the category J is thus the category of elements of F, meaning objects are pairs (d, x) for $x \in F(d)$ and $d \in C$, and morphisms $(d, x) \to (d', x')$ are arrows $f : d \to d'$ such that $F(f)(x) = x'$.

Let $M : J^{\mathrm{op}} \to \mathbf{Set}^C$ be the functor given by $(x, d) \mapsto C(d, -)$; and each arrow $f : d \to d'$ mapped to the precomposition $D(f, -) : g \mapsto g \circ f$. We will write $D(f, -)$ as f^* for the purposes of this proof. This is basically the Yoneda embedding, but applied over the category of elements of F, so we would expect the proof may fall out of the Yoneda lemma. Indeed, the inverse Yoneda isomorphism

$$y_d^{-1} : F(d) \to \mathrm{Nat}(C(d, -), F), \qquad u \mapsto \alpha^u$$

where $\alpha_c^u : D(d, c) \to F(c)$ is the function $g \mapsto F(g)(c)$. We claim this yields a universal co-cone for M over $\mathbf{Set}^D$ with F as the colimit, given by maps $\mu : M(d, x) \to F$ given by $\mu_d^x = y_d^{-1}(x)$.

To show this, let $F(f) : (d, x) \to (d', x')$, and thus the image under M is

$$C(d, -) \xrightarrow{f^*} C(d', -).$$

Naturality of the Yoneda isomorphism in d gives that

$$\begin{array}{ccc} C(d,-) & \xrightarrow{f^*} & C(d',-) \\ \Big\downarrow{\scriptstyle y_d^{-1}} & & \Big\downarrow{\scriptstyle y_{d'}^{-1}} \\ F & = & F \end{array}$$

I don't think this works but I give up this proof is aids. ■

Contravariant functors $F : C^{\mathrm{op}} \to \mathbf{Set}$ are often called *presheaves*; the functor category $\mathbf{Set}^{C^{\mathrm{op}}}$ is often written as $\hat{C}$ to indicate this. The intuition comes from the topological definition of a presheaf falling out when applied to the poset category of open sets of a topological space.

Chapter 4

Adjoints

4.1 Adjunctions

The following notation appears to be fairly dated, but I will use it over the course of this book since Mac Lane uses it.

Definition 4.1: Adjunction

Let C, D be categories. An adjunction from C to D is a triple $(F, G, \varphi) : D \rightharpoonup C$ where F and G are functors

$$C \underset{G}{\overset{F}{\rightleftarrows}} D,$$

and φ is a function which maps pairs $(c, d) \in C \times D$ to an isomorphism

$$\varphi_{(c,d)} : D(F(c), d) \cong C(c, G(d)),$$

which is natural in c and d.

The naturality condition in φ asserts that φ is a natural isomorphism between the compositions of bifunctors

$$\varphi : D(-,-) \circ (F^{\text{op}} \times 1_D) \Rightarrow C(-,-) \circ (1_{C^{\text{op}}} \times G),$$

where we recall that $F^{\text{op}} : C^{\text{op}} \to D^{\text{op}}$. This is necessary due to how the hom functors work, as we well know. More concisely,

$$\varphi : D(F(-), -) \Rightarrow C(-, G(-)).$$

The naturality condition thus asserts that for all $f : c \to c'$ and $g : d \to d'$, the diagram

$$\begin{tikzcd}
D(F(c), d) \arrow[r, "\varphi_{(c,d)}"] \arrow[d, "D(F(f),g)"'] & C(c, G(d)) \arrow[d, "C(f,G(g))"] \\
D(F(c'), d') \arrow[r, "\varphi_{(c',d')}"'] & C(c', G(d'))
\end{tikzcd}$$

commutes. This is equivalently viewed as the book presents it, where we have the commutivity of 2 diagrams, one only considering f and one only considering g. As is, the morphism $C(F(f), g)$ is the map $\varphi \mapsto g \circ \varphi \circ F(f)$, and $D(f, G(g))$ is similar.

With notation as above, we say F is *left-adjoint* to G, and G is *right-adjoint* to F, since F appears on the left and G appears on the right in the hom-sets. We also call F a *left-adjoint functor*, and G a *right-adjoint functor*. We write $F \dashv G$.

Each adjunction gives rise to universal arrows. Let $d = F(c)$. Then the natural isomorphism $\varphi_{(c,d)}$ gives

$$D(F(c), F(c)) \cong C(c, G(F(c))).$$

We then have $1_{F(c)} \in D(F(c), F(c))$, so define $\eta_c = 1_{F(c)}$. By proposition 3.1, we thus have that $(\eta_c, F(c))$ is universal from c to G (since each map $c \to G(d)$ is in bijection with a map $F(c) \to d$), and thus we have an assignment $c \mapsto \eta_c$ to universal arrows *for all* $c \in C$. Moreover, this assignment $c \mapsto \eta_c$ is natural $1_C \Rightarrow G \circ F$, since naturality of φ guarentees that the diagram

$$\begin{tikzcd}
c' \arrow[r, "\eta_{c'}"] \arrow[d, "h"'] & G(F(x')) \arrow[d, "G(F(h))"] \\
c \arrow[r, "\eta_c"'] & G(F(c))
\end{tikzcd}$$

since $\eta_c = \varphi(1_{F(c)})$. By proposition 3.1, we have that the bijection φ can be given by

$$\varphi_{(c,d)}(f) = G(f) \circ \eta_x.$$

I think it's good to show this explicitly here. By naturality of φ, for each $f : F(c) \to d$, the diagram

$$\begin{tikzcd}
D(F(c), F(c)) \arrow[r, "\varphi"] \arrow[d, "D(F(c),f)"'] & C(c, GF(c)) \arrow[d, "C(c,G(f))"] \\
D(F(c), d) \arrow[r, "\varphi"'] & C(c, G(d))
\end{tikzcd}$$

commutes. Chasing the arrow $1_{F(c)}$ around the diagram gives

$$G(f) \circ \varphi(1_{F(c)}) = \varphi(f \circ D(F(c), 1_{F(c)}).$$

By functorality, the left simplifies to $\varphi(f \circ 1_{F(c)}) = \varphi(f)$, and the right is defined as $G(f) \circ \eta_c$. This gives us this equality.

This all follows as well for F. Consider the bijection with $c = G(d)$:

$$\varphi : D(FG(d), d) \cong C(G(d), G(d)).$$

Define by ε_d the image $\varphi^{-1}(1_{G(d)})$, and by much the same arguments, we have that $(\varepsilon_d, G(d))$ is universal from F to d, which is essentially the same as the statement that

$$\varphi^{-1}(f) = \varepsilon_d \circ F(f), \qquad f : c \to G(d).$$

Finally, the map $d \mapsto \varepsilon_d$ is natural $\varepsilon : FG \Rightarrow 1_D$.

Consider the map $1_{G(d)} : G(d) \to G(d)$. We have by the previous equality that

$$\varphi^{-1}(1_{G(d)}) = \varepsilon \circ \varepsilon_d \circ F(1_{G(d)}) = \varepsilon_d \circ 1_{FG(d)} = \varepsilon_d.$$

We thus have

$$1_{G(d)} = (\varphi \circ \varphi^{-1})(1_{G(d)}) = \varphi(\varepsilon_d).$$

Since $\varepsilon_d : FG(d) \to d$, we have that

$$\varphi(\varepsilon_d) = G(\varepsilon_d) \circ \eta_{G(d)}.$$

We typically write this condition diagrammatically as the commutivity of the diagram

$$\begin{tikzcd}
G \arrow[r, "\eta G"'] \arrow[rr, bend left, "1_G"] & GFG \arrow[r, "G\varepsilon"'] & G.
\end{tikzcd}$$

By duality, the entirety of the following statement follows.

Theorem 4.1

An adjunction $(F, G, \varphi) : D \leftharpoonup C$ determines

1. A natural transformation $\eta : 1_C \Rightarrow GF$ such that each component η_c is universal from c to G, and such that for all $f : F(c) \to d$,
$$\varphi(f) = G(f) \circ \eta_c.$$

2. A natural transformation $\eta : FG \Rightarrow 1_D$ such that each component ε_d is universal from F to d, and such that for all $g : c \to G(d)$,
$$\varphi^{-1}(g) = \varepsilon_d \circ F(g).$$

3. The following composites are the identity natural transformations:
$$G \xrightarrow{\eta G} GFG \xrightarrow{G\varepsilon} G$$
$$F \xrightarrow{F\eta} FGF \xrightarrow{\varepsilon F} F$$

We call η the *unit* and ε the *counit* of the adjunction. The results of this theorem are in fact shown to determine adjunctions in various ways. This is summarized by the following theorem.

Theorem 4.2

Each adjunction $(F, G, \varphi) : C \leftharpoonup D$ is completely determined by any of the following.

1. Functors F, G and a natural transformation $\eta : 1_C \Rightarrow GF$ such that each $\eta_x : c \to GF(c)$ is universal from c to G;

2. The functor $G : D \to C$, an object function $F_0 : X \to A$, and a universal arrow $\eta_c : c \to GF_0(c)$ for all $c \in C$;

3. Functors F, G and a natural transformation $\varepsilon : FG \to 1_D$ such that each $\varepsilon_d : FG(d) \to d$ is universal from F to d;

4. The functor $F : C \to D$, an object function $G_0 : D \to C$, and a universal arrow $\varepsilon_d : FG(d) \to d$ for each $d \in D$;

5. Functors F, G and natural transformations $\eta : 1_C \Rightarrow GF$ and $\varepsilon : FG \Rightarrow 1_D$ such that the composites $G\varepsilon \circ \eta G$ and $\varepsilon F \circ F\eta$ are the identity natural transformations.

Proof. Assume (1). Choose $c \in C$ and $d \in D$, and define the bijection $\varphi_{(c,d)} : D(F(c), d) \to C(c, G(d))$ via the previous identification
$$\varphi_{(c,d)}(f) = G(f) \circ \eta_c.$$
Each pair $(\eta_c, F(c))$ is universal from c to G, so proposition 3.1 gives the bijections
$$D(F(c), d) \cong C(c, G(d))$$

for any $d \in D$, natural in d. For the converse, let $f : c \to c'$, and consider the diagram

$$\begin{tikzcd}
D(F(c), d) \arrow[r, "\varphi"] \arrow[d, "D(F(c),f)"'] & C(c, G(d)) \arrow[d, "C(c,G(f))"] \\
D(F(c'), d) \arrow[r, "\varphi"'] & C(c', G(d))
\end{tikzcd}$$

The definition of φ gives

$$f \circ G(g) \circ \eta_c = GF(f) \circ G(g) \circ \eta_c$$

for any $g : F(c) \to d$. idk chase soemthign in the diagram ig?

Assume (2), we show (1). We simply show there exists a unique way to make F_0 into a functor for which η will be natural. Let $h : c \to c'$; universality of η_c guarentees there exists a unique arrow $h' : F_0(c) \to F_0(c')$ making the diagram

$$\begin{tikzcd}
c \arrow[r, "\eta_c"] \arrow[d, "h"'] & GF_0(c) \arrow[d, dashed, "G(h')"] \\
c' \arrow[r, "\eta_{c'}"'] & GF_0(c')
\end{tikzcd}$$

commute. Thus, η is natural if we define $F(h) = h'$, and this is clearly functorial by universality.

(3,4) are dual to (1,2).

Assume (5); define φ by

$$\varphi(f) = G(f) \circ \eta_x, \qquad f : F(c) \to d,$$
$$\varphi^{-1}(g) = \varepsilon_d \circ F(g), \qquad g : c \to G(d).$$

Since G is a functor and η is natural,

$$(\varphi \circ \varphi^{-1})(g) = \varphi(\varepsilon_d \circ F(g)) = G(\varepsilon_d) \circ GF(g) \circ \eta_c = G(\varepsilon_d) \circ \eta_{G(d)} \circ g = g.$$

Therefore, φ is a bijection of hom-sets. It is moreover clearly natural. ■

This theorem is useful, since it allows us to construct adjunctions any time we have a universal arrow from/to every object in a category to/from the same functor. For example, let C have finite products. Then there is a universal arrow from $c \in C$ to Δ for all $c \in C$, and thus the function $(a, b) \mapsto a \times b$ is in fact a functor $C \times C \to C$ by the previous theorem, and $\Delta \dashv \times$. We thus have

$$(C \times C)(\Delta(c), (a, b)) \cong C(c, a \times b) \implies C(c, a) \times C(c, b) \cong C(c, a \times b).$$

Similarly, a category with coproducts admits a coproduct functor which is left-adjoint to Δ. Any limit that can be constructed and always exists is adjoint to some diagonal functor, and it should be clear that proving universality is a simple way of showing the existence of an adjoint.

The final condition makes discussion of adjoints easy, since manipulation of these simple identities, known as the *triangle identities*, is very simple in comparison to some of the alternatives.

Corollary 4.1

Left-adjoints F of a functor G are unique up to isomorphism.

Proof. Adjunctions $F \dashv G$ and $F' \dashv$ give universal arrows $x \to GF(x)$ and $x \to GF(x')$, which must be isomorphic by universality. The proof from her is obvious. ■

Corollary 4.2

A functor $G : D \to C$ has a left adjoint if and only if for all $c \in C$, the functor $C(c, G(d))$ is representable as a functor of $d \in D$.

Proof. This is a restatement of (2) from theorem 4.2. A representation would in this case be a natural isomorphism

$$\varphi : C(c, G(d)) \cong D(F_0(x), a)$$

for an object function F_0. ■

Theorem 4.3

If $G : D \to C$ is additive between *Ab*-categories, and admits a left-adjoint $F : C \to D$, then F is additive, and the adjunction is an isomorphism of abelian groups.

Proof. If η is the unit of the adjunction, then $\varphi(f) = G(f) \circ \eta_c$ for any $f : F(c) \to d$. Additivity gives

$$\varphi(f + f') = G(f + f') \circ \eta_c = G(f)\eta_c + G(f)\eta_c = \varphi(f) + \varphi(f').$$

Next, we have

$$GF(g + g') \circ p\eta_c = \eta_{c'}(g + g')$$

by naturality of η, and thus additivity of F follows easily from that of G. ■

4.2 Examples of Adjoints

Forgetful Functor	*Left Adjoint F*	*Unit of Adjunction*
$U : R\text{-}\mathsf{Mod} \to \mathbf{Set}$	Free R-modules	insertion of generators $j : X \to UF(X)$
$U : \mathbf{Cat} \to \mathbf{Grph}$	Free category	insertion of generators $G \to U(C(G))$
$U : \mathbf{Grp} \to \mathbf{Set}$	Free groups	insertion of generators
$U : \mathbf{Ab} \to \mathbf{Set}$	Free abelian groups	insertion of generators
$U : \mathbf{Ab} \to \mathbf{Grp}$	Quotient by the center $G \mapsto G/Z(G)$	projection onto the quotient $\pi : G \to G/Z(G)$
$U : R\text{-}\mathsf{Mod} \to \mathbf{Ab}$	$A \mapsto R \otimes A$	$A \to U(R \otimes A)$ by $a \mapsto 1 \otimes a$

A completion of this table, and a similar table for (co)limits, can be found on pages 87/88. I do not feel the need to copy down reference material, I would rather compute it myself.

4.3 Reflexive Subcategories

For many forgetful functors, the counit $\eta : FU \Rightarrow 1_A$ maps $a \mapsto \eta_a : F(U(a)) \to a$, which turns out to also be an epimorphism. This is a fact, and the proof requires the following lemma.

Lemma 4.1

Let $f^* : A(a,-) \Rightarrow A(b,-)$ be the natural transformation induced by an arrow $f : b \to a$ in A. Then f^* is monic if and only if f is epi, and f^* is epi if and only if f is a *split* monic.

Proof. By the Yoneda lemma, $f^* \mapsto f$ is a natural bijection, and by a previous exercise a natural transformation is monic/epi if and only if every component is monic/epi.

For $f \in A(a,c)$, by definition $f^*(h) = h \circ f$. Since f^* is a function, assume it is injective, since that is equivalent to monic. We thus have $f^*(h) = f^*(k)$ implies $h = k$, and rewriting this as above gives

$$h \circ f = k \circ f \implies h = k.$$

Thus, f is epi, and the converse is the same. This proves the first statement, so now assume f^* is epi. It is thus true that each component is epi, and thus the map $f^* : A(a,b) \to A(b,b)$ is surjective. There thus exists $h_0 : a \to b$ such that $f^*(h_0) = 1_b$. Expanding this gives $h_0 \circ f = 1_b$, and f has a left-inverse. The converse is trivial. ■

Theorem 4.4

Let $(F, G, \eta, \varepsilon) : X \rightharpoonup A$ be an adjunction.

1. G is faithful if and only if every component ε_a is epi;
2. G is full if and only if every component ε_a is a split monic.

Hence G is fully faithful if and only if each ε_a is an isomorphism $FG(a) \cong a$.

Proof. Note that the arrow function $G_{a,c} : A(a,c) \to X(G(a), G(c))$ turns into a natural transformation by the composition

$$A(a,c) \xrightarrow{G_{a,c}} X(G(a), G(c)) \xrightarrow{\varphi} A(FG(a), c).$$

By the Yoneda lemma, this component of the natural transformation is fully determined by $G(1_a)$ (easily seen by setting $c = a$). This is precisely η_a. Since φ is an isomorphism, this composition is epi if and only if $G_{a,c}$ is epi, and vice versa for mono. This is viewed again as injective vs surjective, hence faithful vs full. Note that we can view this as precomposition by $G_{a,c}$, and thus the proof follows from lemma 4.1. ■

Definition 4.2: Reflective Subcategory

A subcategory A of B is *reflective* if the inclusion functor $i : A \hookrightarrow B$ admits a left-adjoint $F : B \to A$, called the *reflector*. The adjunction $(F, i, \eta, \varepsilon)$ is called a *reflection*. Dually, A is *coreflective* if i has a right-adjoint.

Since the inclusion functor is really an identity, this can be described more succinctly as a functor $R = KF : B \to B$ such that we have a bijection

$$A(R(b), a) \cong B(b, a).$$

Since the inclusion is always faithful, the counit ε is *always* epi. For example, **Ab** is reflective in **Grp**.

4.4 Equivalence of Categories

Categories are isomorphic iff there exist invertible functors between them. In this case, there is an adjunction $(F, G, 1_F, 1_G)$ with $F \circ G = 1$ and $G \circ F = 1$. This is a very strict notion, and it is in fact often more useful to work in a more general notion.

Definition 4.3: Equivalence of Categories

A functor $F : C \to D$ is an *equivalence of categories* when there exists a functor $G : D \to C$ with $FG \cong 1_D$ and $GF \cong 1_C$; that is, they are invertible up to isomorphism. G is also an equivalence of categories here.

An example arises in the following useful definition.

Definition 4.4: Skeleton

A *skeleton* of a category C is any full subcategory A such that each object in C is isomorphic to *exactly one* object in A; in other words, A contains precisely one element from all equivalence classes of C.

It is easily shown that given a skeleton $A \subseteq C$, the inclusion is an equivalence of categories. For each $c \in C$, let $a_c \in A$ have $a_c \cong c$. Define $F : C \to A$ by $c \mapsto a_c$. Since they are isomorphic, by the Yoneda lemma they have a bijection of hom-sets, so there is an apparent way to define action on morphisms as well, although I do not feel like typing up the diagram. Then clearly we have that for any $a \in A$, $F(i(a)) = a \cong a$, and for any $c \in C$, $i(F(c)) = a_c \cong c$.

An *adjoint equivalence* of categories is an adjunction $(F, G, \eta, \varepsilon)$ in which both the unit and counit are natural isomorphisms. It's not hard to show by pulling out the natural isomorphism in adjunctions that you then get the composition $G\eta^{-1} \circ \varepsilon^{-1}G = 1$, and similarly for F, giving a reverse adjunction $(G, F, \varepsilon^{-1}, \eta^{-1})$. Thus, F is both the left- and right-adjoint of G simultaneously, and vice versa.

Theorem 4.5

Let $G : C \to A$. The following are equivalent:

1. G is an equivalence of categories;
2. G is part of an adjoint equivalence;
3. G is fully faithful, and each $c \in C$ is isomorphic to $F(a)$ for some $a \in A$;

Proof. (2) implies (1) by definition, since the unit $\eta : 1_C \Rightarrow GF$ and $\varepsilon : FG \Rightarrow 1_A$ need to be isomorphisms.

(1) implies (3) fairly easily; since $GF \cong 1_C$ by some natural transformation α, we have that each $GF(c) \cong c$ by α for all $c \in C$. Set $a = F(c) \in A$. To show G is fully faithful, α gives a commutative square

$$\begin{tikzcd}
FG(a) \arrow[r, "\alpha_a"] \arrow[d, "FG(f)"'] & a \arrow[d, "f"] \\
FG(a') \arrow[r, "\alpha_{a'}"'] & a'
\end{tikzcd}$$

for all $f : a \to a'$ in A; hence we have

$$f \circ \alpha_a = \alpha_{a'} \circ FG(f) \implies f = \alpha_{a'} \circ FG(f) \circ \alpha_{a'}^{-1}.$$

Apparently this shows G is faithfull, I don't fully see it but sure. This proves T is faithful by symmetry. By commutivity of the above square and the fact that G is faithful, the fact that G is fulll follows very easily.

To show (3) implies (2), we construct from G a left-adjoint F. By theorem **??**, we have that ε is a natural isomorphism. I'll maybe finish this tomorrow. ■

An interesting result is that any full subcategory $A \subseteq C$ satisfying $\forall c \in C \exists a \in A$ such that $c \cong a$ is reflective. This is unsuprising; it amounts to simply saying any full subcategory containing a skeleton is equivalent to C.

4.5 Adjoints for Preorders

Theorem 4.6 (Galois Connections are Adjoint Pairs)

Let P, Q be preorders and $L : P \to Q^{\mathrm{op}}$, $R : L^{\mathrm{op}} \to Q$ two order-preserving functions (functors). Then $L \dashv R$ if and only if for all $p \in P$ and $q \in Q$,

$$L(p) \geq q \qquad \iff \qquad p \leq R(q).$$

In this case, there is precisely one adjunction φ, and we have

$$p \leq RL(p), \qquad LR(q) \geq q, \qquad L(p) \geq LRL(p) \geq L(p),$$

and the same for q.

Proof. The statement that $L(q) \geq p \iff p \leq R(q)$ is equivalent to the existence of single morphisms

$$q \to L(p), \qquad p \to R(q).$$

We thus have the isomorphism

$$\varphi : Q^{\mathrm{op}}(L(p), q) \cong P(p, R(q)).$$

Naturality of φ is trivial. I don't care to show it but the unit is $p \leq RL(p)$ and the counit is the dual, concluding the proof. ■

Pairs of order preserving functions L, R satisfying this condition are known as *Galois Connections* from P to Q. The book discusses how there are galois connections in group actions, and thus Galois theory; and also in logic. I could not care less SKIP

4.6 Cartesian Closed Categories

Much of category theory involves categories with additional structure, such as *closed* categories. We will here define a certain important type of those, and they will be defined in generality later.

Definition 4.5: Cartesian Closed Category

A Cartesian Closed Category is a category C with finite products, such that $\Delta : C \to C \times C$ has a specified right-adjoint $(a, b) \mapsto a \times b$ (that is, a choice for product), and any functor $C \to \mathbf{1}$ and $- \times b : C \to C$ has a right-adjoint $0 \mapsto t$ and $c \mapsto c^b$, respectively, where t is terminal and c^b is notation for some object satisfying

$$\mathrm{Hom}(a \times b, c) \cong \mathrm{Hom}(a, b^c)$$

by the adjunction condition.

It will be shown shortly that $(b, c) \mapsto c^b$ is the object function of a bifunctor, and thus specifying the adjunction amounts to specifying an arrow $e : c^b \times b \to c$ which is natural in c and universal $- \times b$ to c, called the *evaluation map*. This is since in the categories **Set** or **Cat** with c^b given by $\mathrm{Hom}(b, c)$ or $\mathrm{Fun}(b, c)$ respectively, the evaluation map is simply the evaluation $f \mapsto f(x)$.

4.7 Transformation of Adjoints

Let $(F, G, \varphi, \eta, \varepsilon)$ and $(F', G', \varphi', \eta', \varepsilon')$ be adjunctions $X \rightharpoonup A$. A *map* of adjunctions from the first to the second is a pair (K, L) of functors $K : A \to A'$ and $L : X \to X'$ such that the diagram

$$\begin{tikzcd}
A \arrow[r, "G"] \arrow[d, "K"'] & X \arrow[r, "F"] \arrow[d, "L"'] & A \arrow[d, "K"] \\
A' \arrow[r, "G'"'] & X' \arrow[r, "F'"'] & A'
\end{tikzcd}$$

commutes, and such that the diagram

$$\begin{tikzcd}
A(F(x), a) \arrow[r, "\varphi"] \arrow[d, "K"'] & X(x, G(a)) \arrow[d, "L"] \\
A'(KF(x), K(a)) \arrow[d, equal] & X'(L(x), LG(a)) \arrow[d, equal] \\
A'(F'L(x), K(a)) \arrow[r, "\varphi'"'] & X'(L(x), G'K(a))
\end{tikzcd}$$

commutes. Here the arrows given by K and L are the morphism functions those functors induce.

Proposition 4.1

Given adjunctions as above, and functors K, L satisfying the first diagram, the second is equivalent to the condition that

$$L\eta = \eta' L \qquad \text{and} K\varepsilon = \varepsilon' K.$$

Proof. Let the third diagram commute. Set $a = F(x)$, and chase 1_a around the diagram:

$$\begin{tikzcd}
1_a \arrow[r, "\varphi"] \arrow[d, "K"'] & \eta_a \arrow[d, "L"] \\
1_{K(a)} \arrow[d, equal] & L(\eta_a) \arrow[d, equal] \\
1_{F'L(x)} \arrow[r, "\varphi'"'] & \eta'_{L(x)} =
\end{tikzcd}$$

This should probably show the converse as well. ∎

4.8 Composition of Adjoints

Theorem 4.7

Let $(F, G, \eta, \varepsilon)$ be an adjunction from X to A, and $(\overline{F}, \overline{G}, \overline{\eta}, \overline{\varepsilon})$ an adjunction from A to D. Then there is an adjunction

$$(\overline{F}F, G\overline{G}, G\overline{\eta}F \cdot \eta, \overline{\varepsilon} \cdot \overline{F}\varepsilon\overline{G})$$

from X to D, with $\cdot$ vertical composition.

Using this composition, we define the category **Adj** with objects small categories, and morphisms adjunctions. We can also take $\mathbf{Adj}(X, A)$ to be a category using conjugate pairs, which I am too lazy to copy down.

4.9 Subsets and Characteristic Functions

The characteristic function $\mathbb{1}_S$ of a subset $S \subseteq X$ is the function

$$\mathbb{1}_S(x) = 0 \iff x \in S, \qquad \mathbb{1}_S(x) = 1 \iff x \in X \setminus S.$$

This means that the portion of X that maps into $\{0\}$ represents the subset S, and there exists a squrae

$$\begin{tikzcd}
S \arrow[r] \arrow[d] & \{0\} \arrow[d] \\
X \arrow[r] & \{0, 1\}
\end{tikzcd}$$

This classifies subobjects. We generalize this.

Definition 4.6: Subobject Classifier

Let C be a category with terminal object 1. A *subobject classifier* for C is a monomorphism $t : 1 \to \Omega$ such that every monomorphism $m : S \to X$ is a pullback of t in a unique way; such that the square

$$\begin{tikzcd} S \arrow[r] \arrow[d, tail, "m"'] & 1 \arrow[d, tail, "t"] \\ X \arrow[r] & \Omega \end{tikzcd}$$

commutes.

4.10 Categories Like Set

Definition 4.7: Elementary Topos

An elementary **topos** is a category E for which

1. E has all finite limits;
2. E has a subobject classifier;
3. E is cartesian closed.

Recall this implies that an elementary topos has all exponentials; all functors $-\times b$ have a right-adjoint $c \mapsto c^b$, meaning

$$\mathrm{Hom}_E(a \times b, c) \cong \mathrm{Hom}_E(a, c^b).$$

The category **Set** is a topos, but more intetestingly for any C, the category $\mathrm{Fun}(C^{\mathrm{op}}, \mathbf{Set})$ is a topos. That is, the category of C-valued preasheaves on **Set** is a topos. Topoi are extremely important in topological fields.

Chapter 5

Limits

5.1 Creation of Limits

Definition 5.1: Complete

A *complete* category (sometimes *small-complete*) is a category C for which all small limits exist, meaning for all small J and all functors $F : J \to C$, $\varprojlim F$ exists.

We will soon show some important categories, such as **Set**, **Grp**, and **Ab** are complete. We will illustrate the process with an example. Consider the poset category ω of finite ordinals. Note that any for any $F : \omega^{\mathrm{op}} \to \mathbf{Set}$, any limiting cone $\mu : \Delta(\varprojlim F) \Rightarrow F$ satisfies

$$\begin{tikzcd} & & \varprojlim F \arrow[dl, "\mu_3"'] \arrow[d, "\mu_2"] \arrow[dr, "\mu_1"] \arrow[drr, "\mu_0"] & & \\ \cdots \arrow[r, "f_3"'] & F(3) \arrow[r, "f_2"'] & F(2) \arrow[r, "f_1"'] & F(1) \arrow[r, "f_0"'] & F(0) \end{tikzcd}$$

In this case, we can take $\{F(n)\}_{n=0}^{\infty}$ to simply be an ordered sequence of sets, with functions $f_n : F_{n+1} \to F_n$. Note that if we removed the functions f_n, then the product $\prod_{n=0}^{\infty} F_n$ would be a limit of F. However, the projections π_n need not satisfty $\pi_n = f_n \pi_{n+1}$; that is, the diagram need not commute.

Note that elements of the product can be viewed as sequences $\{x_n\}_{n=0}^{\infty}$. For this portion of my notes, to make notation clear, denote by $\mathbf{x}$ the sequence $\{x_n\}_{n=0}^{\infty}$.

To fix the commutivity issue, take L to be the subset of the product such that for all $\mathbf{x} \in L$, we have $f_n(x_{n+1}) = x_n$. Then define functions $\mu_n : L \to F_n$ such that $\mu(\mathbf{x}) = x_n$; that is, $\mu_n = \pi_n|_V$. We then have for all n,

$$(f_n \circ \mu_{n+1})(\mathbf{x}) = f_n(x_n) = x_{n-1} = \mu_{n-1}(\mathbf{x}).$$

Therefore, $\mu : \Delta(L) \Rightarrow F$ is a cone for F. To show universality, let $\tau : \Delta(M) \Rightarrow F$ be another cone for F, for each $\mathbf{m} \in M$ there exists a sequence $\{\tau_n(\mathbf{m})\}_{n=0}^{\infty}$ (denoted here by $\tau(\mathbf{m})$) which satisfies $f_n(\tau_{n+1}(\mathbf{m})) = \tau_n(\mathbf{m})$; that is, the sequence $\tau(\mathbf{m})$ matches. Thus, $\tau(\mathbf{m}) \in L$, and thus we have a function $\mathbf{m} \mapsto \tau(\mathbf{m})$ from $M \to L$. Uniqueness is manifest, and commutivity of the diagram is trivial. Therefore, $L \cong \varprojlim F$.

A sequence $\mathbf{x}$ that "matches" can also be viewed as a cone $\mathbf{x} : \Delta(*) \Rightarrow F$, since functions $\mathbf{x}_n$ amount to identifying elements $x_n \in F_n$, so commutivity of the cone is equivalent to "matching". We can then view L as the set $\mathrm{Cone}(*, F)$ of all cones $* \Rightarrow F$. In fact, this construction is independent of the domain of F.

Theorem 5.1 (Completeness of Set)

Let J be a small category, and $F : J \to \mathbf{Set}$ any functor. Then $\varprojlim F$ exists, and moreover the set $\mathrm{Cone}(*, F)$ of cones $\sigma : * \Rightarrow F$ are a limit for F, with the limiting cone $\nu : \mathrm{Cone}(*, F) \Rightarrow F$ given by

$$\nu_j : \mathrm{Cone}(*, F) \to F_j, \qquad \sigma \mapsto \sigma_j.$$

Proof. Since J is small, $\mathrm{Cone}(*, F)$ is a small set. Let $f : j \to k$ in J, and let σ a cone $* \Rightarrow F$. Then since σ is a cone, we have $\sigma_k = F(f)\sigma_j$. We thus have

$$(F(f) \circ \nu_j)(\sigma) = F(f)(\sigma_j) = F(f) \circ \sigma_j = \sigma_k = \nu_k(\sigma).$$

Therefore, ν is a cone for F. To show universality, let $\mu : \Delta(X) \Rightarrow F$ be a cone. Then for each $x \in X$, there exists a map $x \mapsto \{\mu_j(x)\}_{j \in J}$, and each $\mu_j(x)$ corresponds to a function $* \to \mu_j(x)$. The collection of all these functions forms a cone by the same logic as the above discussion, so we have that there exists a function $X \to \mathrm{Cone}(*, F)$ given by $x \mapsto \{\mu(x)\}$. Uniqueness and commutivity of the diagram are trivial, concluding the proof. ■

By univerality of the limit, we have a natural isomorphism $\mathrm{Cone}(X, F) \cong \mathbf{Set}(X, \mathrm{Cone}(*, F))$ given by $\mu \mapsto (x \mapsto \{\mu(x)\})$. Since a cone itself is a natural transformation $\Delta(*) \Rightarrow F$, this is equivalently written as

$$\mathrm{Nat}(\Delta(X), F) \cong \mathbf{Set}(X, \mathrm{Cone}(*, F)).$$

This is thus an adjunction $\Delta \dashv \mathrm{Cone}(*, -) = \mathrm{Nat}(\Delta(*), -)$ to a representable functor. This is the most interesting part of the proof. This proves in fact that $\mathrm{Cone}(*, F)$ is a limit for F using the universal arrow definition of limit.

Limits in **Grp** and other categories are constructed similarly.

Theorem 5.2

Let $U : \mathbf{Set} \to \mathbf{Grp}$ be the forgetful functor, and let $H : J \to \mathbf{Grp}$ with J small such that HU has a limit L, and $\nu : L \Rightarrow UH$ the limiting cone. Then there is exactly one group structure on L such that morphisms $\nu_j : L \to H(j)$ are group homomorphisms, and L is a limit for H with ν the limiting cone.

Proof. By theorem 5.1, take $L = \mathrm{Cone}(*, UH)$. Let $\sigma, \tau \in \mathrm{Cone}(*, UH)$ be cones, and define the product $\sigma\tau$ of these cones to have components $(\sigma\tau)_j = \sigma_j \tau_j$ (the pointwise product in H_j, aka the product of the elements picked out by the arrows) and the inverse $(\sigma_j)^{-1} = \sigma_j^{-1}$ (the inverse in H_j). Viewing each function as an element of H_j, this defines the subgroup of $\prod_j H_j$ generated by L. Then ν simply becomes the restriction of the projections $\pi_j|_L$, and thus it satisfies the homomorphism condition. Since it's domain is also a group, each ν_j is a group homomorphism. Moreover, requiring that ν_j be a group homomorphism forces this definition of $(\sigma\tau)_j$, showing uniqueness.

For some reason the text fully spells out the obvious fact that any other cone is also a cone in **Set**, so universality follows by universality under U and functorality of U; equivalently, universality in **Set**. ■

We can thus construct limits in **Grp** from limits in **Set** in a unique way. This same argument handles many categories with forgetful functors to set, such as **Ab**, **Ring**, R-Mod. This gives the following definition.

Definition 5.2: Creates Limits

A functor $V : A \to X$ *creates limits* for a functor $F : J \to A$ if

1. For every limiting cone $\tau : x \Rightarrow VF$ in X, there is exactly one pair (a, σ) with $a \in A$, $V(a) = x$, and $\sigma : a \Rightarrow F$ a cone with $V(\sigma) = \tau$;

2. This cone σ is a limiting cone in A.

Similarly, we may define functors to "create products", "create finite limits", "create colimits", etc. Note that a limit is only created in A when the limit of the composite exists in X. Theorem 5.2 can thus be restated as the forgetful functor U creates limits (for all small functors).

5.2 Limits by Products and Equalizers

There is a subtle yet fundamental fact hiding in this discussion of limits. Each limit $F : J \to \mathbf{Set}$ is given by the set $\mathrm{Cone}(*, F) \subseteq \prod_j F_j$. This can be broken into two steps:

1. Construct the product $\prod_j F_j$;

2. Requiring that the elements be a cone is equivalent to requiring that for all $f : j \to k$ in J, $F(f)(x_j) = x_k$. This amounts to saying that $x \in \mathrm{eq}(F(f) \circ \pi_j, \pi_k)$ is an element of the equalizer.

It may be prudent to recall the universal property of equalizers:

$$\begin{tikzcd} E \arrow[r, "\mathrm{eq}"] & \prod_j F_j \arrow[r, shift left, "F(f)\circ\pi_j"] \arrow[r, shift right, "\pi_k"'] & F_k \\ Z \arrow[u, dashed] \arrow[ur] & & \end{tikzcd}$$

More precisely, we require that $x \in \mathrm{eq}(E)$. This construction in fact generalises to arbitrary categories, giving us limits via only products and equalizers.

Theorem 5.3

Let J and C be categories. If C has equalizers of all pairs of arrows, and all products indexed by $\mathrm{Obj}(J)$ and $\mathrm{Mor}(J)$, then C has a limit for all functors $F : J \to C$.

Proof. By assumption, the products

$$\prod_{i\in J} F_i, \qquad \prod_{\alpha\in\mathrm{Mor}(J)} F_{\mathrm{cod}\,\alpha}$$

exist. Since for each $u \in \mathrm{Mor}(J)$, we have $\mathrm{cod}\,u \in J$, there exists a projection $\pi_{\mathrm{cod}\,u} : \prod_{i\in J} F_i \to F_{\mathrm{cod}\,u}$, and thus for each entry in the second product, we have

$$\begin{tikzcd}
\prod_{i\in J} F_i \arrow[r, "\pi_{\operatorname{cod} u}"] & F_{\operatorname{cod} u} \\
\prod_{\alpha\in \operatorname{Mor}\alpha} \arrow[ur, "\pi_{\operatorname{cod} u}"'] &
\end{tikzcd}$$

By the universal property of products, there is a unique morphism $f : \prod F_i \to \prod F_{\operatorname{cod}\alpha}$ making each diagram

$$\begin{tikzcd}
F_{\operatorname{cod} u} & \\
\prod_{\alpha\in \operatorname{Mor}(J)} \arrow[u, "\pi_{\operatorname{cod} u}"] & \prod_{i\in J} F_i \arrow[l, dashed, "f"] \arrow[ul, "\pi_{\operatorname{cod} u}"']
\end{tikzcd}$$

commute. There also exists a composition

$$\prod_{i\in J} F_i \xrightarrow{\pi_{\operatorname{dom} u}} F_{\operatorname{dom} u} \xrightarrow{F(u)} F_{\operatorname{cod} u},$$

and thus there exists a unique morphism $g : \prod F_i \to \prod F_{\operatorname{cod} u}$ making the diagram

$$\begin{tikzcd}
F_{\operatorname{cod} u} & F_{\operatorname{dom} u} \arrow[l, "F(u)"] \\
\prod_{\alpha\in \operatorname{Mor}(J)} \arrow[u, "\pi_{\operatorname{cod} u}"] & \prod_{i\in J} F_i \arrow[l, dashed, "g"] \arrow[u, "\pi_{\operatorname{dom} u}"']
\end{tikzcd}$$

commute. By assumption, there exists an equalizer

$$\operatorname{Eq}(f, g) \xrightarrow{e} \prod_{i\in J} F_i \overset{f}{\underset{g}{\rightrightarrows}} \prod_{\alpha\in \operatorname{Mor}(J)} F_{\operatorname{cod}\alpha}.$$

Therefore, the diagram

$$\begin{tikzcd}
F_{\operatorname{cod} u} & & \\
\prod\limits_{\alpha\in\operatorname{Mor}(J)} F_{\operatorname{cod}\alpha} \arrow[u, "\pi_{\operatorname{cod} u}"] \arrow[d, "\pi_{\operatorname{cod} u}"'] & \prod\limits_{i\in J} F_i \arrow[l, shift right, "f"'] \arrow[l, shift left, "g"] \arrow[ul, "\pi_{\operatorname{cod} u}"'] \arrow[d, "\pi_{\operatorname{dom} u}"] & \operatorname{Eq}(f,g) \arrow[l, "e"'] \\
F_{\operatorname{cod} u} & F_{\operatorname{dom} u} \arrow[l, "F(u)"] &
\end{tikzcd}$$

commutes. The claim is that $\operatorname{Eq}(f,g) \cong \varprojlim F$ by the limiting cone $\nu : \operatorname{Eq}(f,g) \Rightarrow F$ given by $\nu_j = \pi_j e$. Let $\mu : M \Rightarrow F$ be a cone. Since μ has a map to each F_i, there exists a unique induced map

$$\prod_{i\in J} \sigma_i : M \to \prod_{i\in J} F_i.$$

Call this map μ for simplicity. Since $\mu_j = \pi_j \mu$, we have that μ factors through the product, and thus the diagram

$$\begin{tikzcd}
F_{\operatorname{cod} u} & & \\
\prod\limits_{\alpha\in\operatorname{Mor}(J)} F_{\operatorname{cod}\alpha} \arrow[u, "\pi_{\operatorname{cod} u}"] \arrow[d, "\pi_{\operatorname{cod} u}"'] & \prod\limits_{i\in J} F_i \arrow[l, shift right, "f"'] \arrow[l, shift left, "g"] \arrow[ul, "\pi_{\operatorname{cod} u}"'] \arrow[d, "\pi_{\operatorname{dom} u}"] & M \arrow[l, "\mu"'] \\
F_{\operatorname{cod} u} & F_{\operatorname{dom} u} \arrow[l, "F(u)"] &
\end{tikzcd}$$

commutes once again since μ is a cone. Since the equalizer is terminal with respect to this property, there exists a unique morphism $M \to \operatorname{Eq}(f,g)$ making the relevant diagram commute, and thus the equalizer together with ν is a limiting cone. ■

Using the details from the proof, we can classify limits.

Theorem 5.4

The limit of $F : J \to C$ is the equalizer of the morphisms $f, g : \prod_{i\in J} F_i \to \prod_{\alpha\in\operatorname{Mor}(J)} F_{\operatorname{cod}\alpha}$, where

$$\pi_{\operatorname{cod} u} \circ f = \pi_{\operatorname{cod} u}, \qquad \pi_{\operatorname{cod} u} \circ g = F(u) \circ \pi_{\operatorname{dom} u}.$$

The limiting cone ν is $\nu_j = \pi_j e$, for $e : \operatorname{Eq}(f,g) \to \prod_i F_i$ the equalizing morphism.

Corollary 5.1

If C has a terminal object, products of all pairs (finite products), and equalizers of all pairs, then C has finite limits.

Corollary 5.2

If C has small products and equalizers of all pairs, then C is complete.

This second corollary gives another proof that **Set** is complete, and also proves the completeness of quite a few categories. Another important theorem classifies small complete categories.

Proposition 5.1 (Freyd)

Let C be a small category. If C is complete, then C is a preorder for which upper bounds exist for all subsets of elements.

Proof. Suppose for contradiction that C is not a preorder. Then there exist parallel arrows $f, g : a \to b$ with $f \neq g$. Form the product $b^J = \prod_{j \in J} b$ with all entries b. Since there are 2^J ways to map a to each entry in the product, and each is unique, there are at least $2^J = |\mathcal{P}(J)|$ arrows from a to b^J. If $|J| \geq |\mathrm{Mor}(C)|$, then there exists no bijection $\mathrm{Mor}(C)$ to $\mathcal{P}(J)$. Since C is small, so is $\mathrm{Mor}(C)$, so take $J = \mathrm{Mor}(C)$, a contradiction since $\mathrm{Mor}(C)$ must contain at least $|\mathcal{P}(\mathrm{Mor}(C))|$ arrows. ■

5.3 Limits with Parameters

Let $T : J \times P \to X$ be a bifunctor, where we think of the category P as a parameter category, like we might parameterize a function over the unit interval. Suppose for each $p \in P$ that $T(-, p) : J \to X$ has a limit. Then the object function $p \mapsto \varprojlim T(-, p)$ defines a functor. We will prove this by considering the functor category X^P, and the adjoint $S : J \to X^P$ of T giving the isomorphism

$$\mathbf{Cat}(J \times P, X) \cong \mathbf{Cat}(J, X^P).$$

Recall for all $p \in P$ there is a functor $E_p : X^P \to X$ given by $F \mapsto F(p)$, with action on natural transformations given by $E_p(\sigma) = \sigma_p$.

Theorem 5.5

If $S : J \to P^X$ is such that for all $p \in P$, the composite $E_p S$ has a limit L_p with limiting cone $\tau_p : L_p \Rightarrow E_p S$, then there is a unique functor $L : P \to X$ with object function $p \mapsto L_p$ such that $p \mapsto \tau_p$ is a limiting cone

$$\tau : \Delta(L) \Rightarrow S.$$

Proof. Let $h : p \to q$ in P. We write S_p to denote $E_p S$. Taking τ_p and τ_q to be the given cones, let

$h : j \to k$ in J. We thus have the diagram

$$\begin{tikzcd}
 & & L_p \arrow[d, dashed] \arrow[dddll, "\tau_p j"'] \arrow[dddrr, "\tau_p k"] & & \\
 & & L_q \arrow[dl, "\tau_q j"'] \arrow[dr, "\tau_q k"] & & \\
 & S_q(j) \arrow[rr, "S_q(u)"'] & & S_q(k) & \\
S_p(j) \arrow[ur, "S(h)(j)"'] \arrow[rrrr, "S_p(u)"'] & & & & S_p(k) \arrow[ul, "S(h)(k)"]
\end{tikzcd}$$

The diagram commutes by functorality and by cone-ness. The proof from here is obvious. ■

The conclusion can be restated as

$$E_p\left(\varprojlim S\right) = \varprojlim\left(E_p S\right).$$

This means in functor categories, limits may be calculated pointwise.

Corollary 5.3

If X is complete, then so is every functor category $\mathrm{Fun}(-, X)$.

Apparently we then have this.

Theorem 5.6

Write $|P|$ for the discrete category on P. Then the inclusion functor $i : |P| \to P$ induces the functor $i^* : X^P \to X^{|P|}$ which creates limits.

5.4 Preservation of Limits

Definition 5.3: Continuous

A functor $H : C \to D$ is said to *preserve the limits* of functors $F : J \to C$ if for all limiting cones $\nu : L \Rightarrow F$ yield by composition a limit cone $H\nu : H(L) \Rightarrow HF$. Note that composition must preserve both the limit and the limiting cone. A functor is called *continuous* if it preserves all small limits.

Theorem 5.7

Let C be locally small. Then the hom-functors $C(c, -) : C \to \mathbf{Set}$ preserve limits.

Proof. Let $F : J \to C$ a functor with limiting cone $\nu : \varprojlim F \Rightarrow F$. Applying $C(c, -)$ gives a cone ν_* by postcomposition

$$\begin{tikzcd} C(c, \varprojlim F) \arrow[r, "\nu_{*i}"] & C(c, F(i)) \\ X \arrow[u, dashed, "k"] \arrow[ur, "\tau_i"'] & \end{tikzcd}$$

in **Set**, with $i \in J$. For any cone $\tau : X \Rightarrow F$, each $x \in X$ gives a cone $\tau_i(x) : c \to F_i$ in C, and by universality of ν the proof can be concluded easily from here. ■

An alternative proof is as follows. Recall that $\mathrm{Cone}(X, -) \cong \mathbf{Set}(X, \mathrm{Cone}(*, -))$. This gives us

$$\begin{aligned} \mathrm{Cone}(X, C(c, F(-))) &\cong \mathbf{Set}(X, \mathrm{Cone}(*, C(c, F(-)))) \\ &\cong \mathbf{Set}(X, \mathrm{Cone}(c, F)) \\ &\cong \mathbf{Set}(X, C(c, \varprojlim F)). \end{aligned}$$

The adjunction then gives us

$$\varprojlim C(c, F(-)) \cong C(c, \varprojlim F).$$

The theorem also gives us that the contravariant hom functor maps colimits to the corresponding limits in **Set**. This gives us

$$C(\varinjlim F, c) \cong \varprojlim C(F(-), c).$$

Creation and preservation are related.

Theorem 5.8

If $V : A \to X$ creates limits for $F : J \to A$ and the composite $VF : J \to X$ has a limit, then V preserves the limit of F. In particular, if V creates all small limits and X is complete, then A is complete and V is continuous.

Proof. Let $\tau : a \Rightarrow F$ and $\sigma : x \Rightarrow VF$ be limiting cones. Since V creates limits, there is a unique limiting cone $\rho : b \Rightarrow F$ in X such that $V(\rho) = \sigma$ and $V(b) = x$. Since limits are unique up to isomorphism, there exists $\theta : b \cong a$ with $\tau\theta = \rho$ (vertical composition). Thus $V\theta : V(b) = x \cong V(a)$, and $V\tau \circ V\theta = V\rho = \sigma$ by functorality, so V preserves the limit. ■

Definition 5.4: Projective/Injective

An object p in a category C is *projective* if every arrow $h : p \to c$ factors uniquely through an epi $g : b \to c$, meaning the diagram

$$\begin{tikzcd} & p \arrow[dl, dashed, "h'"'] \arrow[d, "h"] \\ b \arrow[r, "g"'] & c \end{tikzcd}$$

commutes. Dually, an object is *injective* if the opposite diagram commutes, with epi replaced by mono.

These requirements are equivalent to the requirement that

- p is projective if and only if for all epi g, $\mathrm{Hom}(p, g) : \mathrm{Hom}(p, b) \to \mathrm{Hom}(p, c)$ is epi in **Set**. That is, $\mathrm{Hom}(p, -)$ preserves epis.
- i is injective if and only if $\mathrm{Hom}(-, q)$ maps monics to epis.

Why have we defined this here? Absolutely no idea.

5.5 Adjoints on Limits

One fact that we somewhat glanced over, but shines new light on our earlier findings that there is an adjunction $\Delta \dashv \mathrm{Cone}(*, -) = \varprojlim$ in **Set**. I want to show that Δ is left-adjoint to $\varprojlim$, provided that (J-ary) limits exist. Any natural transformation $\Delta(c) \Rightarrow F$ is by definition a cone for F, and each cone determines precisely one morphism $c \to \varprojlim F$. Conversely, every morphism $c \to \varprojlim F$ can factor through the limiting cone to produce a cone $\Delta(c) \Rightarrow F$, proving that

$$C^J(\Delta(c), F) \cong C(c, \varprojlim F).$$

Thus, $\Delta \dashv \varprojlim$ in arbitrary categories. This fact is used in an alternative proof of the following theorem. I present neither proof, since I have done the proof before and know how to do it, and also since one of the proofs uses music symbols that I have no idea how to typset. Nevertheless, the following is a fundamental result, and I present the dual as well to highlight this fact.

Theorem 5.9

If a functor $G : A \to X$ is a right-adjoint functor, then it preserves limits. Dually, if $F : X \to A$ is a left-adjoint functor, it preserves all colimits.

5.6 Freyd's Adjoint Functor Theorem

Theorem 5.10 (Existence of Initial Object)

Let D be a complete locally small category. Then D has an initial object if and only if the following property is satisfied.

Solution Set Condition: there exists a collection $\{k_i\}_{i \in I} \subseteq D$ of objects (for I a small set) such that for all $d \in D$ there exists an $i \in I$ and an arrow $k_i \to d$.

Proof. If there exists an initial object 1, then the singleton $\{1\}$ satisfies this property.

Conversely, since D is complete, there exists $w \coloneqq \prod_i k_i$. By composing the existing morphisms with projections, for all $d \in D$ there exists a morphism $w \to d$. By assumption, $D(w, w)$ is small and D is complete, so we can construct the equalizer $e : v \to w$ of the entire set $D(w, w)$. We thus have for all $d \in D$, a morphism $v \to d$.

Suppose $f, g : v \to d$. Let $e_1 : u \to v$ be the equalizer of f, g, forming the diagram

$$\begin{tikzcd}
u \arrow[r, "e_1"] & v \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] \arrow[d, "e"] & d \\
w \arrow[u, dashed, "s"] \arrow[r, "e \circ e_1 \circ s"'] & w \arrow[r, "\pi_i"'] & k_i \arrow[u]
\end{tikzcd}$$

The morphism s is induced by commutivity of the right square and the universality of the equalizer. Since e equalizes all endomorphisms of w, we have

$$ee_1se = 1_we = e1_v.$$

Since e is an equalizer, it is monic (easily shown), and thus $e_1se = 1_v$. Thus, we have

$$fe_1 = ge_1 \implies fe_1se = ge_1se \implies f = g.$$

Thus, the arrow $v \to d$ is unique. ■

Theorem 5.11 (Freyd Adjoint Functor Theorem)

Given a locally small, complete category A, a functor $G : A \to X$ has a left-adjoint if and only if it preserves all small limits and satisfies the following property.

Solution Set Condition: For each $x \in X$, there is a small set I and a family of arrows $\{f_i : x \to G(a_i)\}_{i \in I}$ such that any arrow $h : x \to G(a)$ can be written as a composite $h = G(t) \circ f_i$, for some $i \in I$ and some $t : a_i \to a$.

Proof. If G has a left-adjoint F, then by theorem 5.9 it preserves all limits. Let $x \in X$. The unit η of the adjunction gives $\eta_x : x \to GF(x)$ which is universal from x to G, so for any $h : x \to G(a)$, there exists a unique $t : F(x) \to a$ such that $h = G(t) \circ \eta_x$ by definition of a universal arrow. Take $I = \{\eta_x\}$.

Conversely, it suffices to show for all $x \in X$, there exists an arrow $x \to G(a)$ universal from x to G, by pointwise construction of adjoints ($F(x) = a$, morphisms $x \to G(z)$ map to the unique morphisms $F(x) \to z$). These universal arrows would be initial in the comma category $(x \downarrow G) =: D$, so it suffices to show theorem 5.10 applies. By assumption, since A has small hom-sets, so must D since hom-sets there are arrows in A satisfying commutivity conditions. Recall by definition of the comma category that objects are pairs (a, f) with $f : x \to G(a)$, and morphisms $(a, f) \to (a', f')$ are arrows $k : a \to a'$ in A such that the diagram

$$\begin{tikzcd} & x \arrow[dl, "f"'] \arrow[dr, "f'"] & \\ G(a) \arrow[rr, "G(k)"'] & & G(a') \end{tikzcd}$$

commutes. The solution set condition here states that for any object (a, h), there exists some $i \in I$ and an arrow $t : (a_i, f_i) \to (a, h)$. Since I is small, this is equivalent to the solution set condition from theorem 5.10, so it suffices to show $(x \downarrow G)$ is complete. By corollary 5.2, it suffices to show small products and equalizers of pairs exist. This is proven by using the fact that G preserves small limits to apply the following lemma to the fact that A is complete. ■

Lemma 5.1

If $G : A \to X$ preserves small products (or equalizers), then for each $x \in X$ the projection $Q : (x \downarrow G) \to A$ given by $(s : x \to G(a)) \mapsto a$ creates all small products (or equalizers).

Proof. Let J be a small discrete category, and $\{f_j : x \to G(a_j)\}_{j \in J}$ a family for which the product diagram

$$\pi_j : \prod_{i \in J} a_i \to a_j$$

exists in A. Since G preserves products, the product diagram

$$G(\pi_j) : G\left(\prod_{i \in J} a_i\right) \to G(a_j)$$

exists in X. There thus exists an arrow $f : x \to \prod_i a_i$ such that $\pi_j f = f_j$. We thus have that $\pi_j : f \to f_j$ is a cone in $(x \downarrow G)$, and moreover the projection under Q yields the product diagram $\prod_i a_i \to a_j$. It thus suffices to show $\pi_j : f \to f_j$ is a product diagram, which is easy enough. Thus, Q creates small products.

To show Q creaates equalizers, let $s, t : (a, f) \to (b, g)$ be morphisms in $(x \downarrow G)$, and let e equalize them under Q, where they are viewed as arrows in A:

$$\begin{tikzcd} d \arrow[d, dashed, "r'"'] \arrow[dr, "r"] & & \\ \mathrm{Eq} \arrow[r, "e"'] & a \arrow[r, shift left, "s"] \arrow[r, shift right, "t"'] & b \end{tikzcd}$$

Since G preserves equalizers, $G(e)$ is the equalizer of $G(s)$ and $G(t)$. But since $s, t : f \to g$, we have that $G(s) \circ f = g = G(t) \circ f$ by definition of the comma category, meaning f equalizes $G(s)$ and $G(t)$. By universality of equalizer, we have that there exists a unique $h : x \to G(\mathrm{Eq})$ such that the diagram

$$\begin{tikzcd} x \arrow[d, dashed, "h"'] \arrow[dr, "f"] \arrow[drr, bend left, "g"] & & \\ G(\mathrm{Eq}) \arrow[r, "G(e)"'] & G(a) \arrow[r, shift left, "G(s)"] \arrow[r, shift right, "G(t)"'] & G(b) \end{tikzcd}$$

commutes. We thus have that $e : h \to f$ in $(x \downarrow G)$, in such a way that e is the unique arrow with Q-projection $e : \mathrm{Eq} \to a$.

From here, it suffices to show e is the equalizer of s, t in $(x \downarrow G)$. We know it equalizes it, so we need only show it is universal. Let $k : x \to G(d)$ be an object in $(x \downarrow G)$, and let $r : k \to f$ equalize s, t. Then r equalizes s, t in A, and from here the proof is obvious. ■

Theorem 5.12 (The Representability Theorem)

Let D be locally small and complete. A functor $K : D \to \mathbf{Set}$ is representable if and only if it preserves all small limits and satisfies the following property.

Solution Set Condition: There exists a (small) set $S \subseteq D$ such that for all $d \in D$ and $x \in K(d)$, there is an $s \in S$ and $y \in K(s)$ and an arrow $f : s \to d$ with $K(f)(y) = x$.

Proof. Recall that a representation of a functor K is a universal arrow from the singleton to K, and hence an initial object in $(* \downarrow K)$. Since K is assumed continuous, by lemma 5.1 the category $(* \downarrow K)$ is complete. The solution set condition is a reformulation of that in theorem 5.10, and thus an initial object exists. ■

The adjoint functor theorem gives us adjoints to forgetful functors for many types of algebras, without resorting to explicit constructions. Let Ω be a set of operators, hence a graded set for which each $\omega \in \Omega$ has associated a natural number n called the *arity* of ω. Let E be the set of "identities", such as the rules governing associativity or commutivity or unity. For many (E, Ω)-algebra, the adjoint functor theorem yields a left-adjoint for the forgetful functor.

5.7 Subobjects and Generators

We define subobjects in general categories our best generalization of the inclusion functions. Let A a category, and define the preorder $u \leq v$ on monics if and only if u factors through v; i.e. $u = vh$ for some h. If $u \leq v$ and $v \leq u$, write $u \equiv v$. This forms an equivalence class, so for any $a \in A$ define the subobjects of a as the set of equivalence classes of morphisms into a under $\equiv$. This definition agrees with our general definitions in many categories, such as **Ring**, **Ab**, **Grp**, and R-Mod. However, it does not agree in all categories, such as **Top**.

Lemma 5.2

In any pullback

$$\begin{tikzcd} p \arrow[r, "f'"] \arrow[d, "g'"'] & t \arrow[d, "g"] \\ s \arrow[r, "f"'] & a, \end{tikzcd}$$

f monic implies f' monic, and the same for g.

Proof. Let $h, k : z \to p$ be parallel morphisms such that $f'h = f'k$. Then $gf'h = gf'k$, and by commutivity, $fg'h = fg'k$. If f is monic, then $g'h = g'k$ and $f'h = f'k$. Since p is a pullback, this induces equivalent product diagrams for h and k, and thus $h = k$. Therefore, f' is monic. The same logic shows the same is true for g and g'. ■

The set of all subjects is moreover partially ordered by $\leq$. If $u : s \to a$ and $v : t \to a$ and a has pullbacks, then the pullback gives (as above) another monic $w : p \to a$ such that $w \leq u, v$. Intuitively this would be the *intersection* of the subobjects, the *greatest lower bound* of u and v under p.

Dually, epis $r, s : a \to -$ are equivalent when $r = \theta s$ for some invertible θ. We call equivalence classes of epis the *quotient objects* of a, once again partially ordered by factoring $r \leq s$ if $r = r's$. This definition agrees with the normal definitions of quotients in categories where epis are surjective.

Definition 5.5: Generating Set

A set $S \subseteq C$ in a category C is said to *generate* C when for any parallel morphisms $h, h' : c \to d$ with $h \neq h'$, there exists $s \in S$ and an arrow $f : s \to c$ with $hf \neq h'f$.

Dually, a set $Q \subseteq C$ is a *cogenerating set* when for any pair $h \neq h' : a \to b$, there is some $q \in Q$ and an arrow $g : b \to q$ such that $qh \neq qh'$. A single object is a (co)generator when it's singleton forms a (co)generating set.

Given a functor $G : A \to X$, an arrow $f : x \to G(a)$ is said to *span* a when there is no proper monomorphism $s \to a$ in A for which f factors through $G(s) \to G(a)$.

Lemma 5.3

In a category A, suppose that every set of subobjects of some $a \in A$ has a pullback. Then if $G : X \to A$ preserves these pullbacks, every arrow $h : x \to G(a)$ factors through an arrow $f : x \to G(b)$ which spans b.

Proof. Let $\{u_j : s_j \to a\}_{j \in J}$ be the set of subobjects of a, such that the given h factors as $h = G(u_j) \circ h_j$. Take the pullback $v : b \to a$ of the u_j. Since G preserves pullbacks, it follows that there exists $f : x \to$

$G(b)$, which spans b. ■

5.8 The Special Adjoint Functor Theorem

Theorem 5.13 (Special Initial Object Theorem)

Let D be complete, locally small, and have a cogenerating set Q. Then if for all $d \in D$, all subobjects of d have intersections (pullbacks), then D has an initial object.

Proof. Let $q_0 \coloneqq \prod_{q \in Q} q$ be the product over Q, and take the intersection r of all subobjects of q_0. For any $d \in D$, there is at most one morphism $d \to r$, otherwise their equalizer would give a proper monic to r.

To show r initial, we must construct a unique arrow $r \to d$ for each d. Let H be the set of arrows $h : d \to q \in Q$, and take the small product $\prod_H q$. Let $j : d \to \prod_H q$ be the arrow with components h ($\pi_h \circ j = h$). Since Q cogenerates, j is monic, odl tje rest os witchcraft. ■

Lemma 5.4

Let $G : A \to X$ preserve small limits and pullbacks of families of monics, and $x \in X$. An arrow $h : (f, a) \to (f', a)$ in the comma category $(x \downarrow G)$ is monic if and only if $h : a \to a'$ is monic in A.

Proof. Trivially, if $h : a \to a'$ is monic, then so is $h : f \to f'$. For the converse, consider the pullback of h with h. Note that since h is monic, for any $f, g : z \to x$, if $hf = gh$, then $f = g$. In other words, if the diagram

$$\begin{tikzcd} z \arrow[r, "f"] \arrow[d, "g"'] & a \arrow[d, "h"] \\ a \arrow[r, "h"'] & a' \end{tikzcd}$$

commutes, then we have $f = g$, and thus there is a unique morphism making the diagram

$$\begin{tikzcd} z \arrow[dr, dashed, "f=g"] \arrow[drr, bend left, "f"] \arrow[ddr, bend right, "g"'] & & \\ & a \arrow[r, "1_a"] \arrow[d, "1_a"'] & a \arrow[d, "h"] \\ & a \arrow[r, "h"'] & a' \end{tikzcd}$$

commute. Thus, monic is equivalent to the condition that identities are the pullback of a monic with itself. Recall that by lemma 5.1, the projection $(x \downarrow G) \to A$ creates limits, and in particular creates these pullbacks. Therefore, the projection preserves these pullbacks as well. ■

Theorem 5.14 (The Special Adjoint Functor Theorem)

Let A be complete, locally small, and have a small cogenerating set Q. Let every set of subobjects of any $a \in A$ have a pullback (intersection), and let X be locally small. Then a functor $G : A \to X$ is right-adjoint if and only if G preserves all small limits and all pullbacks of families of monics.

Proof. IF G is right-adjoint, then it indeed preserves limits, and in particular pullbacks of monics. For the converse, it suffices to show once again that for all $x \in X$, there is an initial object in $(x \downarrow G)$. Since X is locally small, the set Q' of all objects $k : x \to G(q)$ with $q \in Q$ is small, and thus cogenerating in $(x \downarrow G)$ fairly obviously: given $s \neq t : (f, a) \to (f', a')$ in $(x \downarrow G)$, there is q_0 and an arrow $h : a \to q_0$ with $hs \neq ht$, and we can regard h as an arrow $(f', a') \to (G(h) \circ f, q_0)$.

Since A complete and G continuous imply $(x \downarrow G)$ is complete by application of lemma 5.1 (projection generates limits), by theorem 5.13 it suffices to construct an intersection for any set of subobjects of any object. Let J index subobjects $h_i : (f_i, a_i) \to (f, a)$ of (f, a). By lemma 5.4, each h_i is also a monic $h_i : a_i \to a$ in A, and by hypothesis they have a pullback $h : b \to a$ in A given by the diagram

$$\begin{tikzcd} & a_i \arrow[dr, "h_i"] & \\ b \arrow[ur, "s_i"] \arrow[rr, "h"'] & & a \end{tikzcd}$$

Since G preserves pullbacks, $G(h) : G(b) \to G(a)$ is a pullback of the family $G(h_i)$ in X. Since each $h_i : f_i \to f$ in $(x \downarrow G)$, we have $G(h_i) \circ f_i = f$ for all $i \in J$, and by universality of pullbacks there is a unique arrow $f_0 : x \to G(b)$ with $G(s_i) \circ f_0 = f_i$:

$$\begin{tikzcd} x \arrow[r, equal] \arrow[d, dashed, "f_0"'] & x \arrow[r, equal] \arrow[d, "f_i"'] & x \arrow[d, "f"] \\ G(b) \arrow[r, "G(s_i)"'] & G(a_i) \arrow[r, "G(h_i)"'] & G(a) \end{tikzcd}$$

Since the projection creates pullbacks, the arrow $h : (f_0, b) \to (f, a)$ is a pullback of the family $\{h_i\}_{i \in J}$, and thus the intersection. ∎

Definition 5.6: Well-Powered

A category A is *well-powered* if the set of subobjects of any $a \in A$ is small; a category is *co-well-powered* when it's set of quotient objects is small.

Corollary 5.4

If A is complete, well-powered, locally small, and has a small cogenerating set, and X is locally small, then a functor $G : A \to X$ is right-adjoint if and only if it is continuous. In particular, any continuous $K : A \to \mathbf{Set}$ is representable.

5.9 Adjoints in Topology

The forgetful functor $U : \mathbf{Top} \to \mathbf{Set}$ has a left-adjoint D, which assigns each set S the discrete topology $\mathcal{P}(S)$. Thus, G preserves all limits (this is why the underlying set of the product space is a cartesian product). I don't feel like copying this down here's two propositions.

Proposition 5.2

If $G : C \to D$ is faithful, D has equalizers, and if for each $x \in C$, $(G \downarrow x) : (C \downarrow x) \to (D \downarrow G(x))$ has a right-adjoint-right-inverse L, then C has equalizers.

Top is well-known to be complete. We can also show more for **Haus**.

Proposition 5.3

Haus, the full subcategory of Hausdorff spaces, is complete and cocomplete. The inclusion **Haus** $\hookrightarrow$ **Top** has a left-adjoint H, and so does the forgetful functor **Haus** $\to$ **Set**.

Chapter 6

Monads and Algebras

6.1 Monads in a Category

Any endofunctor $T : X \to X$ has composites $T^2 \coloneqq T \circ T : X \to X$, and similarly $T^3 = T^2 \circ T$. If $\mu : T^2 \Rightarrow T$ is a natural transformation with components $\mu_x : T^2(x) \to T(x)$, then there is a natural transformation $T\mu : T^3 \Rightarrow T^2$ with components $(T\mu)_x = T(\mu_x) : T^3(x) \to T^2(x)$ by functorality. Similarly, $\mu T : T^3 \Rightarrow T^2$ is also natural with $(\mu T)_x = \mu_{T(x)}$.

Definition 6.1: Monad

A *monad* $T = (T, \eta, \mu)$ in a category X is an endofunctor $T : X \to X$ and two natural transformations

$$\eta : 1_X \Rightarrow T, \qquad \mu : T^2 \Rightarrow T$$

such that the diagrams

$$\begin{tikzcd} T^3 \arrow[r, "T\mu"] \arrow[d, "\mu T"'] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T \end{tikzcd} \qquad \begin{tikzcd} 1_X T \arrow[r, "\eta T"] \arrow[dr, equal] & T^2 \arrow[d, "\mu"] & T 1_X \arrow[l, "T\eta"'] \arrow[dl, equal] \\ & T & \end{tikzcd}$$

commute.

The definition of a monad is basically that of a monoid, but we have replaced sets with functors and $\times$ with $\circ$. We thus call η the *unit* of the monad, and μ the *multiplication* of the monad; the first commutative diagram is then assocaitivity of μ, and the second is the left- and right-unit laws.

Every adjunction gives a monad. Let $(F, G, \eta, \varepsilon)$ be an adjunction from X to A. We have that GF is an endofunctor $X \to X$, the unit a natural transformation $\eta : 1_X \Rightarrow GF$, and the counit $\varepsilon : FG \Rightarrow 1_X$ yields a natural transformation $\mu = G\varepsilon F : GFGF \Rightarrow GF$. We thus have that the associative law becomes

$$\begin{tikzcd} GFGFGF \arrow[r, "GFG\varepsilon F"] \arrow[d, "G\varepsilon FGF"'] & GFGF \arrow[d, "G\varepsilon F"] \\ GFGF \arrow[r, "G\varepsilon F"'] & GF \end{tikzcd}$$

This reduces by dropping the leading G and the trailing F to the diagram

$$\begin{tikzcd} FGFG \arrow[r, "FG\varepsilon"] \arrow[d, "\varepsilon FG"'] & FG \arrow[d, "\varepsilon"] \\ FG \arrow[r, "\varepsilon"'] & 1_A, \end{tikzcd}$$

which commutes by the interchange law for horizontal composition. The unit law becomes

$$G\varepsilon F \circ \eta GF = 1_{GF},$$

and similarly for the other side. These are essentially the triangular identities, so this is indeed a mobnad. We call this the *monad defined by the adjunction* given above. It may be useful to note here that $T\mu = 1_T\mu$.

Dually, a comonad consists of a functor $L : A \to A$ and transformations $\varepsilon : L \Rightarrow 1_A$, $\delta : L \Rightarrow L^2$ satisfying the opposite diagrams. Each adjunction also determines a comonad $(FG, \varepsilon, F\eta G)$ in A.

A morphism of monads $f : (T, \eta, \mu) \to (T', \eta', \mu')$ is a natural transformation $f : T \Rightarrow T'$ that preserves the unity and multiplication, meaning $f \circ \eta = \eta'$, and $f \circ \mu = \mu' \circ ff$, where $\circ$ is vertical composition and juxtaposition is horizontal composition. This then gives the category $\mathbf{Mnd}(X)$ of monads on a category X. This notation may change when I figure out what notation the book will use.

6.2 Algebras for a Monad

Indeed, the converse of the above discussion is true: every monad is induced by a suitable adjunction. There are in fact two known answers; the first constructs from a monad T in X a category X^T of T-algebras, for which an adjunction $X \rightharpoonup X^T$ is then constructed to induce T in X.

Definition 6.2: T-Algebra

Let $T = (T, \eta, \mu)$ be a monad in X. A T-algebra is a pair (x, h) with $x \in X$ and $h : T(x) \to x$, such that the diagrams

$$\begin{tikzcd} T^2(x) \arrow[r, "T(h)"] \arrow[d, "\mu_x"'] & T(x) \arrow[d, "h"] \\ T(x) \arrow[r, "h"'] & x \end{tikzcd} \qquad \begin{tikzcd} x \arrow[r, "\eta_x"] \arrow[dr, "1_x"'] & T(x) \arrow[d, "h"] \\ & x \end{tikzcd}$$

commute. We say x is the underlying object and h is the structure map of the T-algebra. A morphism $f : (x, h) \to (x', h')$ of T-algberas is an arrow $f : x \to x'$ in X such that the diagram

$$\begin{tikzcd} x \arrow[d, "f"'] & T(x) \arrow[l, "h"'] \arrow[d, "T(f)"] \\ x' & T(x') \arrow[l, "h'"] \end{tikzcd}$$

commutes.

Theorem 6.1

Let (T, η, μ) be a monad in X. Then there is an adjunction $(F^T, G^T, \eta^T, \varepsilon^T) : X \rightharpoonup X^T$ given by

$$G : (x, h) \mapsto x, \qquad F : x \mapsto (T(x), \mu_x)$$

$$f : (x, h) \to (x', h') \mapsto f : x \to x', \qquad f : x \to x' \mapsto f : (T(x), \mu_x) \to (T(x'), \mu_{x'});$$

$$\eta^T = \eta, \qquad \varepsilon^T_{(x,h)} = h,$$

where η is once again given as the unit of the monad. Moreover, the monad (T, η, μ) is precisely the monad induced by this adjunction.

Proof. First, we show these maps are indeed functorial. Clearly, G is functorial as it is simply the forgetful functor $X^T \to X$. To show F is functorial, we first show it indeed defines a T-algebra. Taking $x = T(x)$ in definition 6.2 gives the requirement that the diagrams

$$\begin{tikzcd} T^3(x) \arrow[r, "T\mu_x"] \arrow[d, "\mu_x T"'] & T^2(x)) \arrow[d, "\mu_x"] \\ T^2(x)) \arrow[r, "\mu_x"'] & T(x) \end{tikzcd} \qquad \begin{tikzcd} T(x) \arrow[r, "\eta_x T"] \arrow[dr, "1_{T(x)}"'] & T^2(x)) \arrow[d, "\mu_x"] \\ & T(x) \end{tikzcd}$$

commute, which follow from the fact that T is a monad. Functorality then follows by naturality of η and μ.

Next, we show F and G indeed define an adjunction, as described. Note that $G^T F^T(x) = G^T(T(x), \mu_x) = T(x)$, and the same for arrows. Thus, $\eta : 1_X \Rightarrow G^T F^T$ is clearly natural. Conversely, $F^T G^T(x, h) = F^T(x) = (T(x), \mu_x)$. Commutivity of the first square in definition 6.2 gives

$$\begin{tikzcd} T(T(x)) \arrow[r, "\mu_x"] \arrow[d, "T(h)"'] & T(x) \arrow[d, "h"] \\ T(x) \arrow[r, "h"'] & x \end{tikzcd}$$

and thus h is a morphism of T-algebras $(T(x), \mu_x) \to (x, h)$. We thus have a natural transformation

$$\varepsilon^T_{(x,h)} = h : F^T G^T(x, h) \to (x, h),$$

which is natural by definition of a morphism of T-algebras. It at this point suffices to prove the triangle identities

$$G^T \varepsilon^T \circ \eta^T G^T = 1_{G^T}, \qquad \varepsilon^T F^T \circ F^T \eta^T = 1_{F^T}.$$

Let (x, h) be a T-algebra. We compute

$$(G^T \varepsilon^T \circ \eta^T G^T)(x, h) = G^T(\varepsilon^T_{(x,h)}) \circ \eta^T_{G^T(x,h)} = G^T(h) \circ \eta^T_x = h \circ \eta_x.$$

We also have

$$(\varepsilon^T F^T \circ F^T \eta^T)(x) = \varepsilon^T_{F(x)} \circ F^T(\eta^T_x) = \varepsilon^T_{(T(x),\mu_x)} \circ F(\eta_x) = \mu_x \circ T(\eta_x).$$

Diagramattically, this amounts to commutivity of the diagrams

$$\begin{tikzcd} x \arrow[r, "\eta_x"] \arrow[dr, equal] & T(x) \arrow[d, "h"] \\ & x \end{tikzcd} \qquad \begin{tikzcd} T^2(x) \arrow[d, "\mu_x"'] & T(x) \arrow[l, "T\eta_x"'] \arrow[dl, equal] \\ T(x) & \end{tikzcd}$$

The one on the left commutes by definition of a T-algebra (commutative triangle in dfn 6.2), and commutivity of the triangle on the right amounts to the right-unit law in definition 6.1. Thus, the triangle inequalities are satisfied, and $(F^T, G^T, \eta^T, \varepsilon^T)$ is indeed an adjunction.

Finally, we show the induced monad is the monad (T, η, μ). Recall the induced monad takes the form $(G^T F^T, \eta^T, G^T \varepsilon^T F^T)$. By definition, $\eta^T = \eta$, and moreover

$$G^T F^T(x) = G^T(T(x), \mu_x) = T(x), \qquad G^T F^T(f) = G^T(T(f)) = T(f).$$

Thus, $G^T F^T = T$. Finally, note that

$$\left(G^T \varepsilon^T F^T\right)_x = G^T(\varepsilon^T_{F^T(x)}) = G^T(\varepsilon^T_{(T(x),\mu_x)}) = G^T(\mu_x) = \mu_x.$$

Therefore, $G^T \varepsilon^T F^T = \mu$. ■

Here's an example. Given a preorder P, a monad is a closure operation $T : P \to P$ whereby $x \leq T(x)$ for all $x \in P$. A T-algebra is $x \in P$ with a structure map $T(x) \to x$, which must be given by $T(x) \leq x$. Thus, T-algebras in a preorder are simply elements with $x \leq T(x) \leq x$; if $\leq$ is a partial order, we have $T(x) = x$.

Let $G \in \mathbf{Grp}$. Then if X is any set, there is a monad in $\mathbf{Set}$ given by

$$T(X) = G \times X,$$

$$\eta_x : X \to G \times X, \qquad x \mapsto (e_G, x).$$

$$\mu_x : G \times G \times X \to G \times X, \qquad (g_1, (g_2, x)) \mapsto (g_1 g_2, x).$$

A T-algebra then becomes a set X together with a structure map $h : G \times X \to X$ satisfying

$$h(\eta_x(x)) = h(e_G, x) = x, \qquad h(g_1 g_2, x) = h(g_1, h(g_2, x)).$$

Write $g \cdot x$ for $h(g, x)$. These then become

$$e_G \cdot x = x, \qquad (g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x).$$

In other words, h is a group action $h : G \to \mathrm{Aut}_{\mathbf{Set}}(X)$.

Finally, take $T = R \otimes_{\mathbb{Z}} - : \mathbf{Ab} \to \mathbf{Ab}$ for some fixed ring R. Define the unit $A \to T(A) = R \otimes A$ by $a \mapsto 1_R \otimes a$, and define multiplication $R \otimes (R \otimes A) \to R \otimes A$ by $r_1 \otimes (r_2 \otimes a) \mapsto r_1 r_2 \otimes a$. T-algebras then give the standard definition of a left R-module.

6.3 The Comparison with Algebras

Given an adjunction $X \rightharpoonup A$, we can now construct a monad T in X, and the corresponding T-algebras X^T. We can also use this fact to learn about adjunctions, and the category A.

Theorem 6.2 (Comparison of Adjoints with Algebras)

Let $(F, G, \eta, \varepsilon)$ be an adjunction from X to A, and let $T = (GF, \eta, G\varepsilon F)$ be the induced monad. Then there is a unique functor $K : A \to X^T$ with

$$G^T K = G, \qquad KF = F^T,$$

with notation as in theorem 6.1.

Proof. The conclusion amounts to the commutivity of the diagrams

$$\begin{tikzcd} X \arrow[r, dashed, "K"] \arrow[dr, "G"'] & X^T \arrow[d, "G^T"] \\ & A \end{tikzcd} \qquad \begin{tikzcd} X^T & A \arrow[l, dashed, "K"'] \\ X \arrow[u, "F^T"] \arrow[ur, "F"'] & \end{tikzcd}$$

The claim is that $K(a) = (G(a), G\varepsilon_a)$ and $K(f) = G(f)$. Verifying this first requires the verification that $(G(a), G\varepsilon_a)$ is indeed a T-algebra. Note that $G\varepsilon_a : GFG(a) \to G(a)$ since $\varepsilon : FG \Rightarrow 1_A$, and note that $T = GF$. We thus require that the diagrams

$$\begin{tikzcd} GFGFG(a) \arrow[r, "GFG\varepsilon_a"] \arrow[d, "G\varepsilon_a F"'] & GFG(a) \arrow[d, "G\varepsilon_a"] \\ GFG(a) \arrow[r, "G\varepsilon_a"'] & G(a) \end{tikzcd} \qquad \begin{tikzcd} G(a) \arrow[r, "\eta G(a)"] \arrow[dr, "1_{G(a)}"'] & GFG(a) \arrow[d, "G\varepsilon_a"] \\ & G(a) \end{tikzcd}$$

commute. The diagram on the right is simply the triangular identity

$$\eta G \circ G\varepsilon = 1_G,$$

and the diagram on the left is easily shown to be $G\varepsilon\varepsilon$ on both sides.

Since ε is natural, $K(f) = G(f)$ commutes with $G\varepsilon$, and thus is a morphism of T-algebras. Verifying the equalities and functorality amount to simple enough computations, do we focus on showing uniqueness.

The requirement that $G^T K = G$ forces the underlying object to be $G(a)$ for some a. This then immediately gives the definition $K(f) = G(f)$, and thus it remains only to show the structure map is forced. Since we now have two adjunctions with the same unit $(F, G, \eta, \varepsilon)$ and $(F^T, G^T, \eta^T, \varepsilon^T)$, K along with the identity functor 1_X define a map of adjunctions. It follows that $K\varepsilon = \varepsilon^T K$, and thus $K\varepsilon = G\varepsilon$ on arrows. We have by definition of ε^T that

$$\varepsilon^T K(a) = \varepsilon^T (G(a), h) = h.$$

Thus $G\varepsilon_a = \varepsilon^T K(a) = h$ implies that $h = G(\varepsilon_a)$, concluding the proof. ■

6.4 Words and Free Semigroups

For some reason, there's an entire section dedicated to investigating a special case of the comparison functor on semigroups. A semigroup is a set S with an associative binary operation $\cdot$. The free semigroup $W(X)$ on X is the set of all words on X of *positive integer length* (meaning no empty word), with multiplication given by juxtaposition. There is thus a free functor $F(X) = W(X)$, where

$$W(X) \coloneqq \coprod_{n=1}^{\infty} X^n.$$

If $G : \mathbf{Smg} \to \mathbf{Set}$ is the forgetful functor, then $\eta_X(x) = x$ and $\varepsilon_S(s_1, \ldots, s_n) = s_1 \cdot \ldots \cdot s_n$ define an adjunction $(F, G, \eta, \varepsilon)$.

The monad induced here is given by $W = (W, \eta, \mu)$ where

$$W(X) = \coprod_{n=1}^{\infty} X^n,$$

and μ maps any n-tuple of strings to their ordered concatenation. W-algebras then have the form $(S, \nu_1, \ldots)$; that is, sets S with a n-ary operation $\nu_n : S^n \to S$ for each $n \in \mathbb{Z}^+$. The interesting part, however, is that the n-ary operations must take the form

$$\nu_1 = 1_S, \qquad \nu_2 : S^2 \to S, \qquad \nu_{n+1} = \nu_2(\nu_n \times 1_S) : S^{n+1} \to S$$

with $n \geq 2$, where ν_2 is an associative binary operation. This clearly gives that the W-algebras are precisely the same semigroups, but with iterated multiplication ν_2 viewed as an operation ν_n. Similar constructions apply to more familiar monads.

6.5 Free Algberas for a Monad

Let $(F, G, \varphi) : X \rightharpoonup A$ be an adjunction, with $B \subseteq A$ a full subcategory such that $F(X) \subseteq B$. Then there is another adjunction $(F_B, G_B, \varphi_B) : X \rightharpoonup B$ by restricting the codomain of F and the domain of G, without changing φ beyond the obvious. This is seen by the fact that for any $x \in X$ and $b \in B$,

$$B(F_B(x), b) = A(F(x), b) \cong X(x, G(b)) = X(x, G_B(b)).$$

In particular, the induced monad by these adjunctions are *precisely the same*. This means a monad can be induced by many adjunctions, with the "smallest" given by $F(X) = B$. Using our knowledge of free objects, we would expect that we can also do this construction in reverse; for any monad we should be able to construct $F(X)$ directly.

Theorem 6.3 (Kleisi Category for a Monad)

Let (T, η, μ) be a monad in X. For each $x \in X$, define the object x_T, and for each $f : x \to T(y)$, define the arrow $f_T : x_T \to y_T$. Then these objects and morphisms constitute a category X_T with composition $f_T : x_T \to y_T$ and $g_T : y_T \to z_T$ given by

$$g_T \circ f_T = (\mu_z \circ T(g) \circ f)_T,$$

and there exist functors $F_T : X \to X_T$ and $G_T : X_T \to X$ given by

$$F_T(f : x \to y) = (\eta_y \circ k)_T : x_T \to y_T,$$

$$G_T(f_T : x_T \to y_T) = \mu_y \circ T(f) : T(x) \to T^2(y) \to T(y).$$

Then $\varphi_T : f \mapsto f_T$ gives the adjunction $(F_T, G_T, \varphi_T) : X \rightharpoonup X_T$, and the induced monad is precisely T.

Not proving this bullshit. Nor am I proving the next theorem, and the third follows by some details in the proof of the second. How did Kleisi come up with this wtf

Theorem 6.4 (Kleisi Comparison Theorem)

Let $(F, G, \eta, \varepsilon) : X \rightharpoonup A$ be an adjunction and T the induced monad in X. Then there is a unique functor $L : X_T \to A$ with $GL = G_T$ and $LF_T = F$.

Theorem 6.5

Given a monad (T, η, μ) in X, the category of adjunctions (F, G, η, ν) which induce the monad T, whereby morphsisms are maps of adjunctions that are the identity on X, has the Kleisi construction on X^T as an initial object and $(F^T, G^T, \eta, \varepsilon^T)$ a terminal object, where

$$X_T \overset{L}{\dashrightarrow} A \overset{K}{\dashrightarrow} X^T$$

for any A.

6.6 Split Coequalizers

Definition 6.3: Fork/Absolute Coequalizer

Let C be a category. A *fork* in C is a commutative diagram

$$\begin{tikzcd} a \arrow[r, shift left, "\partial_0"] \arrow[r, shift right, "\partial_1"'] & b \arrow[r, "e"] & c. \end{tikzcd}$$

Equivalently, a fork is a cone from $a \rightrightarrows b$ to c, with a coequalizer an initial fork. An *absolute coequalizer* is a coequalizer e which is preserved by all functors, meaning for any $F : C \to X$, the diagram

$$\begin{tikzcd} F(a) \arrow[r, shift left, "F(\partial_0)"] \arrow[r, shift right, "F(\partial_1)"'] & F(b) \arrow[r, "F(e)"] & F(c) \end{tikzcd}$$

is still a coequalizer.

Definition 6.4: Split Fork

A *split fork* in a category C is a fork

$$\begin{tikzcd} a \arrow[r, shift left, "\partial_0"] \arrow[r, shift right, "\partial_1"'] & b \arrow[r, "e"] & c \end{tikzcd}$$

together with two additional arrows

$$c \xrightarrow{s} b \xrightarrow{t} a$$

such that

$$e \circ s = 1_c, \qquad \partial_0 \circ t = 1_b, \qquad \partial_1 t = s \circ e.$$

We say that s and t *split* the fork.

These conditions on a split fork give us that e is a split epi, and s a right-inverse of e. A split fork is equivalently a commutative diagram

$$\begin{tikzcd}
b \arrow[r, "t"] \arrow[rr, bend left, "1_b"] \arrow[d, "e"'] & a \arrow[r, "\partial_0"] \arrow[d, "\partial_1"'] & b \arrow[d, "e"] \\
c \arrow[r, "s"'] \arrow[rr, bend right, "1_c"'] & b \arrow[r, "e"'] & c.
\end{tikzcd}$$

Yet another characterization is taking ∂_1 and e as arrows in the functor category $C^{\mathbf{2}}$, and $(\partial_0, e) : \partial_1 \to e$ a morphism for which there exists a right-inverse $(t, s) \circ (\partial_0, e) = (1, 1)$.

Lemma 6.1

In any split fork (with notation as above), e is the coequalizer of ∂_0 and ∂_1.

Proof. Let $f : b \to d$ equalize the parallel arrows. Then we have

$$fse = f\partial_1 t = f\partial_0 t = f1 = f.$$

Thus, f factors through e via fs. To show uniqueness, take $f = ke$ for $k : c \to d$, and thus $fs = kes = k$. ■

By *split coequalizer*, we now generally refer to the arrow e of the split fork. Since a split fork is defined by equations involving only composites and identities, they are preserved by functors.

Corollary 6.1

Every split fork e is an absolute coequalizer.

Here's a fun example in **Cat**. Let C be a category, and let $\partial_0 = \mathrm{dom}$, $\partial_1 = \mathrm{cod}$. Let $e : C \to \mathbf{1}$. If C has a terminal object a_0, then $s : \mathbf{1} \to C$ given by $0 \mapsto a_0$ is a right-inverse for e. This fork is then split by s and the functor t which maps $c \in C$ to the unique arrow $c \to a_0$.

6.7 Beck's Theorem

Beck's theorem characterizes the categories of T-algebras for which the right-adjunct (forgetful functor) creates suitable coequalizers. Recall that a functor $G : A \to X$ *creates* coequalizers when coequalizers under G in X uniquely induce the corresonding coequalizer in A.

Theorem 6.6 (Beck)

Let $(F, G, \eta, \varepsilon) : X \rightharpoonup A$ be an adjunction, (T, η, μ) the induced monad, X^T the category of T-algebras, and $(F^T, G^T, \eta^T, \varepsilon^T)$ the corresponding adjunction. Then the following are equivalent.

1. The comparison functor $K : A \to X^T$ is an isomorphism;
2. The functor $G : A \to X$ creates coequalizers for all parallel pairs f, g in A for which $G(f), G(a)$ have an absolute coequalizer in X;
3. The functor $G : A \to X$ creates coequalizers for all parallel pairs f, g in A for which $G(f), G(a)$ have a split coequalizer in X.

Proof. By corollary 6.1, (2) implies (3). It suffices to show ($1 \implies 2$) and ($3 \implies 1$).

First, we show ($1 \implies 2$). Let $d_0, d_1 : (x, h) \to (y, k)$ be parallel morphisms for which in X there exists an absolute coequalizer

$$x \underset{d_1}{\overset{d_0}{\rightrightarrows}} y \xrightarrow{e} z.$$

Since $K : A \cong X^T$, it suffices to show there exists a suitable T-algbera structure for z in X^T for which e forms a coequalizer for d_0, d_1, and that this structure is indeed unqiue in the relevant sense. Let $m : T(z) \to z$ be the structure map, and consider the diagram

$$\begin{tikzcd}
T(x) \arrow[r, shift left, "T(d_0)"] \arrow[r, shift right, "T(d_1)"'] \arrow[d, "h"'] & T(y) \arrow[r, "T(e)"] \arrow[d, "k"'] & T(z) \arrow[d, dashed, "m"] \\
x \arrow[r, shift left, "d_0"] \arrow[r, shift right, "d_1"'] & y \arrow[r, "e"'] & z.
\end{tikzcd}$$

Both the d_0 and the d_1 squares on the left commute since d_0 and d_1 are morphisms of T-algebras (definition 6.2), both parallel pairs ar equalized by functorality of T and the fact that e is a coequalizer, and finally since e is an absolute coequalizer, $T(e)$ is also a coequalizer (still in X), and thus there is a unique induced map m making the diagram commute in X. Take this map to be the structure map, which is shown to induce a T-algebra structure by a short diagram chase. This T-algebra structure is indeed unique in the required sense, so we need only show e is an equalizer in X^T. Let $f : (y, k) \to (w, n)$ be any other morphism of T-algebras that equalizes d_0 and d_1. Then $f : y \to w$ in X also equalizes the maps, and thus factors uniquely through the coequalizer e. Hence, e is also a coequalizer in X^T (pending a short verification that the factorization map is a T-algebra morphism).

Now we show ($3 \implies 1$). The conditions that (x, h) be a T-algebra are precisely that

$$T^2(x) \underset{\mu_x}{\overset{T(h)}{\rightrightarrows}} T(x) \xrightarrow{h} x$$

be a fork in X split by

$$x \xrightarrow{\eta_x} T(x) \xrightarrow{\eta_{T(x)}} T^2(x).$$

This is seen immediately by looking at the diagrams in definition 6.2. Additionally, for all $a \in A$, the adjunction $(F, G, \eta, \varepsilon)$ gives a fork

$$FGFG \underset{FG(\varepsilon_a)}{\overset{\varepsilon_{FG(a)}}{\rightrightarrows}} FG(a) \xrightarrow{\varepsilon_a} a$$

by naturality (simple proof), which we call the *canonical presentation of* a. By functorality, applying G to the fork gives a split fork.

Take any other adjunction $(F', G', \eta', \varepsilon') : X \rightharpoonup A$ which induces the same monad, and consider a comparison $M : A' \to A$ which satisfies

$$MF' = F, \qquad GF = G'.$$

Such a comparison is a morphism of adjunctions and thus satisfies $M\varepsilon' = \varepsilon M$. Applying M to the canonical presentation of a' gives

$$MF'G'F'G'(a') \underset{MF'G'(\varepsilon'_a)}{\overset{M\varepsilon'_{F'G'(a')}}{\rightrightarrows}} MF'G'(a') \xrightarrow{M(\varepsilon'_a)} M(a').$$

We have $FGM = MF'G'$ and $MF' = F$. Applying this gives

$$MF'G'F'G' = FGMF'G' = FGFG'.$$

Since $M\varepsilon' = \varepsilon M$, applying this gives

$$M\varepsilon'_{F'G'(a')} = \varepsilon_{MF'G'(a')} = \varepsilon_{FG'(a')}.$$

We also have

$$MF'G'(\varepsilon'_a) = FG'(\varepsilon'_a).$$

Finally, $MF'G'(a') = FG'(a')$, giving the diagram

$$FGFG'(a') \underset{FG'(\varepsilon'_a)}{\overset{\varepsilon_{FG'(a')}}{\rightrightarrows}} FG'(a') \xrightarrow{M(\varepsilon'_a)} M(a').$$

Here, the final arrow satisfies $M(\varepsilon'_a) = \varepsilon_{M(a)}$. Functorality by G gives the fork

$$GFGFG'(a') \underset{GFG'(\varepsilon'_a)}{\overset{G(\varepsilon_{FG'(a')})}{\rightrightarrows}} GFG'(a') \xrightarrow{GM(\varepsilon'_a)} GM(a').$$

Once again applying the fact that $GM = G'$ and $GF = T$, along with some rewriting, gives

$$T^2G'(a') \underset{TG'(\varepsilon'_a)}{\overset{(G\varepsilon F)_{G'(a')}}{\rightrightarrows}} TG'(a') \xrightarrow{G'(\varepsilon'_a)} G'(a'),$$

a fork in X. Recall that forks in X of the form

$$T^2(x) \underset{\mu_x}{\overset{T(h)}{\rightrightarrows}} T(x) \xrightarrow{h} x$$

split via η_x and $T\eta_x$. Note that this fork is a special case of the split fork via the assignments $x = G'(a')$ and $h = G'(\varepsilon'_a)$. Since it splits, it is a coequalizer, and by hypothesis G creates coequalizers in this case. Thus, $M(\varepsilon_a)$ and $M(a')$ are uniquely determined, and thus M is unique if it exists. An existence proof follows immediately by universality of the coequalizers created by G. This argument gives both comparisons $X^T \to A$ and $A \to X^T$, which are inverses. ■

Definition 6.5: Reflection

A functor $G : A \to X$ reflects colimits of $T : J \to A$ when every cone $\lambda : T \Rightarrow \Delta(a)$ for which $G(\lambda) : GT \Rightarrow G\Delta(a)$ is colimiting in X is then colimiting in A.

Similarly, we may define reflection of limits, coequalizers, etc.

6.8 Algebras are T-Algebras

For semigroups, rings, monoids, we already know that the the comparison functor is an isomorphism. This result is in fact exceedingly general, and we generalize it here.

Theorem 6.7 (Algebras are T-algebras)

Let Ω be a set of operators and E a set of identities on Ω. Let $G : (\Omega, E)$-**Alg** $\to$ **Set** the forgetful functor, and T the resulting monad in **Set**. Then $K : (\Omega, E)$-**Alg** $\to \mathbf{Set}^T$ is an isomorphism.

Proof. Let $f, g : A \to B$ morphisms of (Ω, E)-algebras for which the underlying functions have absolute coequalizer e. Let $\omega \in \Omega$ and n-ary operator with actions ω_A, ω_B on A, B, respectively. The diagram

$$\begin{tikzcd} A^n \arrow[r, shift left, "f^n"] \arrow[r, shift right, "g^n"'] \arrow[d, "\omega_A"'] & B^n \arrow[r, "e^n"] \arrow[d, "\omega_B"] & X^n \arrow[d, dashed, "\omega_X"] \\ A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \arrow[r, "e"'] & X \end{tikzcd}$$

commute since f, g are (Ω, E)-algebra morphisms, and since e is an absolute coequalizer, meaning e^n is also a coequalizer, inducing a map ω_X. It then follows by commutivity of the diagram that e is a morphism of (Ω, E)-algebras.

Now we show e is a coequalizer of (Ω, E)-algebras. Let $h : B \to C$ any morphism of (Ω, E)-algebras which coequalizes f and g. Functorality of G gives $hf = hg$ in **Set**, so h factors uniquely as $h = h'e$ by universality. We want to show the diagram

$$\begin{tikzcd} X^n \arrow[r, "h'^n"] \arrow[d, "\omega_X"'] & C^n \arrow[d, "\omega_C"] \\ X \arrow[r, "h'"'] & C \end{tikzcd}$$

commutes. Indeed,

$$h'\omega_X e^n = h'e\omega_B = h\omega_B = \omega_C h^n = \omega_C h'^n e^n$$

since h is a morphism of (Ω, E)-algebras, and by commutivity of a previous diagram. Since e^n is a coequalizer, it is epi, and we have $h'\omega_X = \omega_C h'^n$. Thus, G creates coequalizers for absolute coequalizers, and by Beck's theorem, the comparison functor is an isomorphism. ■

6.9 Compact Hausdorff Spaces

Theorem 6.8

The standard forgetful functor G : **Cmpt Haus** $\to$ **Set** is monadic, meaning it has a left-adjoint, and the comparison functor **Cmpt Haus** $\to \mathbf{Set}^T$ is an isomorphism, for T the monad induced by the adjunction.

Chapter 7

Monoids

7.1 Monoidal Categories

A monoidal category is a "good place to define monoids in". It's a category that has some notion of multiplication, which the book denotes as $\square$, and I will denote $\otimes$. This multiplication should be a bifunctor $\otimes : B \times B \to B$, and should have an identity e. A *strict* monoidal category is where $\otimes$ is strictly associative and e is strictly unital, but this is often too general and we prefer to only require unity and associativity up to isomorphism.

Definition 7.1: Monoidal Category

A monoidal category $B = (B, \otimes, e, \alpha, \lambda, \rho)$ is a category B, a bifunctor $\otimes : B \times B \to B$, an object $e \in B$, and three natural isomorphisms

$$\alpha : \otimes \circ (1_B \times \otimes) \cong \otimes \circ (\otimes \times 1_B), \qquad \alpha_{a,b,c} : a \otimes (b \otimes c) \cong (a \otimes b) \otimes c;$$

$$\lambda : e \otimes - \cong 1_B, \qquad \lambda_a : e \otimes a \cong a;$$

$$\rho : - \otimes e \cong 1_B, \qquad \rho_a : a \otimes e \cong a;$$

such that for all $a, b, c, d \in B$, the pentagon diagram

$$\begin{tikzcd}
a \otimes (b \otimes (c \otimes d)) \arrow[r, "\alpha"] \arrow[d, "1\otimes\alpha"'] & (a \otimes b) \otimes (c \otimes d) \arrow[r, "\alpha"] & ((a \otimes b) \otimes c) \otimes d \arrow[d, "\alpha\otimes 1"] \\
a \otimes ((b \otimes c) \otimes d) \arrow[rr, "\alpha"'] & & (a \otimes (b \otimes c)) \otimes d
\end{tikzcd}$$

and the triangle diagram

$$\begin{tikzcd}
a \otimes (e \otimes c) \arrow[rr, "\alpha"] \arrow[dr, "1\otimes\lambda"'] & & (a \otimes e) \otimes c \arrow[dl, "\rho\otimes 1"] \\
& a \otimes c &
\end{tikzcd}$$

commute. Additionally,

$$\lambda_e = \rho_e : e \otimes e \cong e.$$

We call the commutivity of these diagrams "coherence conditions", and we will see in the next section that they imply all diagrams of these types commute, as we explore coherence more precisely. One remark to make is that since $\otimes$ is a bifunctor, whenever $(ff' \otimes gg')$ is defined, then there is an equality

$$(ff' \otimes gg') = (f \otimes g)(f' \otimes g').$$

A category C with finite products forms a monoidal category by taking $\otimes$ to be the product bifunctor $\times : C \times C \to C$, taking the unit e to be the terminal object t, and taking the natural isomorphisms to be the obvious isomorphisms. When a category with finite products is viewed as monoidal under $\times$, we call it an *cartesian monoidal category*, or simply a *cartesian category*.

Tensor products (as is likely obvious given the choice of notation) also form monoidal categories; very clearly R is a unit for $\otimes_R$, with natural isomorphisms very well-known, so we also have $(R\text{-}\mathsf{Mod}, \otimes_R, R, \alpha, \lambda, \rho)$. The same holds for graded R-modules, R-algebras, (R, R)-bimodules, and more.

Definition 7.2: Strict Morphisms

A *strict morphism* of monoidal categories $T : (B, \otimes, e, \alpha, \lambda, \rho) \to (B', \otimes', e', \alpha', \lambda', \rho')$ is a functor $T : B \to B'$ which preserves $\otimes$, α, λ, and ρ: for any objects $a, b, c \in B$ and morphisms $f, g \in \mathrm{Mor}(B)$,

$$T(a \otimes b) = T(a) \otimes' T(b), \qquad T(f \otimes g) = T(f) \otimes' T(g),$$

$$T\alpha = \alpha' T, \qquad T\lambda = \lambda' T, \qquad T\rho = \rho' T, \qquad T(e) = e'.$$

There is then a category **Moncat** of monoidal categories, with morphisms given by strict morphisms of monoidal categories.

We call (strict) morphisms of monoidal categories "(strict) monoidal functors". Often we will want to relax the definition and require only that there exist a specified natural isomorphism giving $T(a \otimes b) \cong T(a) \otimes' T(b)$, and sometimes we will want to relax this even further. Perhaps the book will investigate this later; otherwise I will investigate later.

7.2 Coherence

A coherence theorem states that all diagrams of a given class commute. Currently, we're concerned with commutivity of diagrams involving α, λ, ρ. We can prove that every "formal" diagram commutes. A formal diagram consists of "binary words", which are $\otimes$-composites which are defined recursively. A binary word of length 0 is the empty word e_0, a binary word of length 1 is a variable placeholder $(-)$, and given binary words v, w of length m, n, respectively, the word $v \otimes w$ has length $m + n$. As an example, $((- \otimes -) \otimes e_0) \otimes -$ is a binary word of length 3, since there are three $(-)$s each having length 1, and one e_0 having length 0.

For each pair of binary words v, w with the same length, define an arrow $v \to w$. We then define the category W of binary words, which is a preorder and a groupoid. Multiplication $(v, w) \mapsto v \otimes w$ makes W a monoidal category, with e_0 as the unit, and α, λ, ρ uniquely determined. By construction, every diagram in W commutes, and thus morphisms $W \to B$ with B monoidal then give the desired commutitive diagrams in B. These morphisms are given by the following theorem, which asserts that W is the free monoidal category on one generator.

Theorem 7.1

For any monoidal category B and any $b \in B$, there is a unique morphism $W \to B$ such that $(-) \mapsto b$.

Proof. Let T be the monoidal functor in question. Definition 7.2 forces $T(e_0) = e$, and otherwise we define the action on elements by substituting b for $(-)$. Given a word v in W, we write v_b to indicate it's image under T. We also set $(v \otimes w)_b = v_b \otimes w_b$, and define recursively from here. These definitions are uniquely determined by b.

To show this recipe preserves the associativity arrows, we construct a graph G_n. Take the vertices of G_n to be all words of length n that do not include e_n, and edges are "basic" arrows. These basic arrows are defined as any arrows that correspond with α in B, all identity arrows, and any tensor of already basic arrows with identity arrows. Intuitively, basic arrows are tensors over any chain of arrows, all of which are identities except exactly one, which must be either α or α^{-1}. Paths $v \to w$ in G_n are thus composable arrows $v_b \to w_b$ of basic arrows in B. Write "directed" for arrows involving α, and "antidirected" for arrows involving α^{-1}. Idk the proof is weird read the book if ur interested. ■

Corollary 7.1

Let B be a monoidal category. Then there is a function can_B which maps pairs of words (v, w) of the same length n to a natural isomorphism $v_B \Rightarrow w_B : B^n \to B$, called the *canonical map* from v_B to w_B. These maps are in such a way that the identity arrow 1_e is canonical, and so are α, λ, ρ, and their inverses.

7.3 Monoids

Definition 7.3: Monoid

Let $(B, \otimes, e, \alpha, \lambda, \rho)$ be a monoidal category. A monoid c in B is a triple (c, μ, η) with $c \in B$, and morphisms

$$\mu : c \times c \to c, \qquad \eta : e \to c$$

such that the diagrams

$$\begin{tikzcd}
c \otimes (c \otimes c) \arrow[r, "\alpha"] \arrow[d, "1_c \otimes \mu"'] & (c \otimes c) \otimes c \arrow[r, "\mu \otimes 1_c"] & c \times c \arrow[d, "\mu"] \\
c \otimes c \arrow[rr, "\mu"'] & & c
\end{tikzcd}$$

$$\begin{tikzcd}
e \otimes c \arrow[r, "\eta \otimes 1_c"] \arrow[dr, "\lambda_c"'] & c \otimes c \arrow[d, "\mu"] & c \otimes e \arrow[l, "1_c \otimes \eta"'] \arrow[dl, "\rho_c"] \\
& c &
\end{tikzcd}$$

commute. A morphism $f : (c, \mu, \eta) \to (c', \mu', \eta')$ is a morphism $f : c \to c'$ in B such that $f\eta = \eta'$, and

$$\begin{tikzcd}
c \otimes c \arrow[r, "\mu"] \arrow[d, "f \otimes f"'] & c \arrow[d, "f"] \\
c' \otimes c' \arrow[r, "\mu'"'] & c'
\end{tikzcd}$$

commutes.

It should at this point in the book be clear what this definition entails. There is a category $\mathbf{Mon}_B$ of monoids in a monoidal category B. Here are some examples.

Monoidal Category	Monoids Therein
$(\mathbf{Set}, \times, 1)$	regular monoids
$(\mathbf{Top}, \times, *)$	topological monoids
$(C^C, \circ, 1_C)$	monads
$(\mathbf{Ab}, \otimes, \mathbb{Z})$	rings
$(R\text{-}\mathsf{Mod}, \otimes_R, R)$	R-algebras
graded modules	graded algebras
$(B^{\mathrm{op}}, \otimes^{\mathrm{op}}, e)$	comonoids in B
$(R\text{-}\mathsf{Mod}^{\mathrm{op}}, \otimes_R^{\mathrm{op}}, R)$	R-coalgebras
$(\mathbf{Cat}, \times, \mathbf{1})$	strict monoidal categories
$(O\text{-}\mathbf{Grph}, \times_O, O \to O)$	Categories

Proposition 7.1 (General Associative Law)

For (C, μ, η) a monoid in B, the iterated products μ_v and μ_w for words v and w of the same length satisfy

$$\mu_w \circ \mathrm{can}_c(v, w) = \mu_v : v_c \to c.$$

Proof. induct on the axioms ■

Theorem 7.2 (Free Monoids)

If a monoidal category B has countable coproducts, and if for all $a \in B$, the functors $- \otimes a$ and $a \otimes -$ preserve coproducts, then the forgetful functor $U : \mathbf{Mon}_B \to B$ has a left-adjoint.

Proof. Preservation of coproducts means that

$$\coprod_n (a \otimes b_n) \cong a \otimes \coprod_n b_n$$

for any countable $\{b_n\}_{n \in \mathbb{N}} \subseteq B$. Construct the set $\{b_n\}$ by

$$b_n \coloneqq a^n \coloneqq \bigotimes_{i=1}^{n} a.$$

Note that there is an injection $j_{m,n}$

$$a^m \otimes a^n \to \coprod_{m,n} (a^m \otimes a^n).$$

Since for any m, n, the canonical map gives an isomorphism $\mathrm{can} : a^m \otimes a^n \to a^{m+n}$, which injects

$$i_{n+m} : a^{m+n} \to \coprod_k a^n k,$$

the diagram

$$\begin{tikzcd}
\coprod_{m,n}(a^m \otimes a^n) \arrow[d, dashed, "\varphi"'] & a^m \otimes a^n \arrow[l, hook', "j_{m,n}"'] \arrow[d, "\text{can}"] \\
\coprod_k a^k & a^{m+n} \arrow[l, hook', "i_{m+n}"]
\end{tikzcd}$$

commutes, with the dashed arrow via universality of coproduct. We also have by left- and right-preservation of coproduct that there is an isomorphism

$$\theta : \left(\coprod_n a^n\right) \otimes \left(\coprod_m a^m\right) \to \coprod_{n,m}(a^n \otimes a^m).$$

This gives the diagram

$$\begin{tikzcd}
\left(\coprod_m a^m\right) \otimes \left(\coprod_n a^n\right) \arrow[r] \arrow[dr, dashed, "\mu"'] & \coprod_{m,n}(a^m \otimes a^n) \arrow[d, dashed, "\varphi"'] & a^m \otimes a^n \arrow[l, hook', "j_{m,n}"'] \arrow[d, "\text{can}"] \\
& \coprod_k a^k & a^{m+n} \arrow[l, hook', "i_{m+n}"]
\end{tikzcd}$$

where μ is simply the composition of φ with the associativity isomorphism. An *annoying* diagram chase that I am NOT typing up shows μ satisfies the associativity axiom of definition 7.3, and defining $\eta_a : e \to \coprod_n a^n$ is simply given by $\eta_a = i_0 : a^0 e \to \coprod_n a^n$. These definitions give a monoid, and $i_1 : a \to U(\coprod_n a^n, \mu, \eta)$ is universal from a to U. By pointwise construction of an adjoint, we are done. ■

7.4 Actions

Definition 7.4: Action

A (left) action of a monoid (c, μ, η) on an object a in the same monoidal category B is an arrow $\nu : c \otimes a \to a$ such that the diagram

$$\begin{tikzcd}
c \otimes (c \otimes a) \arrow[r, "\alpha"] \arrow[d, "1_c \otimes \nu"'] & (c \otimes c) \otimes a \arrow[r, "\mu \otimes 1_a"] & c \otimes a \arrow[d, "\nu"] & e \otimes a \arrow[l, hook', "\eta \otimes 1_a"'] \arrow[dl, "\lambda_a"] \\
c \otimes a \arrow[rr, "\nu"'] & & a &
\end{tikzcd}$$

commutes. A morphism $f : \nu \to \nu'$ of actions on a and a', respectively, is an arrow $f : a \to a'$ such that $\nu'(1_c \otimes f) = f\nu$. These definitions form the category $\mathbf{Lact}_c$ of left-actions from c to objects in B.

Right-actions are defined similarly. There is a forgetful functor $\mathbf{Lact}_c \to B$ given by $(\nu : c \to a) \mapsto a$ and $f \mapsto f$.

7.5 The Simplicial Category

Definition 7.5: The Simplicial Category

The simplicial category $\mathbf{\Delta}$ takes as objects finite ordinals $[n-1] =: n = \{0, \dots, n-1\}$, and as morphisms weakly monotonic functions, meanining functions $f : n \to m$ such that $i \le j$ implies $f(i) \le f(j)$.

The simplicial category is very important, hence why I dedicate an entire definition to it. Regular addition is turned into a bifunctor $+ : \mathbf{\Delta} \times \mathbf{\Delta} \to \mathbf{\Delta}$ via the following identification on arrows. Take $f : n \to n'$ and $g : m \to m'$. Define $f + g : m + n \to m' + n'$ as the map

$$(f+g)(x) = \begin{cases} f(x), & 0 \le x < m \\ g(x) - (m-1), & m \le x < n \end{cases}.$$

That is, we kind of just place f and g side by side. We then have that $(\mathbf{\Delta}, +, 0)$ is a *strict* monoidal category, and since 1 is terminal, there is a monoid $(1, \mu, \eta)$ defined in the only possible way; all diagrams commute manifestly since 1 is terminal. This monoid is universal in the following sense.

Proposition 7.2

Let (c, μ', η') be a monoid in a *strict* monoidal category $(B, \otimes, e)$. Then there is a unique monoidal functor $F : (\mathbf{\Delta}, +, 0) \to (B, \otimes, e)$ such that $F(1) = c$, $F\mu = \mu'$, and $F\eta = \eta'$.

Proof. Write $\mu^{(k)}$ for the unique arrow $k \to 1$, and note that $\mu^{(0)} = \eta$, $\mu^{(1)} = 1_1$, and $\mu^{(2)} = \mu$. This continues:

$$\mu^{(3)} = \mu(\mu + 1) = \mu(1 + \mu).$$

Commutivity (and associativity) here follow since 1 is terminal. We thus obtain

$$\mu^{(n)} \left(\sum_i \mu^{(k_1)} \right) = \mu^{(\sum_i k_i)}.$$

Moreover, given any $f : m \to n$, we can define m_i as the ordinal number pf the subset $f^{-1}(i)$ in m. We then very simply have

$$f = \sum_{i=0}^{n-1} \mu^{(m_i)}, \qquad \sum_{i=0}^{n-1} m_i = m.$$

Now consider the functor $F : \mathbf{\Delta} \to B$. Requiring $F(1) = c$ and $F(0) = e$ uniquely determines the object function. Using the same representation as the earlier part I didn't copy, wehre $\mu^{(n)}$ in a monoidal category means the n-fold multiplication with parethenases all at the front, we have that requiring $F\mu = \mu'$ and $F\eta = \eta'$ implies $F(\mu^{(n)}) = \mu'^{(n)}$. Since compositions in $\mathbf{\Delta}$ correspond to associativity under $+$, F is a functor by associativity of $\otimes$. ∎

This gives a complete characterization of $\mathbf{\Delta}$; it's objects are the free monoid (in **Set**) under $+$, and it's arrows are compositions and associations of η and μ. There is another characterization though, which is extremely important.

Definition 7.6

Let $i \in n$, and let $\delta_i^n : n \to n+1$ be the unique injective monotone map $n \to n+1$ that *does not* map to i:

$$\delta_n^i(x) = \begin{cases} x, & x < i \\ x+1, & x \geq i \end{cases}.$$

Also let $\sigma_n^i : n+1 \to n$ be the unique surjective monotone function $n+1 \to n$ that double counts i:

$$\sigma_n^i(x) = \begin{cases} x, & x \leq i \\ x-1, & x > i \end{cases}.$$

These are often referred to as face maps δ and degeneracy maps σ, or sometimes these names are only attached to the induced maps in a simplicial set or object. Often the n is omitted and is inferred from context. Notation for these maps varies, and a more common notation is d^i and s^i for δ^i and σ^i, respectively.

Clearly, any monic factors via δ maps, and any epi factors via σ maps, and any monotone map $f : n \to n'$ factors as $f = hg$ for h epi $n'' \to n'$ and g mono $n \to n'$. This gives a description of all maps in $\mathbf{\Delta}$ as composites of δs and σs. We make precise this fact.

Lemma 7.1

Any arrow $f : n \to n'$ in $\mathbf{\Delta}$ has a unique representation as

$$f = \delta_{i_1} \circ \cdots \circ \delta_{i_k} \circ \sigma_{j_1} \circ \cdots \circ \sigma_{j_h},$$

where $n - h + k = n'$, and

$$n' > i_1 > \cdots > i_k \geq 0, \qquad 0 \leq j_1 < \cdots < j_h < n-1.$$

Proof. Any monotone function is obviously uniquely determined by it's image, and hence the elements it "skips", and the elements it counts multiple times. ■

In particular, any composite of δ and σ may be put in this form. Some (simple to verify) identities are

$$\delta_i \delta_j = \delta_{j+1}\delta_i, \qquad i \leq j;$$

$$\sigma_j \sigma_i = \sigma_i \sigma_{j+1}, \qquad i \leq j;$$

$$\sigma_j \delta_i = \begin{cases} \delta_i \sigma_{j-1}, & i < j \\ 1, & j \leq i \leq j+1 \\ \delta_{i-1}\sigma_j, & i > j+1 \end{cases}.$$

These relations in particular suffice to put any composite of δs and σs into that of lemma 7.1. The book dedicates a proposition to this fact.

The category $\mathbf{\Delta}$ can be interpreted geometrically via affine simplicies, giving a (faithful) functor $\Delta : \mathbf{\Delta} \to \mathbf{Top}$ representing $\mathbf{\Delta}$ as a subcategory of $\mathbf{Top}$. We define this functor as such. Take $\Delta(0) = \varnothing$ viewed as a topological space, and take $\Delta(n+1)$ to be the standard n-simplex

$$\Delta(n+1) = \left\{ (t_0, \ldots, t_n) \mid t_i \geq 0 \forall i, \sum_{i=0}^{n} t_i = 1 \right\}.$$

We generally write Δ_n instead of $\Delta(n)$. A more common and more modern notation is to write Δ^n for the standard n-simplex and to write $[n]$ instead of $n+1$ for the object of $\mathbf{\Delta}$. Action on morphisms $f : [n] \to [m]$

is given by $\Delta(f) : \Delta^n \to \Delta^m$:

$$\Delta(f)(t_0, \dots, t_n) = (s_0, \dots, s_m), \qquad s_j = \sum_{f(i)=j} t_i.$$

In this notation, $\Delta_{n+1} = \Delta^n$ has dimension n and $n+1$ vertices, and $\Delta(f)$ is the unique affine map that sends the vertex i (given by $\mathbf{t}^i = 1, \mathbf{t}^{j \neq i} = 0$) of Δ^n to the vertex $f(i)$ of Δ^m. It can be verified this is functorial with basic affine geometry, which I do not know.

Of particular importance is the category $\mathbf{\Delta}^+$, which is the full subcategory of $\mathbf{\Delta}$ which omits 0. Topologists often work in $\mathbf{\Delta}^+$ instead of $\mathbf{\Delta}$, and cycle notation so $0 = \{0\}$; in general $n = [n]$. This is often a much more convenient place to work with simplicial objects, but we distinguish $\mathbf{\Delta}$ from $\mathbf{\Delta}^+$ in this for various categorical reasons, even though it is nonstandard.

Definition 7.7: Simplicial Object

A simplicial object S in a category X is a contravariant functor $S : \left(\mathbf{\Delta}^+\right)^{\mathrm{op}} \to X$, and a *morphism* of simplicial objects $S \to S'$ in X is a natural transformation $\theta : S \Rightarrow S'$. Simplicial objects in **Set** are called *simplicial sets*, and the category of simplicial sets is denoted **sSet**.

When considering a simplicial object S in X, we generally use the notation $[n] \mapsto S_n$, $\delta_i \mapsto d_i$, and $\sigma_i \mapsto s_i$. This gives the more traditional description of a simplicial object, being a list $\{S_i\}_{i \in \mathbb{N}}$ of objects of X, together with collections of arrows $d_i : S_n \to S_{n-1}$ and $s_i : S_n \to S_{n+1}$ for $i \in [n]$, called the *face* and *degeneracy* maps, respectively. These maps must also satisfy the relations dual to those explored earlier; in particular,

$$d_i d_{j+1} = d_j d_i, \qquad i \leq j;$$

$$s_{j+1} s_i = s_i s_j, \qquad i \leq j;$$

$$d_i s_j = \begin{cases} s_{j-1} d_i, & i < j \\ 1, & j \leq 1 \leq j+1 \\ s_j d_{i-1}, & i > j+1 \end{cases}.$$

We call S_n the object of n-simplicies.

For example, if Y is an affine simplex with linearly ordered vertices, then $d_i(Y)$ is the ith face, which is (defined as) the face that does not include the ith vertex. Similarly, $s_i(Y)$ is the degenerate simplex with the vertex i doubled.

An *augmented* simplicial object in X is a contravariant functor $S' : \mathbf{\Delta}^{\mathrm{op}} \to X$ (note here we use $\mathbf{\Delta}$ instead of $\mathbf{\Delta}^+$). An *augmentation* of a simplicial object S is an object $S_{-1} \in X$ and an arrow $\varepsilon : S_0 \to S_{-1}$ in X with $\varepsilon d_0 = \varepsilon d_1 : S_1 \to S_{-1}$; thus we may extend S to an augmented simplicial object S' with $S'(\delta_0) = \varepsilon$ and $S'(0) = S_{-1}$.

Given a simplicial object S in an abelian category A, We can construct a chain complex $(S_\bullet, \partial_\bullet)$ with the boundary morphisms given by

$$\partial_n = \sum_{i=0}^{n} (-1)^i d_i : S_n \to S_{n-1}.$$

The identity $d_i d_{j+1} = d_j d_i$ for $i \leq j$ gives $\partial_{n-1} \partial_n = 0$ after some annoying expansions. We can then compute the homologies

$$H_n(S_\bullet) = \frac{\operatorname{Ker} \partial_n}{\operatorname{Im} \partial_{n-1}}.$$

Note that $\operatorname{Ker} \partial_0 = S_0$, since $\partial_0 : S_0 \to S_{-1} = 0$, giving us

$$H_0(S_\bullet) = S_0 / \operatorname{Im} \partial_1.$$

Consider the composition

$$S: \qquad \mathbf{\Delta}^{\mathrm{op}} \times \mathbf{Top} \xrightarrow{\mathrm{Hom}(\Delta(-),-)} \mathbf{Set} \xrightarrow{F^{\mathbb{Z}}} \mathbf{Ab},$$

where $F^{\mathbb{Z}}$ (or F for simplicity) is the free abelian group, viewed as a functor. Thus for each topological space X there exists an augmented simplicial object $S(-, X)$ in $\mathbf{Ab}$. Since arrows in $\mathrm{Hom}(\Delta^n, X)$ are singular n-simplicies (whatever that means), so $S(X)_{n+1} = F(\mathrm{Hom}(\Delta^n, X))$ is the free abelian group generated by singular n-simplicies. The associated chain complex is called the *singular chain complex*, and the homolohies are the *singular homology* of X. This construction works for any concrete abelian category A for which there exists a free functor $F : \mathbf{Set} \to A$, but $\mathbf{Ab}$ is the obvious choice to use here.

To summarize,

- $\mathbf{\Delta}$ is the category with objects finite ordinals, and arrows weakly order-preserving functions;
- $\mathbf{\Delta}$ is a full subcategory of $\mathbf{Cat}$, when viewing ordinals as poset categories, since monotonic functions then become functors;
- $\mathbf{\Delta}$ is a strictly monoidal category and contains the universal monoid $(1, \mu : 2 \to 1, \eta : 0 \to 1)$, and it's arrows are iterated multiplications $\mu^{(m_0)} + \cdots \mu^{(m_{n-1})}$;
- $\mathbf{\Delta}$ is a subcategory of $\mathbf{Top}$ when viewing $n + 1 = [n]$ as the standard n-simplex Δ^n.

Simplicial objects are extremely important in algebraic topology.

7.6 Monads and Homology

Let $L = (L, \varepsilon, \delta)$ be a comonad, meaning $L : A \to A$, $\varepsilon : L \to 1_A$, and $\delta : L \to L^2$ satisfy

$$(\delta L) \circ \delta = (L\delta) \circ \delta, \qquad (\varepsilon L) \circ \delta = 1_L = (L\varepsilon) \circ \delta.$$

$\mathbf{\Delta}^{\mathrm{op}}$ contains the universal comonoid, and since A^A is a strict monoidal category, there is a unique monoidal functor $\mathbf{\Delta}^{\mathrm{op}} \to A^A$ mapping $(1, 1 \to 2, 1 \to 0) \mapsto (L, \varepsilon, \delta)$. Note in particular that this is an augmented simplicial object in A^A, with $\delta = s_0$ the degeneracy arrow, ε the augmentation, and $n \mapsto L^n$ the object function. Faces and degenerates are given by

$$d_i^n = L^i \varepsilon L^{n-1}, \qquad s_i^n = L^i \delta L^{n-i-1}.$$

Assuming that A is an *Ab*-category gives us that we have a chain complex $L^\bullet(a)$:

$$L(a) \xleftarrow{\partial_2} L^2(a) \xleftarrow{\partial_3} L^3(a) \xleftarrow{\partial_4} \cdots$$

with the augmentation $\varepsilon_a : L(a) \to a$ giving a resolution of a, for any $a \in A$. The book now gives a very long example considering cohomologies of some group Π.

7.7 Closed Categories

Definition 7.8: Symmetric

A monoidal category B is *symmetric* when there exists a natural isomorphism $\gamma_{a,b} : a \otimes b \to b \otimes a$ such that

$$\gamma_{a,b} \circ \gamma_{b,a} = 1, \qquad \rho_b = \lambda_b \circ \gamma_{b,e} : b \otimes e \cong b,$$

and such that the diagram

$$\begin{tikzcd}
a \otimes (b \otimes c) \arrow[r, "\alpha"] \arrow[d, "1\otimes\gamma"'] & (a \otimes b) \otimes c \arrow[r, "\gamma"] & c \otimes (a \otimes b) \arrow[d, "\alpha"] \\
a \otimes (c \otimes b) \arrow[r, "\alpha"'] & (a \otimes c) \otimes b \arrow[r, "\gamma\otimes 1"'] & (c \otimes a) \otimes b1
\end{tikzcd}$$

commutes.

This condition is a coherence condition; a symmetric monoidal category is then a monoidal category for which $\otimes$ is commutative up to natural isomorphism. Note that cartesian monoidal categories, in particular, are symmetric.

Definition 7.9: Closed Category

An *closed category* V is a symmetric monoidal category for which each functor $- \otimes b : V \to V$ has a specified right-adjoint $(-)^b : V \to V$; we call $(-)^b$ the internal hom-functor.

Alternative notations for $(-)^b$ include $[b, -]$. An example is $(\mathbf{Ab}, \otimes, -)$ with $(-)^B := \mathrm{Hom}(B, -)$; this in fact holds in general R-modules.

We have seen already that *Ab*-categories are categories for which hom-sets are abelian groups. We can generalize this further; for a monoidal category V, a V-category R is a collection of objects; for each pair $r, s \in R$ an *object* $R(r, s) \in V$; to each triple $r, s, t \in R$ an arrow known as composition

$$\circ : R(s, t) \otimes R(r, s) \to R(r, t);$$

to each $r \in R$ an arrow $e \to R(r, r)$. These are subject to the usual unital and associative axioms for categories. Note that this does not define morphisms in the usual way; although we obtain hom-objects this does not actually give us arrows. The way we obtain arrows is by defining a functor $U : V \to \mathbf{Set}$, and taking the hom-sets to be $U(R(-, -))$. In particular, we take $U = V(e, -)$. We then say that R is a V-enriched category. Since $\mathbf{Ab}(\mathbb{Z}, A) \cong A$, this recovers the usual notion of *Ab*-categories as *Ab*-enriched categories.

If the enriching category V is *closed*, basically all of basic category theory applies to it's enriched categories.

Chapter 9

Special Limits

9.1 Filtered Limits

Recall a partial order is a preorder for which $a \leq b$ and $b \leq a$ implies $b = a$. A preorder does not have this quality. A preorder P is *directed* when any $p, q \in P$ have an upper bound in P, meaning there is some r with $p, q \leq r$. Induction guarentees that the same holds for finite sets. A directed preorder is often called a *directed set* or a *filtered set*. Using some inspiration from poset categories, we can generalize this notion to *any* category.

Definition 9.1: Filtered Category

A *filtered* category is a category J such that for all $j, j' \in J$, there exists $k \in J$ and arrows $j \to k$ and $j' \to k$.

1. for all $j,' j \in J$, there exists $k \in J$ and arrows $j \to k$ and $j' \to k$;
2. for any parallel arrows $u, v : i \to j$ in J, there is $k \in J$ and an arrow $w : j \to k$ such that $wu = wv$:

$$i \underset{v}{\overset{u}{\rightrightarrows}} j \xrightarrow{w} k$$

   In other words, all parallel arrows have arrows that equalize them (but not universaly; this does not guarentee equalizers exist).

These conditions state that the set $\{j, j'\}$, viewed as a discrete category, always has a co-cone with vertex k, and that any diagram $i \rightrightarrows j$ also always has a cone k. Any finite diagram in a filtered category is thus the base of at least one cone. Some authors (including the author himself in another text) call this "cofiltered". The terminology varies.

A *filtered colimit* is the colimit of a functor $F : J \to C$ with J filtered. A filtered limit is likewise a limit over a filtered category. There are some important theorems attached to filtered (co)limits, and we explore them here.

Theorem 9.1

A category C with finite coproducts and all small colimits over *directed preorders* has all small coproducts.

Proof. Given a set J, we want to show $\varinjlim F$ exists for any functor $F : J \to C$. Let J^+ be the preorder with objects all *finite* subsets $S \subseteq J$. To show J^+ is filtered, note that if $A, B \subseteq J$ are finite, then

$A \cup B \subseteq J$ is finite with $A, B \subseteq A \cup B$. Define $F^+ : J^+ \to C$ as the map

$$S \mapsto \coprod_{s \in S} F(s).$$

We define the map on morphisms as follows. Let $u : S \subseteq T$; define $F^+(u)$ as the unique arrow

$$\begin{tikzcd}
F^+(S) = \coprod_{s \in S} F(s) \arrow[r, dashed] & \coprod_{t \in T} F(t) = F^+(T) \\
F(s) \arrow[u, "i_s"] \arrow[r, equal] & F(s) \arrow[u, "i'_s"']
\end{tikzcd}$$

induced by universality of the coproduct over S, since $F^+(T)$ has inclusions for all $F(s)$. Universality makes F^+ functorial. There exists an inclusion $J \hookrightarrow J^+$ given by $j \mapsto \{j\}$, and since $\coprod \{j\} = j$ is simply the limit over $\mathbf{1}$, we have that F^+ agrees with F on J.

Let $\theta : F^+ \Rightarrow G$ be a natural transformation to any functor $G : J^+ \to C$. Since $\{s\} \subseteq S$ for all $s \in S$, we obtain arrows

$$G(\{s\} \subseteq S) : G\{s\} \to G(S)$$

for all $s \in S \in J^+$. Naturality of θ gives

$$\begin{tikzcd}
\coprod_{s \in S} F(s) \arrow[r, "\theta_S"] & G(S) \\
F(s) \arrow[u, "i_s"] \arrow[r, "\theta_s"'] & G\{s\} \arrow[u, "G(\{s\} \subseteq S)"']
\end{tikzcd}$$

since $F^+(\{s\} \subseteq S)$ is immediately found to be the injection. By universality of coproduct, θ_S is uniquely determined by the values of θ_s. In particular, any cocone $\nu^+ : F^+ \Rightarrow \Delta(c)$ is determined entirely by it's values $\nu^+(j) : F(j) \to c$ for $j \in J$, and the collection of all values $\nu^+(j)$ form a cocone $\nu^+|_J =: \nu : F \Rightarrow \Delta(c)$. Since J^+ is a finite directed preorder, by hypothesis $\varinjlim F^+$ exists, so it suffices to show ν is limiting if and only if ν^+ is limiting. This is immediate for components of a natural transformation of cones over J^+ on non-singletons are unique by universality of coproduct, so uniqueness of the morphism of cocones is equivalent to uniqueness of the natural transformation on objects $j \in J$; this is precisely the uniqueness of the morphisms of cocones over J. ∎

This proof allows us to prove that a category is (co)complete by proving only that finite (co)products, small (co)equalizers, and small (co)limits over directed preorders exist. Note here that directed preorders are indeed a class of filtered categories. The book continues by proving **Grp** is cocomplete via showing the forgetful functor creates filtered colimits. I decided not to type this up, although as usual I still worked through it independently.

9.2 Interchange of Limits

We have seen previously that a bifunctor $F : P \times J \to X$ with X cocomplete defines functors $p \mapsto \varinjlim F(p, -)$ such that the limiting cocones $\mu_p : F(p, -) \Rightarrow \varinjlim F(p, -)$ are natural in p. One can then prove rather easily that

$$\varinjlim_J \varinjlim_P F \cong \varinjlim_P \varinjlim_J F.$$

Here, we define the colimit over P of F as the limit of $F(-,j)$ for some fixed $j \in J$. This necessitates a choice of j, which is clarified by taking the limit over j of the functor $j \mapsto \varinjlim F(-,j)$.

Dually, limits commute. What is not so easily shown is commutivity of limits with colimits. For any choice of (p,j), there is a commutative diagram

$$\begin{tikzcd}
F \arrow[d, "\mu_{p,j}"'] & & \varprojlim_P F(-,j) \arrow[ll, "\nu_{p,j}"'] \arrow[r, "\mu_j"] \arrow[d, dashed, "\alpha_j"] & \varinjlim_J \varprojlim_P F \arrow[d, dashed, "\kappa"] \\
\varinjlim_J F(p,-) \arrow[rr, "\nu_p"'] & & \varprojlim_P \varinjlim_J F \arrow[r, equal] & \varprojlim_P \varinjlim_J F
\end{tikzcd}$$

We call $\kappa : \varinjlim_J \varprojlim_P F \to \varprojlim_P \varinjlim_J F$ the canonical morphism. It is not necessarily an isomorphism, but we have some criteria for when it is.

Theorem 9.2

If P is finite and J is small and filtered, and $F : P \times J \to \mathbf{Set}$, then the canonical arrow is an isomorphism.

9.3 Final Functors

Often we can compute limits over subcategories. Given a subcategory $I \subseteq J$ and a functor $F : J \to C$, we can compute the limit of the restriction $F|_I : I \to C$. This idea gives rise to the notion of a final functor.

Definition 9.2: Final Functor

A functor $L : J' \to J$ is *final* if for each $k \in J$, the comma category $(k \downarrow L)$ is nonempty and connected.

Here a connected category means for any $a, b \in X$, there is a zigzag from a to b. A zigzag is a finite chain of objects $\{x_i\}$ such that there is a morphism between each x_i and x_{i+1} (the morphism can go either direction). A subcategory is final when the inclusion is final.

Let $L : J' \to J$ final and $F : J \to X$. Suppose that the colimits $\varinjlim F$ and $\varinjlim FL$ exist. If $\mu' : FL \Rightarrow \varinjlim FL$ and μ are the colimiting cones, then since $L(J') \subseteq J$, we have that there is some $\mu_{L(j')}$ for all $j' \in J'$. Thus, $\varinjlim F$ is also a cone for FL, and by universality of colimit there is a unique morphism $h : \varinjlim FL \to \varinjlim F$ with $h\mu'_{j'} = \mu_{L(j')}$ for all $j' \in J'$.

Theorem 9.3

If $L : J' \to J$ is final, and $F : J \to X$ is a functor such that $x = \varinjlim FL$ exists, then $\varinjlim F$ exists, and h (as described above) is an isomorphism.

Proof. Let $\mu : FL \Rightarrow \varinjlim FL$ be a colimiting cone. For each $k \in J$, choose an arrow $u : k \to L(j')$ and define $\tau_k = F(u) \circ \mu_{j'}$ for some $j' \in J'$. Since μ is a cone and $(k \downarrow L)$ is connected, this definiton is independent of choice. It follows that $\tau : F \Rightarrow x$ is a cone with base F, and universality of μ shows that τ is also limiting. Thus, $x = \varinjlim F$; h is obviously an isomorphism. ■

The dual of a final functor is an inital functor, and the dual of theorem 9.3 is useful for limits.

9.4 Diagonal Naturality

Definition 9.3: Dinatural

Let C, B be categories and $S, T : C^{\text{op}} \times C \to B$ functors. A dinatural transformation $\alpha : S \stackrel{\cdot}{\to} T$ is a function α which assigns to each $c \in C$ a morphism $\alpha_c : S(c,c) \to T(c,c)$ called the *component* of α at c, such that for all $f : c \to c'$ in C, the diagram

$$\begin{tikzcd}
& S(c,c) \arrow[r, "\alpha_c"] & T(c,c) \arrow[dr, "T(1,f)"] & \\
S(c',c) \arrow[ur, "S(f,1)"] \arrow[dr, "S(1,f)"'] & & & T(c,c') \\
& S(c',c') \arrow[r, "\alpha_{c'}"'] & T(c',c') \arrow[ur, "T(f,1)"'] &
\end{tikzcd}$$

commutes.

Intuitively, a dinatural transformation is naturality on the diagonal. The commutative hexagon gives a sort of naturality condition for the diagonal element (f, f). Defining naturality directly on the diagonal morphisms wouldn't work due to the contravariance of the left entry, meaning we would no longer be working with only the diagonal objects. The commutative hexagon is our alternative definition.

Clearly any natural transformation $\tau : S \Rightarrow T$ gives rise to a dinatural transformation in the obvious way, by taking the diagonal elements $\alpha_c \coloneqq \tau_{c,c}$. We can also consider functors T which are dummy in one variable, meaning T is a composition

$$T : C^{\text{op}} \times C \xrightarrow{\pi_C} C \xrightarrow{T_0} T$$

for π_C the projection, and T_0 some functor. More precisely, any functor $T_0 : C \to B$ may be viewed as a bifunctor dummy in one variable by precomposition with the relevant projection. A functor dummy in both variables would then just be a constant object.

Consider S dummy in the second variable and T dummy in the first. A dinatural transformation $\alpha : S \stackrel{\cdot}{\to} T$ is given by

$$\begin{tikzcd}
& S(c,c) = S_0(c) \arrow[r, "\alpha_c"] & T_0(c) = T(c,c) \arrow[dr, "T(1,f) = T_0(f)"] & \\
S(c',c) = S_0(c') \arrow[ur, "S(f,1) = S_0(f)"] \arrow[dr, "S(1,f) = S_0(1)"'] & & & T_0(c') = T(c,c') \\
& S(c',c') = S_0(c') \arrow[r, "\alpha_{c'}"'] & T_0(c') = T(c',c') \arrow[ur, "T(f,1) = T_0(1)"'] &
\end{tikzcd}$$

Dropping the identity morphisms, this simply becomes a generalization of the naturality condition from contravariant S to covariant T:

$$\begin{tikzcd} S_0(c') \arrow[r, "S_0(f)"] \arrow[d, "\alpha_{c'}"'] & S_0(c) \arrow[d, "\alpha_c"] \\ T_0(c') & T_0(c) \arrow[l, "T_0(f)"] \end{tikzcd}$$

We call α a natural transformation from contravariant to covariant, dually covariant to contravariant. One can also construct regular natural transformations between covariant and contravariant functors by taking them *both* to be dummy in the first or second entries, respectively.

If $T = b$ is dummy in both variables $(c, d) \mapsto b$, a dinatural transformation $\alpha : S \dot{\to} b$ becomes

$$\begin{tikzcd} S(c', c) \arrow[r, "S(f,1)"] \arrow[d, "S(1,f)"'] & S(c, c) \arrow[d, "\alpha_c"] \\ S(c', c') \arrow[r, "\alpha_{c'}"'] & b \end{tikzcd}$$

for each $f : c \to c'$ in C. Dinatural transformations $\alpha : S \dot{\to} b$ are called *extranatural* transformations, *wedges*, or sometimes *supernatural* transformations. Extranaturality generalizes beyond this, but not in this text (or at least this part). Generally a wedge refers to the exact diagram given here.

Evaluation is an example. Fix some set A. Let X a set, and define $V_X : \mathrm{Hom}(X, A) \times X \to A$ be the evaluation function $(f, x) \mapsto f(x)$. Note here that $\mathrm{Hom}(-, A) \times -$ is a functor $\mathbf{Set}^{\mathrm{op}} \times \mathbf{Set} \to \mathbf{Set}$, and $(gf)(x) = g(f(x))$ implies that the extranaturality square commutes, making V an extranatural transformation

$$V : \mathrm{Hom}(-, A) \times - \dot{\to} A.$$

Since V is also natural in A, we say V is *dinatural* in X and *natural* in A.

We use the various types of dinatural transformations together, and frequently drop the "di". Thus for $S : C^{\mathrm{op}} \times C \times A \to B$ and $T : A \times D^{\mathrm{op}} \times D \to B$ functors, a natural transformation $\gamma : S \dot{\to} T$ is a function mapping each triple $(c, a, d) \in C \times A \times D$ to a morphism

$$\gamma(c, a, d) : S(c, c, a) \to T(a, d, d)$$

such that γ is natural in a, and dinatural in c and d.

Compositions of dinatural transformations need not be dinatural, but composing on the left and right with natural transformations preserves dinaturaliy. If $\alpha : S \dot{\to} T$ is dinatural, $\sigma : S' \Rightarrow S$ is natural, and $\tau : T \Rightarrow T'$ is natural, then there is a dinatural transformation $S' \dot{\to} T'$ with arrows given by the composition

$$S'(c, c) \xrightarrow{\sigma_{c,c}} S(c, c) \xrightarrow{\alpha_c} T(c, c) \xrightarrow{\tau_{c,c}} T'(c, c).$$

Here's another case.

Proposition 9.1

Let $R : C \to B$, $S : C \times C^{\mathrm{op}} \times C \to B$, and $T : C \to B$ be functors. Let ρ and σ be functions defined for all $c, d \in C$ as arrows

$$\rho(c, d) : R(c) \to S(c, d, d), \qquad \sigma(d, c) : S(d, d, c) \to T(c),$$

natural in c and dinatural in d. Then the composition $\alpha_c = \sigma(c, c) \circ \rho(c, c)$ is natural $R \Rightarrow T$.

Proof. Naturality in c gives us that the vertical composition $\sigma(d,-)\circ\rho(-,d)$ is natural $R \Rightarrow T$ for any fixed $d \in C$. Dinaturality allows d to vary with the restriction that d be equivalent to the other component; meaning we can define $\sigma(c,c)\circ\rho(c,c)$ as the natural transformation. ■

9.5 Ends

Definition 9.4: End

An *end* of a functor $S : C^{\mathrm{op}} \times C \to X$ is a universal dinatural transformation from a constant e to S. That is, an end is a pair (e, ω) with $e \in X$ and $\omega : e \stackrel{\cdot}{\to} S$ a wedge with the property that for any wedge $\beta : x \stackrel{\cdot}{\to} S$, there is a unique arrow $h : x \to e$ in X such that $\beta_a = \omega_a h$ for all $a \in C$. By common abuse of language, we generally refer to the object e as the end of S, and we write it with integral notation

$$e = \int_c S(c,c) = \int_{c\in C} S(c,c) = \int_C S.$$

By definition of a wedge, with notation as above this implies for all $f : b \to c$ in C, there is a commutative diagram

$$\begin{tikzcd}
 & x \arrow[d, dashed, "h"] \arrow[ddl, bend right, "\beta_b"'] \arrow[ddr, bend left, "\beta_c"] & \\
 & \int_{c\in C} S(c,c) \arrow[dl, "\omega_b"'] \arrow[dr, "\omega_c"] & \\
S(b,b) \arrow[dr, "S(1,f)"'] & & S(c,c) \arrow[dl, "S(f,1)"] \\
 & S(b,c) &
\end{tikzcd}$$

Ends are a particularly important class of limit, and just from looking at the diagram it's apparent they show some similarity with pullbacks. We call ω the ending wedge or the universal wedge. As always, any two ends are isomorphic.

Natural transformations give examples of ends. Let $U, V : C \to X$. We thus end up with a composition

$$\mathrm{Hom}(U(-), V(-)) : C^{\mathrm{op}} \times C \xrightarrow{U^{\mathrm{op}} \times V} X^{\mathrm{op}} \times X \xrightarrow{\mathrm{Hom}(-,-)} \mathbf{Set}.$$

A wedge $\tau : Y \stackrel{\cdot}{\to} \mathrm{Hom}(U(-), V(-))$ for $Y \in \mathbf{Set}$ induces for any $f : b \to c$ the commutative diagram

$$\begin{tikzcd}
Y \arrow[r, "\tau_b"] \arrow[d, "\tau_c"'] & \mathrm{Hom}(U(b), V(b)) \arrow[d, "V(f)^*"] \\
\mathrm{Hom}(U(c), V(c)) \arrow[r, "U(f)_*"'] & \mathrm{Hom}(U(b), V(c))
\end{tikzcd}$$

where $U(f)_*$ is precomposition by $U(f)$, and likewise for $V(f)^*$. This states that for each $y \in Y$, there is an equivalence

$$V(f) \circ \tau_b(y) = \tau_c(y) \circ U(f).$$

Diagramatically, this gives

$$\begin{tikzcd}
U(b) \arrow[r, "\tau_b(y)"] \arrow[d, "U(f)"'] & V(b) \arrow[d, "V(f)"] \\
U(c) \arrow[r, "\tau_c(y)"'] & V(c)
\end{tikzcd}$$

which implies that $\tau_{(-)}(y) : U \Rightarrow V$ is natural for any fixed $y \in Y$. We thus have an assignment $y \mapsto \tau_{(-)}(y)$ which gives a function $\eta : Y \to \mathrm{Nat}(U, V)$. Moreover, we have an assignment $\omega : \mathrm{Nat}(U, V) \stackrel{\cdot}{\to} \mathrm{Hom}(U(-), V(-))$ given by $\omega_c(\lambda) = \lambda_c$. I will show this is a wedge in a second. For now, note that $\omega_c(\eta(y)) = \tau_c(y)$, and thus the diagram

$$\begin{tikzcd}
Y \arrow[r, "\tau_c"] \arrow[d, "\eta"'] & \mathrm{Hom}(U(c), V(c)) \arrow[d, "1"] \\
\mathrm{Nat}(U, V) \arrow[r, "\omega_c"'] & \mathrm{Hom}(U(c), V(c))
\end{tikzcd}$$

commutes for each c. Moreover, this definition of η is manifestly unique, meaning that with the assumption that ω is an end, we have found that

$$\mathrm{Nat}(U, V) = \int_{c \in C} \mathrm{Hom}(U(c), V(c)).$$

Finally, to show ω is a wedge, chase a natural transformation λ around the square

$$\begin{tikzcd}
\mathrm{Nat}(U, V) \arrow[r, "\omega_b"] \arrow[d, "\omega_c"'] & \mathrm{Hom}(U(b), V(b)) \arrow[d, "V(f)^*"] \\
\mathrm{Hom}(U(c), V(c)) \arrow[r, "U(f)_*"'] & \mathrm{Hom}(U(b), V(c))
\end{tikzcd}
\qquad
\begin{tikzcd}
\lambda \arrow[r, mapsto, "\omega_b"] \arrow[d, mapsto, "\omega_c"'] & \lambda_b \arrow[d, mapsto, "V(f)^*"] \\
\lambda_c \arrow[r, mapsto, "U(f)_*"'] & V(f) \circ \lambda_b = \lambda_c \circ U(f)
\end{tikzcd}$$

Since $f : b \to c$, this is precisely the naturality condition, and thus ω is indeed a wedge.

Ends are manifestly limits. We offer an explicit limitting construction for completeness. Let C be a category, and define the category $\mathcal{C}$ to have as objects not only all objects of C, but also all *morphisms* of C. Given an object/arrow z in C, we denote it's correspondence in $\mathcal{C}$ by $\overline{z}$. Arrows in $\mathcal{C}$ are defined as follows. For each non-identity $f : a \to b$ in C, there are two arrows

$$\overline{a} \longrightarrow \overline{f} \longleftarrow \overline{b}$$

in $\mathcal{C}$. Identities also exist. It's impossible to compose two non-identity arrows in $\mathcal{C}$.

Given a functor $S : C^{\mathrm{op}} \times C \to X$, there exists a functor $\overline{S} : \mathcal{C} \to X$ given by

$$\begin{tikzcd}
\overline{b} \arrow[rr] \arrow[d, mapsto] & & \overline{f} \arrow[d, mapsto] & & \overline{c} \arrow[ll] \arrow[d, mapsto] \\
S(b, b) \arrow[rr, "S(1_b, f)"'] & & S(b, c) & & S(c, c) \arrow[ll, "S(f, 1_c)"]
\end{tikzcd}$$

Proposition 9.2

With notation as above, the limit of $\overline{S}$ exists if and only if the end of S exists, and there is an isomorphism

$$\theta : \int_{c\in C} S(c,c) \cong \varprojlim \overline{S},$$

which is unique in the sense that the diagram

$$\begin{tikzcd} \int_{c\in C} S(c,c) \arrow[r, "\omega_c"] \arrow[d, dashed, "\theta"'] & S(c,c) \\ \varprojlim \overline{S} \arrow[ur, "\lambda_c"'] & \end{tikzcd}$$

commutes for all $c \in C$, for λ the limiting cone and ω the universal wedge.

Proof. The cone condition of limit gives for any $f : b \to c$ in C the diagram

$$\begin{tikzcd} & \varprojlim \overline{S} \arrow[dl, "\lambda_b"'] \arrow[d, "\lambda_{b,c}"] \arrow[dr, "\lambda_c"] & \\ S(b,b) \arrow[r, "S(1,f)"'] & S(b,c) & S(c,c) \arrow[l, "S(f,1)"] \end{tikzcd}$$

Here the morphism $\lambda_{b,c}$ is forced by commutivity of the diagram to simply be the diagonal element of the diagram

$$\begin{tikzcd} \varprojlim \overline{S} \arrow[r, "\lambda_b"] \arrow[d, "\lambda_c"'] & S(b,b) \arrow[d, "S(1,f)"] \\ S(c,c) \arrow[r, "S(f,1)"'] & S(b,c) \end{tikzcd}$$

which is precisely the requirement that λ be a wedge. Universality of limit is then equivalent to universality of wedge, and also by universality, θ exists and is unique for any universal wedge ω. ■

Corollary 9.1

If C is small and X is complete, every functor $S : C^{\mathrm{op}} \times C \to X$ has an end in X.

The converse is also true; not only can we reduce ends to limits, but we can also reduce limits to ends.

Proposition 9.3

Let $T : C \to X$ be a functor. Define by S the composition functor

$$C^{\mathrm{op}} \times C \xrightarrow{\pi_C} C \xrightarrow{T} X$$

for π_C the projection. Then $(e, \tau : e \Rightarrow T)$ is a limit of T if and only if $(e, \tau : e \ddot{\to} S)$ is an end of S.

Proof. I prove universal wedges are limits; the proof is the exact same in the converse direction. Let the end of S exist. Let $\tau : e \ddot{\to} S$ be the universal wedge, giving for all $f : b \to c$ in C the diagram

$$\begin{tikzcd} e \arrow[r, "\tau_b"] \arrow[d, "\tau_c"'] & S(b,b) \arrow[d, "S(1,f)"] \\ S(c,c) \arrow[r, "S(f,1)"'] & S(b,c). \end{tikzcd}$$

Applying the fact that $S(x, y) = T(y)$ by definition, commutivity of this diagram is equivalent to the diagram

$$\begin{tikzcd} e \arrow[r, "\tau_b"] \arrow[d, "\tau_c"'] & T(b) \arrow[d, "T(f)"] \\ T(c) \arrow[r, "T(1)=1"'] & T(c) \end{tikzcd}$$

commuting. Dropping the identity morphism and rewriting in a more conventional way, this diagram reads

$$\begin{tikzcd} & e \arrow[dl, "\tau_b"'] \arrow[dr, "\tau_c"] & \\ T(b) \arrow[rr, "T(f)"'] & & T(c) \end{tikzcd}$$

which commutes for all $f : b \to c$. This is precisely the condition that e be a cone. Thus, cones for T are precisely wedges for S, and thus universality of τ as a cone is equivalent to universality of τ as a wedge. ■

The conclusion gives us

$$\int_{c \in C} S(c,c) \cong \varprojlim T.$$

We write

$$\int_{c \in C} T(c) \cong \int_{c \in C} S(c,c) \cong \varprojlim T,$$

and leave the dummy variable implicit. Ends are thus another characterization of limits, and quite a powerful one.

Some terminology that is likely expected. A functor $H : X \to Y$ *preserves the end* of a functor $S : C^{\mathrm{op}} \times C \to X$ when for any end $\omega : e \ddot{\to} S$, $H\omega : H(e) \ddot{\to} HS$ is an end; more commonly this is written

$$H\left(\int_{c \in C} S(c,c)\right) \cong \int_{c \in C} HS(c,c).$$

Similarly, H *creates* the end of S when to each end $\nu : y \ddot{\to} HS$ in Y, there is a unique wedge $\omega : e \ddot{\to} S$ with

$H\omega = \nu$ such that ω is an end. Since ends are precisely limits, all our preservation and creation theorems necessarily apply. For example, right-adjoint functors and hom-functors preserve ends.

9.6 Coends

Definition 9.5: Coend

Dual to the notion of an end is that of a coend. A coend of a functor $S : C^{\mathrm{op}} \times C \to X$ is a pair (d, ζ) such that $d \in X$ and ζ is a universal wedge $\zeta : S \ddot{\to} d$. We write

$$d = \int^{c \in C} S(c, c)$$

for the ending object, and refer to d simply as the coend of S.

An example of coends is tensor products. Take R to be an *Ab*-category with one element, otherwise known as a ring. A *left* R-module B is an additive functor $K : R \to \mathbf{Ab}$ mapping $* \mapsto B$; scalar multiplication by $r \in R$ is defined as $b \mapsto K(r)(b)$. A right R-module A is then an additive functor $R^{\mathrm{op}} \to \mathbf{Ab}$ mapping $* \mapsto A$. With $\otimes$ the tensor product of abelian groups, then $R \mapsto A \otimes B$ is a bifunctor $R^{\mathrm{op}} \times R \to \mathbf{Ab}$, and the coend is precisely the tensor product over R:

$$\int^{r \in R} A \otimes B \cong A \otimes_R B.$$

Indeed, the wedge condition for ζ is that the diagram

$$\begin{tikzcd} A \otimes B \arrow[r, "1_A \otimes r"] \arrow[d, "r \otimes 1_B"'] & A \otimes B \arrow[d, "\zeta"] \\ A \otimes B \arrow[r, "\zeta"'] & A \otimes_R B \end{tikzcd}$$

commutes for all r; this is precisely the requirement that $\zeta(ar \otimes b) = \zeta(a \otimes rb)$.

The purpose of this is to point out a more general construction. Let $(B, \otimes, e, \alpha, \lambda, \rho)$ be a monoidal category, and let $S : P \to B$ and $T : P^{\mathrm{op}} \to B$ be functors. Then there is a "tensor product" of functors

$$T \otimes_P S \coloneqq \int^{p \in P} T(p) \otimes S(p).$$

The tensor product is an object of B. An example is given by the simplicial category. Take the functor $\Delta : \mathbf{\Delta} \to \mathbf{Top}$ given by $[n] \mapsto \Delta^n$; recall that simplicial sets are precisely functors $S : \mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Set}$. The copower $S \cdot X$ given by

$$\coprod_{s \in S} X.$$

Since $\mathbf{Top}$ has small coproducts, the assignment $([n], [m]) \mapsto S[n] \cdot \Delta^m$ is a functor $\mathbf{\Delta}^{\mathrm{op}} \times \mathbf{\Delta} \to \mathbf{Top}$, and the coend

$$\int^{[n] \in \mathbf{\Delta}} S[n] \cdot \Delta^n$$

is the usual geometric realization of the simplicial set S, given by pasting n-simplicies according to face and degeneracy operators.

9.7 Ends with Parameters

Proposition 9.4 (End/Limit of a Natural Transformation)

Let $\gamma : S \Rightarrow S'$ be a natural transformation with $S, S' : C^{\mathrm{op}} \times C \to X$, such that there exist ends (e, ω) and (e', ω') for S and S', respectively. Then there is a unique arrow $e \to e'$ in X, denoted $\int_{c\in C} \gamma_{c,c}$, such that for all $c \in C$, the diagram

$$\begin{tikzcd}
\int_{c\in C} S(c,c) \arrow[r, "\omega_c"] \arrow[d, dashed, "\int_{c\in C}\gamma_{c,c}"'] & S(c,c) \arrow[d, "\gamma_{c,c}"] \\
\int_{c\in C} S'(c,c) \arrow[r, "\omega'_c"'] & S'(c,c)
\end{tikzcd}$$

commutes.

Proof. It suffices to show the composition $\gamma_{c,c} \circ \omega_c$ is a wedge $\int_{c\in C} S(c,c) \ddot{\to} S'(c,c)$; the existence of the unique morphism then follows by universality of ω' with respect to wedges for S'. Indeed, recall composing a dinatural transformation with a natural transformation in this manner yields another dinatural transformation, so the composition yields a dinatural transformation $\gamma \circ \omega : \int_{c\in C} S(c,c) \ddot{\to} S'$, and since the end is simply an object in X, this is precisely the definition of a wedge with base S' and vertex $\int_{c\in C} S(c,c)$. ■

We call the arrow $\int_c \gamma_{c,c}$ the *end* of the natural transformation γ. Universality of wedges gives the following identity for composition with another natural transformation $\gamma' : S' \Rightarrow S''$:

$$\int_{c\in C} (\gamma' \circ \gamma)_{c,c} = \left(\int_{c\in C} \gamma'_{c,c}\right) \circ \left(\int_{c\in C} \gamma_{c,c}\right).$$

This rule allows us to show that an end/limit involving a parameter $p \in P$ is a functor of that parameter in a certain sense.

Theorem 9.4 (Parameter Theorem for Ends and Limits)

Let $T : P \times C^{\mathrm{op}} \times C \to X$ be a functor such that $T(p, -, -)$ has an end $\omega^p : \int_{c\in C} T(p,c,c) \ddot{\to} T(p,-,-)$ for each $p \in P$. Then there is a unique functor $U : P \to X$ such that $U(p) = \int_{c\in C} T(p,c,c)$, such that the components of the wedge ω^p define for each $c \in C$ a transformation $\omega_c^p : U(p) \to T(p,c,c)$ natural in p, meanining

$$\omega_c^{(-)} : U \Rightarrow T(-,c,c).$$

Proof. Functorality gives that any $s : p \to q$ in P induces a natural transformation $T(s,-,-) : T(p,-,-) \Rightarrow T(q,-,-)$. The arrow function of U must then be

$$s \mapsto \int_{c\in C} T(s,c,c),$$

as defined in proposition 9.4. The composition rule shows that U is functorial, and the uniqueness of the arrow shows U is unique. ■

We write the functor U as
$$U = \int_{c\in C} T(-,c,c).$$
This makes sense given our previous notation:
$$p \mapsto \int_{c\in C} T(p,c,c), \qquad s \mapsto \int_{c\in C} T(s,c,c).$$
Note here the c, c in the natural transformation denotes the component at (c,c). This notation suggests that U itself is an end, and indeed it is if we view it as an object in X^P. Given the functor $T : P \times C^{\mathrm{op}} \times C \to X$, define the functor $T^{\#} : C^{\mathrm{op}} \times C \to X^P$ by $(x,y) \mapsto T(-,x,y)$, for both objects and morphisms.

Theorem 9.5 (Parameter Theorem Continued)

With notation as above, and under the same hypotheses for T as theorem 9.4, the functor $T^{\#}$ has the end
$$\omega^{\#} : \int_{c\in C} T(-,c,c) \ddot{\to} T^{\#},$$
where $(\omega_c^{\#})_p = \omega_c^p$ for all $p \in P$ and $c \in C$.

Proof. By the previous theorem, since ω was natural in p, there is for each $c \in C$ an arrow (natural transformation) of X^P
$$\omega_c^{\#} : \int_{c\in C} T(-,c,c) \Rightarrow T(-,c,c)$$
with component at p given by ω_c^p. $\omega^{\#}$ is a wedge, and is universal since each component is universal. ■

9.8 Iterated Ends and Limits

Proposition 9.5

Let $S : P^{\mathrm{op}} \times P \times C^{\mathrm{op}} \times C \to X$ be a functor such that each functor $S(p,q,-,-)$ has an end $\int_{c\in C} S(p,q,c,c)$, where $p, q \in P$. By the parameter theorems, regard these ends as a bifunctor $P^{\mathrm{op}} \times P \to X$, and S as a bifunctor $S : (P \times C)^{\mathrm{op}} \times (P \times C) \to X$. Then there is an isomorphism
$$\theta : \int_{(p,c)\in P\times C} S(p,c,p,c) \cong \int_{p\in P} \left(\int_{c\in C} S(p,p,c,c) \right).$$
Moreover, the double end on the left exists if and only if the end over P on the right exists.

Proof. For each $p, q \in P$, there is an end
$$\omega_{p,q} : \int_{c\in C} S(p,q,c,c) \ddot{\to} S(p,q,-,-).$$
Fix $x \in X$, and consider a P-indexed family of morphisms
$$\left\{ \rho_p : x \to \int_{c\in C} S(p,p,c,c) \right\}_{p\in P}.$$

The wedges $\omega_{p,q}$ give a $P \times C$-indexed family $\{\xi_{p,c}\}_{(p,c)\in P\times C}$ defined by the composition

$$\xi_{p,c} : x \xrightarrow{\rho_p} \int_{c\in C} S(p,p,c,c) \xrightarrow{(\omega_{p,p})_c} S(p,p,c,c).$$

We then have that for any $f : c \to c'$ in C, the diagram

$$\begin{tikzcd}
x \arrow[dr, "\rho_p"] \arrow[drr, bend left, "\xi_{p,c}"] \arrow[ddr, bend right, "\xi_{p,c'}"'] & & \\
& \int_{c\in C} S(p,p,c,c) \arrow[r, "(\omega_{p,p})_c"] \arrow[d, "(\omega_{p,p})_{c'}"'] & S(p,p,c,c) \arrow[d, "S(p,p,1_c,f)"] \\
& S(p,p,c',c') \arrow[r, "S(p,p,f,1_{c'})"'] & S(p,p,c,c')
\end{tikzcd}$$

commutes since $\omega_{p,p}$ is a wedge. Since $\omega_{p,p}$ is universal, every such $(P \times C)$-indexed family is such a composite for a unique family ρ. Now ρ or $\xi_{-,c}$ are extranatural in p (here we fix some $c \in C$ for ξ) if and only if one of the squares

$$\begin{tikzcd}
x \arrow[r, "\rho_p"] \arrow[d, "\rho_q"'] & \int_{c\in C} S(p,p,c,c) \arrow[d, "\int_{c\in C} S(1_p,s,c,c)"] \\
\int_{c\in C} S(q,q,c,c) \arrow[r, "\int_{c\in C} S(s,1_q,c,c)"'] & \int_{c\in C} S(p,q,c,c)
\end{tikzcd}
\qquad
\begin{tikzcd}
x \arrow[r, "\xi_{p,c}"] \arrow[d, "\xi_{q,c}"'] & S(p,p,c,c) \arrow[d, "S(1_p,s,c,c)"] \\
S(q,q,c,c) \arrow[r, "S(s,1_q,c,c)"'] & S(p,q,c,c)
\end{tikzcd}$$

commutes for each $s : p \to q$ in P. Consider the cube obtained by gluing these squares with the relevant component of ω:

$$
\begin{tikzcd}
 & x \arrow[rr, "\xi_{p,c}"] \arrow[dd] \arrow[dl, equal] & & S(p,p,c,c) \arrow[dd, "S(1_p,s,c,c)"] \\
x \arrow[rr, "\rho_p"'] \arrow[dd, "\rho_q"'] & & \displaystyle\int_{c\in C} S(p,p,c,c) \arrow[ur, "\omega_{p,p,c}"] \arrow[dd, "\int_{c\in C} S(1_p,s,c,c)"] & \\
 & S(q,q,c,c) \arrow[rr, "S(s,1_q,c,c)"'] & & S(p,q,c,c) \\
\displaystyle\int_{c\in C} S(q,q,c,c) \arrow[ur, "\omega_{q,q,c}"] \arrow[rr, "\int_{c\in C} S(s,1_q,c,c)"'] & & \displaystyle\int_{c\in C} S(p,q,c,c) \arrow[ur, "\omega_{p,q,c}"] &
\end{tikzcd}
$$

The side squares all commute by various results: the ones with ends of natural transformations commute by proposition 9.4 since $\omega_{p,p}$ is an end for $S(p,p,-,-)$ and $\omega_{p,q}$ is an end for $S(p,q,-,-)$; on the left square ρ is blocking the name of the vertical arrow, but the square reads $\xi_{q,c} = \omega_{q,q,c} \circ \rho_q$, which is true by definition; the top square commutes by the same logic. Thus, the front square commutes if and only if the back square commutes. Thus ρ_p is a wedge in p if and only if ξ is a wedge in (p,c). In particular, wedges $x \overset{..}{\to} \int_{p\in P} S(-,-,c,c)$ correspond precisely to wedges $x \overset{..}{\to} S$, and commutivity of the cube preserves universality. Thus the double end exists if and only if the end over P exists, and then universality guarentees equivalence up to isomorphism. In particular, the isomorphism is the unique arrow making the diagram

$$
\begin{tikzcd}
\displaystyle\int_{(p,c)\in P\times C} S(p,p,c,c) \arrow[d, dashed, "\theta"'] \arrow[drr, bend left, "\xi_{p,c}"] & & \\
\displaystyle\int_{p\in P}\left(\int_{c\in C} S(p,p,c,c)\right) \arrow[r, "\rho_p"'] & \displaystyle\int_{c\in C} S(p,p,c,c) \arrow[r, "\omega_{p,p,c}"'] & S(p,p,c,c)
\end{tikzcd}
$$

commute by universality. ■

Remark

This reduces the problem of a double end to that of iterated ends, but IF AND ONLY IF *all* pairs (p,q) have an end for $S(p,q,-,-)$.

Corollary 9.2 (Fubini's Theorem for Ends)

Let $S : P^{\mathrm{op}} \times P \times C^{\mathrm{op}} \times C \to X$ be a functor such that the ends $\int_{c \in C} S(p,q,c,c)$ and $\int_{p\in P} S(p,p,b,c)$ exist for all $b, c \in C$ and $p, q \in P$. By the parameter theorems, regard S as a bifunctor in p, q or b, c, respectively; then there is an isomorphism

$$\theta : \int_{p\in P} \left(\int_{c\in C} S(p,p,c,c) \right) \cong \int_{c\in C} \left(\int_{p\in P} S(p,p,c,c) \right).$$

The exterior ends exist if and only if the other exists, and the isomorphism is as described in proposition 9.5.

This theorem was not proven by Fubini, to noone's subprise. This proposition proves by proposition 9.3 the same fact for limits:

$$\varprojlim_P \varprojlim_C F(p,c) \cong \varprojlim_{P\times C} F(p,c) \cong \varprojlim_C \varprojlim_P F(p,c).$$

Duality guarentees the same for colimits and coends.

Chapter 10

Kan Extensions

This section was heavily supplimented with Riehl's *Category Theory in Context* since I was not a massive fan of Mac Lane's presentation of the Kan extension.

10.1 Adjoints and Limits

If C is a category which is J-complete or cocomplete, then limit or colimit provides a right or left adjoint for $\Delta : C \to C^J$, respectively:

$$\begin{tikzcd} & C \arrow[d, "\Delta"] & \\ & C^J \arrow[u, shift left=8, "\varprojlim"'] \arrow[u, shift right=8, "\varinjlim"] & \end{tikzcd}$$

This notion as a converse; left-adjoints can be interpreted as limits. For $\mathbf{0}$ the empty category, an initial object 0 is precisely

$$0 = \varinjlim(\mathbf{0} \to C) = \varprojlim(1_C : C \to C).$$

The equality on the right is a little new. Although it's not too difficult to work out, it's easier seen as a special case of this lemma.

Lemma 10.1

If $\lambda : d \Rightarrow 1_C$ is a cone and $F : J \to C$ is a functor such that $\lambda F : d \Rightarrow F$ is a limiting cone for F, then d is initial in C.

Proof. First, I think it's useful to reconcile the theorem statement. Note that λF is the horizontal composition $\lambda \cdot 1_F$ given by composing along the diagram

$$\begin{tikzcd} J \arrow[r, bend left, "F"] \arrow[r, bend right, "F"'] \arrow[r, phantom, "\Downarrow 1_F"] & C \arrow[r, bend left, "\Delta(d)"] \arrow[r, bend right, "1_C"'] \arrow[r, phantom, "\Downarrow \lambda"] & C. \end{tikzcd}$$

This gives us that

$$\lambda F : \Delta(d) \circ F \to 1_C \circ F.$$

Clearly, the composite $1_C \circ F = F$, but the theorem assumes $\Delta(d) \circ F = \Delta(d)$. Indeed, $\Delta(d)(x) = d$ and

$\Delta(d)(f) = 1_d$ implies that

$$(\Delta(d) \circ F)(x) = \Delta(d)(F(x)) = d, \qquad (\Delta(d) \circ F)(f) = \Delta(d)(F(f)) = 1_d.$$

Since λ is a cone, for each $f : d \to c$ in C, the diagram

$$\begin{tikzcd} & d \arrow[dl, "\lambda_d"'] \arrow[dr, "\lambda_c"] & \\ d \arrow[rr, "f"'] & & c \end{tikzcd}$$

commutes. Similarly, since λF is a cone, there exists an arrow $\lambda_{F(i)} : d \to F(i)$ for each $i \in J$, and thus the diagrams

$$\begin{tikzcd} & d \arrow[dl, "1_d"'] \arrow[dr, "\lambda_{F(i)}"] & \\ d \arrow[rr, "\lambda_{F(i)}"'] & & F(i) \end{tikzcd} \qquad \begin{tikzcd} & d \arrow[dl, "\lambda_d"'] \arrow[dr, "\lambda_{F(i)}"] & \\ d \arrow[rr, "\lambda_{F(i)}"'] & & F(i) \end{tikzcd}$$

commute since λ is a cone. Since λF is limiting, the bottom two triangles give $\lambda_d = 1_d$, and thus the top triangle gives $f = \lambda_c$. Thus each arrow $f : d \to c$ in C is equal to the unique arrow $\lambda_c : d \to c$, and thus d is initial. ■

This reduces initial objects to limits, and recall that a functor $G : A \to X$ that preserves limits has a left-adjoint precisely when each $x \in X$ gives a comma category $(x \downarrow G)$ with an initial object.

Theorem 10.1 (Criterion for Existence of an Adjoint)

A functor $F : A \to X$ has a left-adjoint if and only if

1. G preserves all limits;
2. For each $x \in X$, the limit $\varprojlim(\Pi_A^x : (x \downarrow G) \to A)$ exists in A.

The left adjoint is then constructed pointwise as

$$F(x) = \varprojlim(\Pi_A^x).$$

Proof. If G is adjoint, the comma category has an initial object $\eta_x : x \to GF(x)$, and thus the limit exists. For the converse, the projection creates the relevant limits, so by lemma 10.1, this is equivalent to Freyd's adjoint functor theorem. ■

10.2 Weak Universality

Definition 10.1: Weakly Universal

An arrow is *weakly universal* if the induced arrow is not unique. Explicitly, Given a functor $G : A \to X$ and an object $x \in X$, a weakly universal arrow from x to G is a pair $(r, w : x \to G(r))$ with $r \in A$ and w an arrow in X, such that for all $f : x \to G(a)$, there exists a (not necessarily unique) arrow $f' : r \to a$ such that the diagram

$$\begin{tikzcd} x \arrow[r, "w"] \arrow[dr, "f"'] & G(r) \arrow[d, "\exists f'"] & r \arrow[d, "\exists f'"] \\ & G(a) & a \end{tikzcd}$$

commutes.

We can thus define weak limits, weak (co)products, and so on. The book offers an application by reframing Freyd's theorem for the existence of an initial object as follows:

Proposition 10.1

A locally small, complete category D has an initial object if and only if it has a (small) set $S \subseteq D$ which is weaky initial: for all $d \in D$ there is $s \in S$ and an arrow $s \to d$.

The book offers a refinement of the previous proof.

10.3 The Kan Extension

Definition 10.2: Kan Extension

Given functors $F : C \to E$ and $K : C \to D$, a *left Kan extension* of F along K is a functor $\mathrm{Lan}_K F : D \to E$ and a natural transformation $\eta : F \Rightarrow \mathrm{Lan}_K F \circ K$ which is initial with respect to this property; hence for any other pair $G : D \to E$ and $\gamma : F \Rightarrow G \circ K$, the natural transformation γ factors uniquely through η as the horizontal composite indicated in the third diagram

$$\begin{tikzcd} C \arrow[rr, "F"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Downarrow\eta"] & E \\ & D \arrow[ur, dashed, "\mathrm{Lan}_K F"'] & \end{tikzcd} \quad \begin{tikzcd} C \arrow[rr, "F"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Downarrow\gamma"] & E \\ & D \arrow[ur, "G"'] & \end{tikzcd} \quad = \quad \begin{tikzcd} C \arrow[rr, "F"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Downarrow\eta"] & E \\ & D \arrow[ur, bend left, "\mathrm{Lan}_K F"] \arrow[ur, bend right, "G"'] \arrow[ur, phantom, "\Downarrow\exists!"] & \end{tikzcd}$$

Dually, a *right Kan extension* of F along K is one where the natural transformation ε points in the opposite direction, such that ε is final with respect to diagrams

$$\begin{tikzcd} C \arrow[rr, "F"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Uparrow\varepsilon"] & E \\ & D \arrow[ur, dashed, "\mathrm{Ran}_K F"'] & \end{tikzcd}$$

If $K : C \to D$ embeds C as a full subcategory of D, then the left and right Kan extensions become more intuitive, since F does indeed extend along K to a larger category in a universal way. I indicate the extension functor with a dashed arrow since universality of the transformation determines the functor uniquely up to (natural) isomorphism, as is always the case with universals. More precisely, there is a natural isomorphism

$$\mathrm{Nat}(F, - \circ K) \cong \mathrm{Nat}(\mathrm{Lan}_K F, -).$$

One can also think of a Kan extension of F along K as the closest approximation of F as a composition beginning with K, in that it may only differ by a universal natural transformation. These definitions can be restated in any 2-category, and we generally drop the prefix "Kan" when working outside of **Cat**. The Kan extension problem is closely related to that of adjoints for functors.

Proposition 10.2

Fix $K : C \to D$ and a category E. If the left and right Kan extensions of all functors $F : C \to E$ along K exist, then these define left- and right-adjoints to the precomposition functor $K^* : E^D \to E^C$:

$$E^D(\mathrm{Lan}_K F, -) \cong E^C(F, (-) \circ K), \qquad E^C((-) \circ K, F) \cong E^D(-, \mathrm{Ran}_K F).$$

Proof. The natural transformation in a right Kan extension is universal from K^* to F, and we already know that if all objects in a category have universal arrows from K^* to them then the arrows constitute a left-adjoint to K^*. Duality of this fact gives us the right adjoint, and so we have $\mathrm{Lan}_K(-) \dashv K^* \dashv \mathrm{Ran}_K(-)$. ■

Theorem 10.2 (Pointwise Construction)

Given functors $F : C \to E$ and $K : C \to D$, if for every $d \in D$ the colimit of $F \circ \Pi^d$ exists, for $\Pi^d : (K \downarrow d) \to C$ the projection of the comma category, then F has a left Kan extension given pointwise by

$$\mathrm{Lan}_K F := \varinjlim \left(F \circ \Pi^d\right).$$

Dually, for $\Pi_d : (d \downarrow K) \to C$ the projection, if the limits exist then the right Kan extension is given pointwise as

$$\mathrm{Ran}_K F(d) := \varprojlim \left(F \circ \Pi_d\right).$$

In both case, action given on morphisms $g : d \to d'$ is given by the unique arrow $\varinjlim F\Pi^d \to \varinjlim F\Pi^{d'}$ that commutes with the limiting cones; respectively the limit and Π_d for the right Kan extension.

Proof. First, we show the arrow can be defined as such. Any morphism $g : d \to d'$ in D induces a functor $g_* : (K \downarrow d) \to (K \downarrow d)$ by postcomposition; since postcomposition doesn't change the starting objects this commutes with the projections:

$$\begin{tikzcd} (K \downarrow d) \arrow[rr, "g_*"] \arrow[dr, "\Pi^d"'] & & (K \downarrow d') \arrow[dl, "\Pi^{d'}"] \\ & C & \end{tikzcd}$$

Take $\lambda^{d'} : F\Pi^{d'} \Rightarrow \varinjlim F\Pi^d$ to be the colimiting cone; by commutivity of the diagram above this gives a cocone $\lambda^{d'} g_*$ for $F\Pi^{d'} g = F\Pi^d$. More precisely, we get a cocone

$$\lambda^{d'} g_* : F\Pi^d \Rightarrow \Delta\left(\varinjlim F\Pi^{d'}\right) \circ g_* = \Delta\left(\varinjlim F\Pi^d\right),$$

where I have explicitly mentioned the diagonal functor to motivate the equality. Since $\varinjlim F\Pi^d$ is intial with respect to cocones for $F\Pi^d$, there is a unique morphism

$$\begin{tikzcd}
F\Pi^d(f) \arrow[rr, "F\Pi^d(\varphi)"] \arrow[dr, "\lambda^d_f"'] \arrow[ddr, bend right, "\lambda^{d'}_{gf}"'] & & F\Pi^d(h) \arrow[dl, "\lambda^d_h"] \arrow[ddl, bend left, "\lambda^{d'}_{gh}"] \\
& \mathrm{Lan}_K F(d) \arrow[d, dashed, "\mathrm{Lan}_K F(g)"] & \\
& \mathrm{Lan}_K F(d') &
\end{tikzcd}$$

where I have replaced the colimits with the images under $\mathrm{Lan}_K F$ as defined. Uniqueness implies functorality.

To complete the proof, it suffices to show $\mathrm{Lan}_K F$ is *indeed* a Kan extension, meaning we must recover the unit $\eta : F \Rightarrow \mathrm{Lan}_K F \circ K$ from the colimiting cocones. Define the component $\eta_c : F(c) \to \mathrm{Lan}_K F K(c)$ by

$$\eta_c := \lambda^{K(c)}_{1_{K(c)}} : F(c) \to (\mathrm{Lan}_K F \circ K)(c).$$

To show naturality, let $h : c \to c'$ in C. Construct the comma category $(K \downarrow K(c'))$, and consider the objects

$$(c, K(h)) : \qquad K(c) \xrightarrow{K(h)} K(c')$$

$$(c', 1_{K(c')}) : \qquad K(c') \xrightarrow{1_{K(c')}} K(c').$$

Since the diagram

$$\begin{tikzcd}
K(c) \arrow[rr, "K(h)"] \arrow[dr, "K(h)"'] & & K(c') \arrow[dl, "1_{K(c')}"] \\
& K(c') &
\end{tikzcd}$$

commutes, we have that $h : (c, K(h)) \to (c', 1_{K(c')})$. The cone $\lambda^{K(c')}$ then gives the diagram

$$\begin{tikzcd}
F\Pi^{K(c')}(c, K(h)) = F(c) \arrow[rr, "F\Pi^{K(c')}(K(h))=F(h)"] \arrow[dr, "\lambda^{K(c')}_{K(h)}"'] & & F(c') = F\Pi^{K(c')}(c', 1_{K(c')}) \arrow[dl, "\lambda^{K(c')}_{1_{K(c')}}"] \\
& \varinjlim F\Pi^{K(c')} = \mathrm{Lan}_K F(c'). &
\end{tikzcd}$$

The rightmost triangle of a previous diagram gives for any $g : d \to d'$, for all objects $(c, f) \in (K \downarrow d)$, the diagram

$$\begin{tikzcd}
& F\Pi^d(c, f) = F(c) \arrow[dl, "\lambda^d_f"'] \arrow[dr, "\lambda^{d'}_{gf}"] & \\
\mathrm{Lan}_K F(d) \arrow[rr, dashed, "\mathrm{Lan}_K F(g)"'] & & \mathrm{Lan}_K F(d')
\end{tikzcd}$$

commutes. Take $d = K(c)$, $g = K(h)$, and $(c, 1_{K(c)}) \in (K \downarrow K(c))$. Since $K(h) \circ 1_{K(c)} = K(h)$, the diagram then reads

$$\begin{tikzcd} & F(c) \arrow[dl, "\lambda^d_{1_{K(c)}}"'] \arrow[dr, "\lambda^{K(c')}_{K(h)}"] & \\ \mathrm{Lan}_K F(K(c)) \arrow[rr, dashed, "\mathrm{Lan}_K F(h)"'] & & \mathrm{Lan}_K F(K(c')). \end{tikzcd}$$

Combining this diagram with a previous one along the diagonal gives

$$\begin{tikzcd} F(c) \arrow[r, "F(h)"] \arrow[d, "\eta_c = \lambda^{K(c)}_{1_{K(c)}}"'] \arrow[dr, "\lambda^{K(c')}_{K(h)}"] & F(c') \arrow[d, "\lambda^{K(c')}_{1_{K(c')}} = \eta_{c'}"] \\ \mathrm{Lan}_K F(K(c)) \arrow[r, "\mathrm{Lan}_K F(K(h))"'] & \mathrm{Lan}_K F(K(c')). \end{tikzcd}$$

This gives that $\eta : F \Rightarrow \mathrm{Lan}_K F \circ K$ is natural.

Finally, we show η is universal. Let (G, γ) satisfy the extension criterion. Note that for all $f : K(c) \to d$ in D, there is an arrow $\lambda^d_f : F(c) \to \mathrm{Lan}_K F(d)$. Note also that there is an arrow

$$F(c) \xrightarrow{\gamma_c} GK(c) \xrightarrow{G(f)} G(d).$$

Notably, $G(-) \circ \gamma_{(-)}$ is a co-cone, and by universality there exists α_d making the diagram

$$\begin{tikzcd} F(c) \arrow[r, "\lambda^d_f"] \arrow[d, "\gamma_c"'] & \mathrm{Lan}_K F(d) \arrow[d, dashed, "\alpha_d"] \\ GK(c) \arrow[r, "G(f)"'] & G(d) \end{tikzcd}$$

commute. Here, we assume there exists some $c \in C$ and some $f : K(c) \to d$ for each $d \in D$; if there exists some d for which there exist no such f, then $K \downarrow d) = \mathbf{0}$ is the empty category, and $\mathrm{Lan}_K F(d) = \varinjlim(\mathbf{0} \to E)$ is the colimit of the empty functor, and hence an initial object, meaning $\alpha_d : \mathrm{Lan}_K F(d) \to G(d)$ is unique by universality in E.

To show α is natural $\mathrm{Lan}_K F \Rightarrow G$, let $g : d \to d'$ in D, and consider the diagram

$$\begin{tikzcd} & F(c) \arrow[dl, "\lambda^d_f"'] \arrow[dr, "\lambda^{d'}_{gf}"] & \\ \mathrm{Lan}_K F(d) \arrow[rr, "\mathrm{Lan}_K F(g)"] \arrow[d, "\alpha_d"'] & & \mathrm{Lan}_K F(d') \arrow[d, "\alpha_{d'}"] \\ G(d) \arrow[rr, "G(g)"'] & & G(d'). \end{tikzcd}$$

Here we once again assume such an f exists; if no such f exists then the diagram commutes immediately simply by uniqueness. Otherwise, note by definition that $\alpha_d \circ \lambda^d_f = G(f) \circ \gamma_c$, and also that $\alpha_{d'} \circ \lambda^{d'}_{gf} =$

$G(gf) \circ \gamma_c = G(g) \circ G(f) \circ \gamma_c$. Expanding the pentagon along these equalities gives

$$\begin{tikzcd}
F(c) \arrow[r, equal] \arrow[d, "\gamma_c"'] & F(c) \arrow[d, "\gamma_c"] \\
GK(c) \arrow[d, "G(f)"'] & GK(c) \arrow[d, "G(f)"] \\
G(d) \arrow[d, equal] & G(d) \arrow[d, "G(g)"] \\
G(d) \arrow[r, "G(g)"'] & G(d').
\end{tikzcd}$$

Since this commutes, so does the outer pentagon, and since the top triangle commutes by definition, the pentagon diagram commutes fully, rendering α natural. The commutivity of this square in fact forces α to be defined in this way, making it unique.

Finally, it remains to be shown $\alpha K \circ \eta = \gamma$. This is in fact a special case of the definition of α; for each $c \in C$, consider the object $(c, 1_{K(c)}) \in (K \downarrow K(c))$. This gives the diagram

$$\begin{tikzcd}
F(c) \arrow[r, "\lambda^{K(c)}_{1_{K(c)}}", "=:\eta_c"'] \arrow[d, "\gamma_c"'] & \mathrm{Lan}_K F(K(c)) \arrow[d, "\alpha_{K(c)}"] \\
GK(c) \arrow[r, equal, "G(1_{K(c)})"'] & GK(c)
\end{tikzcd}$$

which commutes as previously stated. ■

Corollary 10.1

If C is small and D is locally small, then

- if E is complete, any left Kan extension along any $K : C \to D$ exists;
- If E is cocomplete, any right Kan extension along any $K : C \to D$ exists.

Corollary 10.2

If $K : C \to D$ is fully faithful, then the (co)unit $\varepsilon : \mathrm{Lan}_K F \circ K \cong F$ is an isomorphism.

Proof. If $c \in C$, $\mathrm{Lan}_K F(c)$ is a limit over the comma category $(K \downarrow K(c))$. Since K is fully faithful, each $f : K(d) \to K(c)$ can be written as some $f = K(h)$ for a unique $h : d \to c$. Thus $1 : K(c) \to K(c)$ is initial in this category, we have

$$\mathrm{Lan}_L F(K(c)) = \varinjlim F\Pi^{K(c)} = F\Pi^{K(c)}(c, 1_{K(c)}) = F(c).$$

Hence $\eta = 1$. ■

10.4 Kan Extensions as Coends

Theorem 10.3 (Kan Extensions as Coends)

Let $F : C \to E$ and $K : C \to D$ be functors such that for all $c, c' \in C$ and all $d \in D$, the copowers $\mathrm{Hom}_C(K(c'), d) \cdot T(c)$ in E. Then if for all $d \in D$ the following coend exists, then F has a left Kan extension, with object function given by

$$\mathrm{Lan}_K F(d) = \int^{c \in C} \mathrm{Hom}_C(K(c), d) \cdot F(c).$$

Proof. By universality, it suffices to show there is a natural isomorphism of sets $\mathrm{Nat}(\mathrm{Lan}_K F, -) \cong \mathrm{Nat}(F, (-) \circ K)$. By the parameter theorem, there is indeed a functor

$$d \mapsto \int^{c \in C} \mathrm{Hom}_C(K(c), d) \cdot F(d).$$

We compare it with another functor $G : D \to E$. Recall that for any $U, V : C \to D$, we have

$$\mathrm{Nat}(U, V) = \int_{c \in C} \mathrm{Hom}_D(U(c), V(c)).$$

Applying this here, along with the Yoneda lemma, gives natural isomorphsms

$$\begin{aligned}
\mathrm{Hom}_E(F(c), SK(c)) &\cong \mathrm{Nat}(\mathrm{Hom}_D(K(c), -), \mathrm{Hom}_E(F(c), S(-)) \\
&\cong \int_{d \in D} \mathrm{Hom}_{\mathbf{Set}}(\mathrm{Hom}_C(K(c), d), \mathrm{Hom}_E(F(c), S(d))).
\end{aligned}$$

Here we must take **Set** sufficiently large to avoid foundational problems; I discard the special notation because I don't feel like writing it. This will be important later.

For now, we have the following sequence of isomorphisms:

$$\begin{aligned}
\mathrm{Nat}(\mathrm{Lan}_K F, S) &\cong \int_{d \in D} \mathrm{Hom}_E(\mathrm{Lan}_K F(d), S(d)) \\
&\cong \int_{d \in D} \mathrm{Hom}_E\left(\int^{c \in C} \mathrm{Hom}_C(K(c), d) \cdot F(c), S(d)\right) \\
&\cong \int_{d \in D} \int_{c \in C} \mathrm{Hom}_E(\mathrm{Hom}_C(K(c), d) \cdot F(c), S(d)) \\
&\cong \int_{d \in D} \int_{c \in C} \mathrm{Hom}_{\mathbf{Set}}(\mathrm{Hom}_D(K(c), d), \mathrm{Hom}_E(F(c), S(d))) \\
&\cong \int_{c \in C} \int_{d \in D} \mathrm{Hom}_{\mathbf{Set}}(\mathrm{Hom}_D(K(c), d), \mathrm{Hom}_E(F(c), S(d))) \\
&\cong \int_{c \in C} \mathrm{Hom}_E(F(c), SK(c)) \\
&\cong \mathrm{Nat}(F, SK).
\end{aligned}$$

Some explanation of the various steps is in order. The first isomorphism is the above definition of Nat via ends; the second is the definition of $\mathrm{Lan}_K F$; the third is by continuity of $\mathrm{Hom}(-, X)$; the fourth will be explained later; the fifth is fubini, which applies since both ends exist; the sixth follows by our earlier discussion, and the final step is once again the definition of Nat via ends. Note all of these steps are natural in S, which suffices to prove the theorem.

It now suffices to show the fourth isomorphism. Explicitly writing the copower as a coproduct for clarity gives, by continuity of Hom,

$$\mathrm{Hom}_E\left(\coprod_{\mathrm{Hom}_C(K(c),d)} F(c), S(d)\right) \cong \prod_{\mathrm{Hom}_C(K(c),d)} \mathrm{Hom}_E(F(c), S(d)).$$

Recall that repeated products in **Set** are defined (for finite products, up to isomorphism) as exponentials, equivalently as hom-sets,

$$\prod_{\mathrm{Hom}_C(K(c),d)} \mathrm{Hom}_E(F(c), S(d)) \coloneqq \mathrm{Hom}_{\mathbf{Set}}(\mathrm{Hom}_C(K(c), d), \mathrm{Hom}_E(F(c), S(d))).$$

This shows the given isomorphism. ■

The dual statement gives

$$\mathrm{Ran}_K F(d) = \int_{c \in C} F(c)^{\mathrm{Hom}_C(L(c),d)}$$

valid when powers (repeated products) and the end exist.